UNIVERSIDADE FEDERAL DE SÃO JOÃO DEL-REI – UFSJ
CAMPUS CENTRO-OESTE DONA LINDU – CCO
PROGRAMA DE PÓS-GRADUAÇÃO EM BIOTECNOLOGIA

Molecular Architect (MolAr), um workflow para Triagem Virtual de Alto Desempenho

Eduardo Habib Bechelane Maia

Divinópolis, 03 de dezembro de 2019

UNIVERSIDADE FEDERAL DE SÃO JOÃO DEL-REI – UFSJ
CAMPUS CENTRO-OESTE DONA LINDU – CCO
PROGRAMA DE PÓS-GRADUAÇÃO EM BIOTECNOLOGIA

Molecular Architect (MolAr), a workflow for virtual High-Throughput Screening (vHTS)

Eduardo Habib Bechelane Maia

Divinópolis, 03 de dezembro de 2019

UNIVERSIDADE FEDERAL DE SÃO JOÃO DEL-REI – UFSJ
CAMPUS CENTRO-OESTE DONA LINDU – CCO
PROGRAMA DE PÓS-GRADUAÇÃO EM BIOTECNOLOGIA

Molecular Architect (MolAr), a workflow for virtual High-Throughput Screening (vHTS)

Final thesis presented to the graduate Program in Biotechnology, in the Federal University of São João del-Rei as a partial fulfilment of the requirements for the Doctor degree in Biotechnology.

Advisor: Prof. Dr. Alex Gutterres Taranto

Co-Advisor: Prof. Dr. Alisson Marques da Silva

UNIVERSIDADE FEDERAL DE SÃO JOÃO DEL-REI – UFSJ

CAMPUS CENTRO-OESTE DONA LINDU – CCO

PROGRAMA DE PÓS-GRADUAÇÃO EM BIOTECNOLOGIA

Ficha catalográfica elaborada pela Divisão de Biblioteca (DIBIB)
e Núcleo de Tecnologia da Informação (NTINF) da UFSJ,
com os dados fornecidos pelo(a) autor(a)

M217m

Maia, Eduardo Habib Bechelane.
Molecular Architect (MolAr), um workflow para Triagem Virtual de Alto Desempenho / Eduardo Habib Bechelane Maia ; orientador Alex Guterres Taranto; coorientador Alisson Marques da Silva. -- Divinópolis, 2019.
154 p.

Tese (Doutorado - Programa de Pós-Graduação em Biotecnologia - Doutorado) -- Universidade Federal de São João del-Rei, 2019.

1. Virtual Screening. 2. Structure-based Virtual Screening. 3. Computer-Assisted Drug Design. I. Taranto, Alex Guterres, orient. II. Silva, Alisson Marques da, co-orient. III. Título.

UNIVERSIDADE FEDERAL DE SÃO JOÃO DEL-REI – UFSJ

CAMPUS CENTRO-OESTE DONA LINDU – CCO

PROGRAMA DE PÓS-GRADUAÇÃO EM BIOTECNOLOGIA

UNIVERSIDADE FEDERAL DE SÃO JOÃO DEL-REI – UFSJ
INSTITUÍDA PELA LEI Nº 10.425, DE 19/04/2002 – D.O.U. DE 22/04/2002
PROGRAMA DE PÓS-GRADUAÇÃO EM BIOTECNOLOGIA

Universidade Federal de São João del-Rei

ATA DE DEFESA DE TESE

DISCENTE: EDUARDO HABIB BECHELANE MAIA

NÍVEL: DOUTORADO

DATA DA DEFESA: 19 DE NOVEMBRO DE 2019

HORÁRIO DE INÍCIO: 13H30 **LOCAL:** SALA 201 C / UFSJ / CCO

MEMBROS DA BANCA	FUNÇÃO	TÍTULO	INSTITUIÇÃO DE ORIGEM
ALEX GUTTERRES TARANTO	Presidente	Doutora	UFSJ
ALISSON MARQUES DA SILVA	Titular	Doutora	CEFET-MG
THIAGO DE SOUZA RODRIGUES	Titular	Doutora	CEFET-MG
ANGELO AMANCIO DUARTE	Titular	Doutor	UEFS
LUIZ GUILHERME MACHADO DE MACEDO	Titular	Doutor	UFSJ
LUIS FERNANDO SOARES	Titular	Doutor	UFSJ

TÍTULO DA DISSERTAÇÃO:
"Molecular Architect (MoLar), um workflow para Triagem Virtual de Alto Desempenho."

Em sessão pública, após apresentação da tese durante 55 minutos, o discente foi argüido oralmente pelos membros da banca por 120 minutos. Em seguida, reunida em sessão secreta, às 17:00 horas, a banca considerou o discente acima identificado APROVADO.

(Aprovado ou Não Aprovado)

Na forma regulamentar, foi lavrada a presente ata, que é abaixo assinada pelos membros da banca, na ordem acima relacionada e pelo discente.

Local e data: Divinópolis, 19 de novembro de 2019

Presidente: _____ Membro 1: _____

Membro 2: _____ Membro 3: _____

Membro 4: _____ Membro 5: _____

Candidato: _____

Obs.: O aluno deverá encaminhar à coordenação do curso, no prazo máximo de 90 dias os exemplares definitivos da dissertação ou tese.

UNIVERSIDADE FEDERAL DE SÃO JOÃO DEL-REI – UFSJ

CAMPUS CENTRO-OESTE DONA LINDU – CCO

PROGRAMA DE PÓS-GRADUAÇÃO EM BIOTECNOLOGIA

To my family

Acknowledgements

Firstly, I would like to express my sincere gratitude to my advisor Prof. Alex Gutterres Taranto for the continuous support of my Ph.D study. Thank you for your encouragement, patience, dedication and example. I am honored to have had you as my advisor.

Special thanks to my Co-advisor Prof. Alisson Marques da Silva, for his constant disposition, guidance, encouragement, availability and friendship.

I thank my fellow labmates in for the stimulating discussions, and for all the funny moments we have had in the lab in the last years. In particular, I am grateful to Caique, Jéssica, Renata and Felipe for their patience and assistance, especially at the beginning of my doctoral journey.

I would like to thank my family: my parents and to my brother and sister for supporting me spiritually throughout writing this thesis and my life in general.

I am grateful to my wife, Ellen and my son, Mateus, whose company I had to sacrifice several times during these years of hard work. I owe them many hours of company, jokes and laughter.

Last but not the least, I am grateful to God who enlightened my path and put these extraordinary people in my life.

Resumo

A descoberta de novos medicamentos é um desafio para a indústria farmacêutica e o uso do Desenho de Fármacos Assistido por Computador (DFAC) pode ser destacado, pois essa técnica permite um melhor direcionamento dos experimentos, reduzindo o tempo e os custos envolvidos na pesquisa. Uma das utilizações mais comuns do DFAC é a realização da Triagem Virtual (TV). Essa técnica permite, através da aplicação de modelos teóricos, uma melhor seleção de moléculas com características físico-químicas desejadas. Existem softwares gratuitos utilizados para esse fim, mas geralmente são difíceis de usar, não executam todo o processo de TV, o que exige a combinação de vários softwares diferentes, além de muitas vezes não possuírem uma interface gráfica e exigirem conhecimentos avançados de computação, como linguagens de programação. Nesse contexto, esta tese apresentará o desenvolvimento do MolAr, um workflow com interfaces simples e intuitivas que atuam de maneira integrada para executar todo o processo de VS de forma automatizada. O MolAr foi avaliado em três estudos de caso. No primeiro, o MolAr foi utilizado para realizar um estudo de modelagem por homologia e de TV para determinar se a isatina e a convolutamidina, bem como seus derivados, têm algum efeito inibitório sobre a ciclooxigenase-1, ciclooxigenase-2 e o opióide Mu. O segundo estudo de caso verificou a viabilidade do uso de MolAr em sistemas de DNA-ligante. Verificou-se que no estudo de caso avaliado, tanto o Autodock Vina quanto o DOCK 6 tiveram bons resultados nessa tarefa, mas o uso da triagem virtual de consensus foi capaz de enriquecer os resultados. O terceiro estudo de caso demonstrou que a abordagem para executar o DOCK 6 usando a conformação final do ligante fornecida pela Autodock Vina melhorou o desempenho do DOCK 6 durante a TV e, consequentemente, melhorou os resultados da TV. Por fim, o MolAr já está sendo utilizado por pesquisadores de outros centros de pesquisa, o que demonstra que alcançou o objetivo proposto de auxiliar a execução de todo o processo de VS de maneira efetiva.

Palavras-chave: *Modelagem da Homologia, Triagem Virtual, Triagem Virtual Baseada em Estrutura, Funções de Pontuação, Design de Medicamentos Assistidos por Computador*

Abstract

Discovery new drugs is a major challenge for pharmaceutical industry and the use of Computer-Assisted Drug Design (CADD) can be highlighted. These methods allow a better addressing of the experiments, which can reduce the time and costs involved in research of new drugs. One of the most common uses in CADD is to perform screening of virtual compound libraries, also known as Virtual Screening (VS). This technique allows, through the application of theoretical models, a better selection of molecules with desired physical chemical characteristics and able to become drugs. There are many free tools used for this purpose, but they are often difficult to use and usually do not have a graphical interface. In addition, several tools must be used to perform the entire VS process, requiring the user to process the results of one software program to be used in other software, which is a source of human error. Furthermore, there are still some programs that require knowledge of advanced computational skills, such as programming languages. In this context, this thesis will show the development of MolAr, a workflow with simple and intuitive interfaces that act in an integrated manner to perform the entire VS process in an automated form. MolAr was evaluated in three case studies. In the first one, MolAr was used to perform a homology modeling and Virtual Screening study to determine whether Isatin and Convolutamidine, as well as its derivatives, have an inhibitory effect on cyclooxygenase-1, cyclooxygenase-2 and Mu opioid molecular target. The second case study verified the feasibility of using MolAr in DNA-Ligand systems. It was verified that in the case study evaluated, both the Autodock Vina and the DOCK 6 presented good presented good reliability, when considering the values of the AUC-ROC, but the use of consensus virtual screening was able to enrich the results. The third case study demonstrated that the approach to carry out DOCK 6 using the final ligand conformation provided by Autodock Vina improved DOCK 6 performance during VS and consequently improved the VS results. Finally, the software is already being used by many researchers from various research centers, which demonstrates that it achieved the proposed goal of assisting the execution of the whole VS process in an easy way.

Keywords: *Homology Modeling, Virtual Screening, Structure-based Virtual Screening, Scoring Functions, Computer-Assisted Drug Design*

List of Abbreviations

ANNs	Artificial neural networks
AUC-ROC	Area under the curve ROC
BraMMT	Brazilian Malaria Molecular Targets
CADD	Computer Aided Drug Design or Computer-Assisted Drug Design
CD	Consensus docking
CSA	Conformational space annealing
CVS	Consensus Virtual Screening
DOPE	Discrete Optimized Protein Energy
EBI	European Institute of Bioinformatics
EMBL	European Molecular Biology Laboratory
FF	Force field
INPI	Instituto Nacional de Propriedade Intelectual
IUPAC	International Union of Pure and Applied Chemistry
IVS	Inverse Virtual Screening
KNN	k-nearest neighbors
MDDR	MDL Drug Data Report
MIFs	Molecular interaction fields
MolAr	Molecular Architect
MQE	Model quality estimation
NMR	Nuclear Magnetic Resonance
OOMT	Our Own Molecular Target
OSS	Open source license
PDB	Protein Data Bank
PM7	Parametric Method 7
QSAR	Quantitative structure-activity relationship
R&D	Research and Development
RMSD	Root mean square deviation
ROC	Receiver operating characteristic
SA	Simulated annealing
SBDD	Structure-based Drug Design
SBVS	Structure-based virtual screening
TBVS	Target-based virtual screening
UFSJ	Federal University of São João del-Rei
vHTS	Virtual high throughput screening
VS	Virtual Screening

List of Figures

Figure 1 – General VS scheme 20
Figure 2 – Identification of a ligand candidate by a typical scoring function 23
Figure 3 - Consensus Docking approach.....26
Figure 4 - RMSD between the 2YND protein and a model generated by homology modeling using 4C68 and 3IU1 proteins as template 35
Figure 5 - ROC Curve example of the 4O1Z protein using 5 active ligands and 250 decoys 37
Figure 6 - Ramachandran plot of the protein 4PH9. In this figure, 2 amino acids are in disallowed regions and so they are highlighted in red.....38
Figure 7 - MolAr methodology workflow.....54
Figure 8 – Homology Modeling Workflow performed by MolAr 55
Figure 9 - Missing residues workflow performed by MolAr.....57
Figure 10 – Multi chain modeling performed by MolAr.....57
Figure 11 - Octopus Workflow performed by MolAr 59
Figure 12 - DOCK 6 Workflow performed by MolAr 59
Figure 13 – Consensus Virtual Screening Workflow performed by MolAr 61
Figure 14 - License setup screen displayed during MolAr installation setup 63
Figure 15 - MolAr main screen 64
Figure 16 - Target Builder menu.....65
Figure 17: MolAr homology modeling screen (a) Sequence Data; (b) Simulation Parameters; (c) Other Options.....66
Figure 18 – Select Templates Screen displayed by MolAr during the homology modeling process 67
Figure 19 – Homology modeling result screen.....68
Figure 20 - Ramachandran plot of the model selected on Figure 19.....69
Figure 21 – Multi chain modeling screen 70
Figure 22 - Multi chain modeling result screen 70
Figure 23 - Missing gaps screen.....71
Figure 24 - Missing gaps result screen 71
Figure 25 - Octopus submenu 72
Figure 26 - Run Mopac screen 72
Figure 27 - Octopus main screen 73
Figure 28 - OOMT database.....74

Figure 29 – Contents of the folder 1AGW of OMMT database....74
Figure 30 - conf file for 1AGW protein....74
Figure 31 – Database Manager: (a) Create database main screen (b) Configure database screen; (c) Fix database screen; (d) Edit database screen....75
Figure 32 - Redock basic screen....76
Figure 33 - Redock advanced screen....77
Figure 34 - Virtual screening with DOCK 6 main screen....78
Figure 35 - Virtual Screening with DOCK 6 main screen after pressing the advanced options button....78
Figure 36 – VS with DOCK 6 result screen....78
Figure 37 - Filter being performed by MolAr in the ligand column....79
Figure 38 - ROC Curve for 4O1Z VS....79
Figure 39 - CVS main screen....80
Figure 40 - CVS result screen....81
Figure 41 – View sub menu (a) Open PDB file screen with Jmol; (b) Open PDB file screen with PyMol; (c) Visualization of the 3D structure of 2YND protein in JMol; (d) Visualization of the 3D structure of 2YND protein in Pymol....82
Figure 42 - Ramachandran plot screen....83
Figure 43 - Ramachandran plot of the 4XWK protein using MolAr....83
Figure 44 - RMSD calculation between two proteins after homology modeling....84
Figure 45 - Adjust the protonation state screen....85
Figure 46 - Adjust protein protonation state....85
Figure 47 - Adjust ligand protonation state....86
Figure 48 – ROC Curve main screen....86
Figure 49 - ROC Curve for the protein 4PH9 generated by the ROC curve screen....87
Figure 50 - General Structure of (a) Isatin (b) Convolutamydine....88
Figure 51 - ROC Curves for COX-1, COX-2 and Mu opioid targets....91
Figure 52 - 1VZK (a) 3D Structure view (b) Interactions in 2D....96
Figure 53 – ROC curves (a) ROC curve obtained after performing VS with Autodock Vina; (b) ROC curve obtained after performing VS with DOCK 6; (c) ROC curve obtained after performing consensus virtual screening. 97
Figure 54 – DUD-E experiments workflow....105

List of tables

Table 1 – Virtual Screening Software programs 44
Table 2 - Programs installed by MolAr on Linux and why they are needed 52
Table 3 - Libraries installed by MolAr on Linux and why they are needed.....53
Table 4 – Program installed by MolAr on Windows and why they are needed 53
Table 5 – Isatin and Isatin derivatives.....88
Table 6 - Convolutamydine and synthetic analogues obtained from isatins 88
Table 7 - Virtual Screening results for COX-1 and COX-2 92
Table 8 - Virtual Screening results for Mu Opioid receptor.....94
Table 9 - Position of the active ligands identified by DOCK 6, Autodock Vina and Consensus 98
Table 10 - Subset of Targets chosen in DUD38 99
Table 11 - Interactions of crystallographic ligands with the targets used in this case study.100
Table 12 - Comparison between running DOCK 6 on ligands conformation provided by Autodock Vina or running DOCK 6 on original ligands.....105

Summary

1. INTRODUCTION 11

2. LITERATURE REVIEW 17

2.1 *De novo* drug design 17

2.2 Computer-Assisted Drug Design (CADD) 17

2.3 Virtual Screening (VS) 19
2.3.1 Structure-based virtual screening (SBVS) 21

2.4 Scoring Functions 23

2.5 Consensus Docking 25

2.6 Virtual Databases 27

2.7 Development of Models for Drug Design by Homology Modeling 29
2.7.1 SWISS-MODEL 31
2.7.2 MODELLER 32

2.8 Methods of evaluating the quality of a model and simulation 33
2.8.1 Root Mean Square Deviation (RMSD) 33
2.8.2 ROC Curve and AUC 35
2.8.3 Ramachandran Plot 37

2.9 Protein Protonation State 38

2.10 VIRTUAL SCREENING ALGORITHMS 40

2.11 VS SOFTWARE PROGRAMS 42

3. JUSTIFICATIONS 49

4. OBJECTIVES 50

4.1 Overall Objective 50

4.2 Specific Objectives 50

5. METHODOLOGY 51

5.1 Homology Modeling 55

5.2 VS with Autodock Vina 58

5.3 VS with DOCK 6 59

5.4 Consensus Virtual Screening (CVS) 61

6. RESULTS AND DISCUSSION 62

6.1 Installation setup 63

6.2 MolAr 63
6.2.1 Target Builder Menu 64
6.2.2 Docking Menu 71
6.2.3 Tools Menu 81

6.3 Case studies 87
6.3.1 Evaluating of Pharmacological effects of Isatine and Convolutamine on cyclooxygenase-1, cyclooxygenase-2 and Mu opioid molecular target 87
6.3.2 Investigation of the best in silico model for DNA-Ligand systems 95
6.3.3 Evaluation of the consensus virtual screening approach implemented by MolAr 98

7. FINAL CONSIDERATIONS AND FUTURE WORK 107

REFERENCES 110

APPENDIX 131

Appendix A – INPI RUNMOPAC registration 131

Appendix B – INPI MolAr registration 132

Appendix C – INPI patent application 133

Appendix D – Published paper 136

Appendix E – Paper submitted to Frontiers in Chemistry 137

Appendix F – Download history of the first software version 138

Appendix G – MolAr default parameters 143

1. Introduction

The increased cost and evolution of medicines available in the last century have led to an improvement in the quality of life of the world population. However, while on average the quality of life has been improved, a third of the population is still without access to essential medicines, which means that more than 2 billion people cannot afford to buy basic medicines (CHAN, 2017; LEISINGER; GARABEDIAN; WAGNER, 2012). This problem is even worse in some places in Africa and Asia, where more than 50% of the people face problems obtaining medicines (LEISINGER; GARABEDIAN; WAGNER, 2012). Moreover, throughout the world, more than 18 million deaths occurred every year could be avoided, as well as tens of millions of deaths related to poverty and lack of access to essential medicines. (SRIDHAR, 2008). The price of many medicines is inaccessible to limited-income populations and middle-income countries (STEVENS; HUYS, 2017) and this is probably the biggest factor that makes access to medicines difficult.

While there is a need to increase the population's access to medicines, the pharmaceutical industry is facing unprecedented challenges in its business model (PAUL et al., 2010). Nevertheless, the drug development process is time-consuming, laborious and expensive. Developing a new drug for treating the Alzheimer's disease, for example, can cost as much as $ 5.6 billion, from the start of research to the launch of the new drug (CUMMINGS; REIBER; KUMAR, 2018). The development of a new drug has an average cost between 1 and 2 billion USD and could take 10 to 17 years (LEELANANDA; LINDERT, 2016), since it has to move through all phases for new drug development, from target discovery to drug registration. Even so, Arrowsmith (ARROWSMITH, 2012) shows that the probability of a drug candidate reach the market after entering at phase I of the clinical trials fell down from 10% in the 2002-2004 period to about 5% between 2006 and 2008. So, it represents a 50% decrease in just four years. In the period from 2001 to 2011, although there was an increase in the revenues of pharmaceutical companies, the Research and Development (R&D) contribution to this revenue decreased (ARROWSMITH, 2012). In addition, annual approval of lead drugs has not been satisfactory in the past decade (ZHAO; GUO, 2009). Thus, the increase in available resources have not mean an increase in the number of new drugs. Over the period of 1960 and 2000, the costs of developing new drugs have been

increased nearly 9 fold (MORGAN et al., 2011). Morgan (MORGAN et al., 2011) explains that it can be attributed to failure of drugs at later stages and higher costs at each stage in the drug development process. When a drug candidate proves inefficient in the late stages of the development flow, there is a higher waste of money because the aggregate cost of discovering a new drug grows in the final stages of the drug development cycle, given the effort has already been made in its development.

Thus, a high failure rate of new drug candidates in clinical trials (ARROWSMITH, 2012) demands the development of ways to select the compounds most likely to show bioactivity, since fewer drugs would fail in late stages of the development flow. So, to decrease the cost of discovering a new drug, it is necessary to increase the probability that a new compound will pass through all stages and reach the end of the development cycle.

According to Zhao and Guo (ZHAO; GUO, 2009), a small number of drugs with new mechanisms of action are discovered every year (two to six per year). Many drugs with the same mechanism of action are developed in parallel and the main differences between them come from different optimization protocols and chemo-types of the molecule in question.

Several approaches have been used as a solution to the low number of new drugs arriving in the market. Among them, it could be cited de novo drug design (DEVI; SATHYA; COUMAR, 2015; JAIN; AGRAWAL, 2004), drug repositioning (PADHY; GUPTA, 2011), the discovery of new drugs based on similarities (WERMUTH, 2006) and me too (also known as follow-on) (ZHAO; GUO, 2009).

One form to minimize cost and time in drug development process that has been used extensively recently is the Computer Aided Drug Design (CADD) methodologies also known as Molecular Modeling. Oglic (OGLIC et al., 2018) defines CADD as a cyclic process of developing new drugs in which all stages of design and analysis are performed by computer programs, operated by medicinal chemistries.

In the last decades, a great number of emerging technologies have been applied in research projects and development of new medicines (RODRIGUES et al., 2012). In vitro and in vivo experiments are still essential in the development of new drugs. However, in some cases there is little experimental information, since they are generally of high operational cost and time consuming (VERLI, 2014). In this context,

the development of drugs using computational simulation (*in silico*) techniques, which employ Molecular Modeling techniques, has gained prominence, since it enables the researcher to simulate and predict several essential factors, such as toxicity, activity, bioavailability and efficacy, before even the compound undergoes in vitro tests. Thus, it allows a greater planning and targeting, which can reduce the time and the expenses with research (FERREIRA; GLAUCIUS; ANDRICOPULO, 2011). In addition, due to the high cost of experimental methods, the prior use of in silico techniques is a good alternative for reducing costs in the development of a new drug.

Computational chemists increasingly apply CADD. Thus, they use theoretical methods and computational techniques to model or reproduce the behavior of molecules, which makes them capable of solving a series of chemical and biological problems. This has enabled the prediction of physical-chemical properties and biological activity (LIONTA et al., 2014). In this way, computational tools were created and improved to aid in the discovery and improvement of new chemical compounds that have biological activity against a molecular target.

CADD represents a fast and valid way for scientific research of new compounds with pharmacological potential (FERREIRA; GLAUCIUS; ANDRICOPULO, 2011; RIPPHAUSEN; NISIUS; BAJORATH, 2011). It makes possible to analyze a large number of molecules in a short period of time, demonstrating how they bind to receptors of pharmacological interest, even before their synthesis. In summary, the rational drug design process begins with the identification of molecular targets for a given compound (natural or synthetic), followed by careful molecular analysis of the major molecular recognition forces between ligand and receptor, design of the most promising molecules, and finally the experimental assay (KONG et al., 2011).

The large number of compounds makes it impossible to experimentally analyze all possible ligands of a given molecular target. Thus, in order to evaluate many molecules simultaneously, it is necessary to employ fast methods to select the most promising compounds and to increase the probability of identification of a drug candidate. Virtual Screening (VS) is a fast and cost effective approach used for screening a virtual database and currently, it is one of the main techniques used in the development of new drugs (LOURENÇO et al., 2015). VS uses computational methods and considering the number of potential biological targets for new drugs, it plays a

prominent role among current strategies for the identification of promising bioactive substances (CARREGAL et al., 2017). VS is a theoretical approach that allows the identification of new hits in large compound libraries, from the three-dimensional structure of the molecular target using docking software. These software programs measure the affinity of a small molecule (the ligand) to a molecular target to determine the interaction energy of the resulting complex. Furthermore, from the complex between ligand and target, the visualization software may exhibit the intermolecular interaction that is responsible for molecular recognition.

In this scenario, there are proprietary and free software. Top proprietary software such as Sybyl-X (TRIPOS, 2011), MOE (CHEMICAL COMPUTING GROUP, 2004), OpenEye (HAWKINS; SKILLMAN; NICHOLLS, 2007), Schrodinger (DELANO, 2002), Discovery Studio (SAN DIEGO: ACCELRYS SOFTWARE INC., 2015), as well as the free ones Autodock Vina (TROTT; OLSON, 2009) and DOCK 6 (ALLEN et al., 2015) assist researchers in the process of VS and already use the most modern algorithms and computational techniques.

In general, to use this proprietary software, it is necessary to pay for an annual license even when the license is to be used in a single machine. For example, in a price survey, conducted in January 2019, with some companies developing proprietary software used in VS, the value, in dollars, of the annual license for a single machine of MOE is $2,550, OpenEye is $10,900 and Sybyl-X is $ 21,000. The amount of money needed can increase even more depending on the license options, which makes it difficult to access these tools.

On the other hand, existing free software does not perform the whole VS process in and automated way (e.g. Homology Modeling, construct loop regions, adjust protonation state, run the virtual screening). So, it is necessary to combine several free software if researcher want to use them to perform the whole VS process, such as Modeller (WEBB; SALI, 2016), Pymol (DELANO, 2002), MGLTools (MORRIS et al., 2009), Chimera (PETTERSEN et al., 2004), Autodock Vina (TROTT; OLSON, 2009) and DOCK 6 (ALLEN et al., 2015). In this way, much more human intervention is required than if the whole process were automated, which makes the whole process more prone to human errors.

Another problem using free software is the complexity of their interfaces, which makes them very difficult to use. Their interfaces are mainly non-intuitive, with confusing or nonexistent error messages and therefore difficult to use, not serving users expectations. In many of the free software the problem is even worse, as they have no graphical interface and all its use is done by command line. In some cases, such as the MODELLER (ŠALI, 2013), user need prior knowledge in computer programming languages.

Thus, improving software accessibility can make technologies more available to chemists and biologists with minimal computing skills. Hence, making the use of software easier, which on the other hand would require specialized computer skills such as software maintenance, bioinformatics techniques knowledge and the manipulation and maintenance of large biological databases can be of great value in the success of the VS process, since automation procedures ensure that the methods used can be easily reproduced, excluding the variability of a manual process performed by a human. This allows the whole process to be evaluated and compared with greater reliability and accuracy. Thus, the development of a new free and easy-to-use software would greatly contribute to the development of new drugs.

This thesis will present a new workflow called Molecular Architect (MolAr), which act in an integrated form to carry out the whole VS process. It removes from the researcher the need to have advanced computational skills (installation of tools and libraries, programming languages, bash scripts, etc.), thus facilitating the execution of the VS simulations through the simplification and automation of the whole process. MolAr allows virtual screening using AutoDock Vina and DOCK 6. In addition, MolAr workflow proposes that in consensus virtual screening between DOCK 6 and AutodoDock Vina, DOCK 6 uses the best ligand conformations defined by Autodock Vina. Thus, when using conformations already optimized by AutoDock Vina, DOCK 6 presents improved results in VS. Therefore, it is expected a decrease in human errors when performing the VS process using MolAr plus a faster speed in performing VS experiments, because the whole process was simplified, integrated and automated.

In section 2, we present an overview of the challenges involved in the development of new drugs, as well as the main databases and approaches used in molecular docking, major scoring functions used and some docking programs involved

in the process of VS. Section 3 and 4 describe, respectively, the justifications for this project and the objectives that we aim to achieve with its development. In section 5, the software development methodology is explained. Next, in section 6, The development of MolAr and its features is explained. As a differential to other existing free VS software, MolAr has simple and intuitive interfaces and it allows the execution of the entire virtual screening process, assisting users to perform the VS process, in an automated and integrated manner. Thus, it removes from the researcher the need to have advanced computational skills (installation of programs and libraries, programming languages, bash scripts, etc.). Moreover, MolAr facilitates the execution of the VS simulations through the simplification and automation of the whole process. Therefore, it is expected a decrease of human errors in the execution of VS process and a greater speed in its execution, because the whole process was simplified, integrated and automated. The developed Workflow has already been used in researches carried out by Federal University of São João del-Rei (UFSJ) researchers, and we present, in section 6.3, three cases studies where we use MolAr. The three case studies performed shows that MolAr is a promising software and it has achieved its goal of assisting researchers to carry out the VS process, eliminating the need for advanced computing knowledge and speeding up the process. Section 7 presents some final considerations and possible future work. Finally, the appendix shows the INPI approved registry of the RUNMOPAC and MolAr (Appendix A and B) and the INPI patent application (Appendix C). Appendix D shows the first page of the paper published in the Journal of Molecular Modeling and Appendix E presents the first page of the submitted paper to Frontiers of Chemistry. Then, Appendix F list the MolAr and Octopus downloads already made, where it can be verified that the developed software has already been downloaded in several different countries.

2. Literature Review

In this study, as the focus of this thesis is the development of a workflow to automate VS, the entire literature review was carried out considering the steps necessary to carry out the entire VS process.

2.1 *De novo* drug design

The term “de novo” means, in Latin, “from the beginning”, “a fresh” or “a new”. In other words, the term de novo drug design involves designing an entirely new molecule from the scratch. This approach is a particularly challenging task, because the chemical space that can be explored is vast (DENG et al., 2013; TODA; GOTO; HIRAYAMA, 2008). It allows the chemical space to be better explored and it opens a possibility to find new structures that do not exist in any database of molecules (DEVI; SATHYA; COUMAR, 2015). In de novo drug design, the 3D structure of the molecular target is used to design newer molecules.

Structure-based Drug Design (SBDD) development is a process employed in de novo drug design in which the 3D structure of the molecular target is used to design newer molecules. In SBDD, previous ligands are not considered. In this case, the molecules are commonly constructed within the binding site, with each suggestion being evaluated using a scoring function. Another way to do this is studying the binding site to determine where functional groups can bind strongly. So, it is possible to connect these functional groups and then convert them into real molecules (JAIN; AGRAWAL, 2004). Structure based drug design will be explained in detail in section 2.3.1.

2.2 Computer-Assisted Drug Design (CADD)

Researchers are constantly investing in the development of new methods to increase the efficiency of the drug discovery process (HILLISCH; PINEDA; HILGENFELD, 2004). One approach used to increase effectiveness in the development of new drugs is the use of Computer-Assisted drug design techniques

(CADD, well known as an in-silico method), which uses computational chemistry approach for the drug discovery process. CADD is a cyclic process of developing new drugs in which all stages of design and analysis are performed by computer programs, operated by medicinal chemistries (OGLIC et al., 2018). The International Union of Pure and Applied Chemistry (IUPAC) defines CADD (VAN DE WATERBEEMD et al., 1997) as all computer-assisted techniques used to discover, design and optimize biologically active compounds with a putative use as drugs.

The primary purpose of CADD is to verify the interaction of a ligand with its molecular target, how intensely this affinity will be, and the conformational geometry adopted by the complex after this bonding. Generally, this activity involves the use of techniques of mechanics and molecular dynamics as well as ab initio methods of quantum chemistry. In addition, scoring functions are used to estimate the binding affinity between a ligand and a target. These methods can use a variety of computer science techniques, including linear regression (RUIZ-CARMONA et al., 2014), machine learning (LIU; WANG, 2015), neural networks (TAYARANI et al., 2016), or other statistical techniques (HARRISON, 2010), to predict the binding affinity between a ligand and target.

Strategies for CADD may vary, depending on what information about the target and ligand are available. In the early stage of the drug development process, it is normal for little or no information to exist about the target, ligands, and structure. CADD techniques are able for obtaining this information, such as which proteins can be targeted in pathogenesis and what are the possible active ligands that can inhibit the proteins. Kapetanovic (KAPETANOVIC, 2008) briefly notes that CADD comprises: i) making the drug discovery and development process faster with the contribution of *in silico* simulations; ii) the optimization and identification of new drugs using the computational approach to identify chemical and biological information about possible ligands and/or molecular targets; iii) the use of simulations to eliminate compounds with undesirable properties and to select the candidates with more chances for success. Recent software uses empirical molecular mechanics, quantum mechanics and, more recently, statistical mechanics. This last advance allowed the explicit effects of solvents to be incorporated (DAS; SAHA, 2017).

CADD has gained prominence, since it allows the simulation and prediction of several essential factors, such as toxicity, activity, bioavailability and efficacy, even before the compound is submitted to *in vitro* tests, thus allowing greater planning and direction of the research (FERREIRA; GLAUCIUS; ANDRICOPULO, 2011). Better planning of the research means, in this case, fewer *in vitro* and *in vivo* experiments, reduced runtime and overall research costs. Moreover, CADD enables the analysis of a large number of molecules in a short period of time, demonstrating how they bind to targets of pharmacological interest even before their synthesis.

CADD represents a fast and valid methodology for scientific research of new compounds with pharmacological potential (FERREIRA; GLAUCIUS; ANDRICOPULO, 2011; RIPPHAUSEN; NISIUS; BAJORATH, 2011). Thus, CADD enables the analysis of a large number of molecules in a short period of time, demonstrating how they bind to targets of pharmacological interest even before their synthesis. In general, drug design process starts with the identification of molecular targets for a given compound (natural or synthetic), followed by a careful molecular analysis of the major molecular recognition forces between the ligand and the receptor, design of the most promising molecules, and finally the experimental assays (TODA; GOTO; HIRAYAMA, 2008).

Alternatively, CADD simulations require a high computational cost, taking up to weeks if long jobs are run for molecular dynamics simulations. Therefore, it is a continuous challenge to find viable solutions that reduce the simulation runtime and at the same time, increase the accuracy of the simulations (DUFFY et al., 2012; KONG et al., 2011; RIPPHAUSEN; NISIUS; BAJORATH, 2011). In this context, VS is one promising approach.

2.3 Virtual Screening (VS)

Popular VS techniques had their origins in the 1980s, but VS first appeared in publication in 1997 (HORVATH, 1997). In recent times, the use of VS techniques has proven to be an excellent alternative to high throughput screening, especially in terms of cost-effectiveness and probability of finding the most appropriate result through a large virtual database (SURABHI; SINGH, 2018).

VS is an *in-silico* technique used in the drug discovery process. IUPAC defines VS as computational methods that rank the molecules in a database by their forecast of continuous, categorical biological or chemical properties (MARTIN et al., 2016). During VS, large databases of known 3D structures are automatically evaluated using computational methods (MAIA et al., 2017). VS has worked like a funnel (Figure 1), selecting more promising molecules for *in vitro* assays to be performed. With the use of VS, it is expected to identify those molecules that are more susceptible to bind to the molecular target, typically in general, a protein or enzyme receptor. Therefore, VS assists in identifying the most promising hits able to bind to the target, in general, a protein or enzyme receptor, and only the most promising molecules are synthesized. In addition, VS identifies compounds that may be toxic or have unfavorable pharmacodynamic and pharmacokinetic properties. Thus, VS techniques have played a prominent role among strategies for the identification of new bioactive substances (BERMAN et al., 2013) .

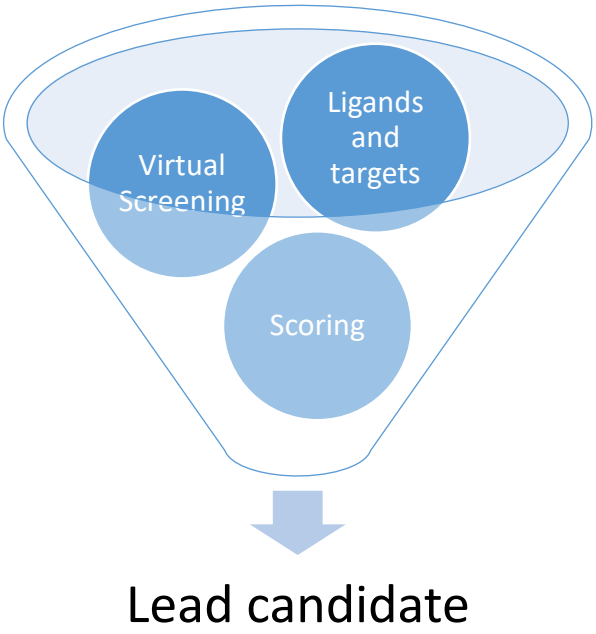


Figure 1 – General VS scheme

VS for drug discovery is becoming an essential tool to assist in fast and cost-effective lead discovery and drug optimization (ZHANG et al., 2018). This technique can aid in the discovery of bioactive molecules, since they allow the selection of the compounds in a structure database most likely to show biological activity against a

target of interest. After identification of bioactivity, the molecules undergo biological assays. In addition, there are VS techniques using machine learning methods that predict compounds of specific pharmacodynamic, pharmacokinetic or toxicological properties based on their structural and physicochemical properties derived from the ligand structure (MA et al., 2009). Hence, VS tools have played a prominent role among strategies for the identification of new bioactive substances, since they increase the speed of the drug discovery process as long as they automatically evaluate large compound libraries through computational simulations (MAITHRI G et al., 2016).

Structure-based virtual screening (SBVS) is a robust, useful and the most promising *in silico* technique for drug design (LIONTA et al., 2014). Therefore, this research will address the SBVS, although there are other types of VS such as ligand-based virtual screening (BANEGAS-LUNA; CERÓN-CARRASCO; PÉREZ-SÁNCHEZ, 2018) and fragment-based virtual screening (WANG et al., 2015).

2.3.1 Structure-based virtual screening (SBVS)

Structure-based virtual screening (SBVS), also known as target-based virtual screening (TBVS), attempts to predict the best interaction mode of two molecules to form a stable complex. Thus, the likelihood of coupling of candidate ligands and target protein considers the affinity degree of the complex. SBVS methods require that the 3D structure of the target protein be known so that the interactions between the target and each chemical compound can be predicted *in silico* (LIU et al., 2018). In this strategy, the compounds are selected from a database and classified according to their affinity for the receptor site.

Among the techniques of SBVS, molecular docking has been noteworthy due to low computational cost and the good results achieved (MENG et al., 2011). This technique emerged in the 1980s, when Kuntz et al. (KUNTZ et al., 1982) designed and tested a set of algorithms that could explore the geometrically feasible alignments of a ligand and target. They showed that structures close to the correct structures could be obtained after docking. In addition, their tests detected other possible structures to be tested. However, although the approach was promising, it was only in the 1990s that it became widely used when there was an improvement of the techniques used in

conjunction with an increase in computational power and greater access to the structural data of target molecules. During the execution of a SBVS, the evaluated molecules are sorted according to their affinity against the receptor site. Hence, it is possible to identify ligands that are more likely to present some pharmacological activity with the molecular target. Score functions have been used to verify the likelihood of a binding site describing the affinity between the ligand and target. In this process, a reliable scoring function is a critical component of the docking process (LEELANANDA; LINDERT, 2016).

The use of SBVS has advantages and disadvantages. Among the advantages are the following:

I. There is a decrease in the time and cost involved in the screening of millions of small molecules.
II. There is no need for the physical existence of the molecule, so it can be tested computationally even before being synthesized.
III. There are several tools available to assist SBVS.

The disadvantages can be highlighted as:

I. Some tools work best in specific cases, not in more general cases (LIONTA et al., 2014).
II. It is difficult to accurately predict the correct binding position and classification of compounds due to the difficulty of parameterizing the complexity of ligand-receptor binding interactions.
III. It can generate false positives and false negatives.

Despite the disadvantages noted above, many studies using SBVS have been developed in recent years (CARPENTER et al., 2018; CARREGAL et al., 2017; DUTKIEWICZ; MIKSTACKA, 2018; MUGUMBATE et al., 2017; NUNES et al., 2019; SURABHI; SINGH, 2018; WÓJCIKOWSKI; BALLESTER; SIEDLECKI, 2017), which shows that although it has disadvantages, the reduction of time and cost obtained in the process make the use of SBVS in the development of drugs still widely used.

2.4 Scoring Functions

Molecular docking software uses scoring functions to estimate the force of noncovalent interactions between a ligand and a molecular target using mathematical methods. A scoring function is one of the most important components in SBVS (HUANG; GRINTER; ZOU, 2010) as it is primarily responsible for predicting the binding affinity between a target and its ligand candidate. Thus, the scoring functions are the main reason for the failure of docking tools (TEN BRINK; EXNER, 2009). Therefore, despite the wide use, the estimation of the interaction force between a ligand and a molecular target remains a major challenge in VS.

Figure 2 illustrates a selection from 2 possible ligands by using a typical scoring function. Among ligands I and II, the scoring function selected the one that best interact with the molecular target. Thus, the ligand with the highest binding affinity to the target can be select for further testing.

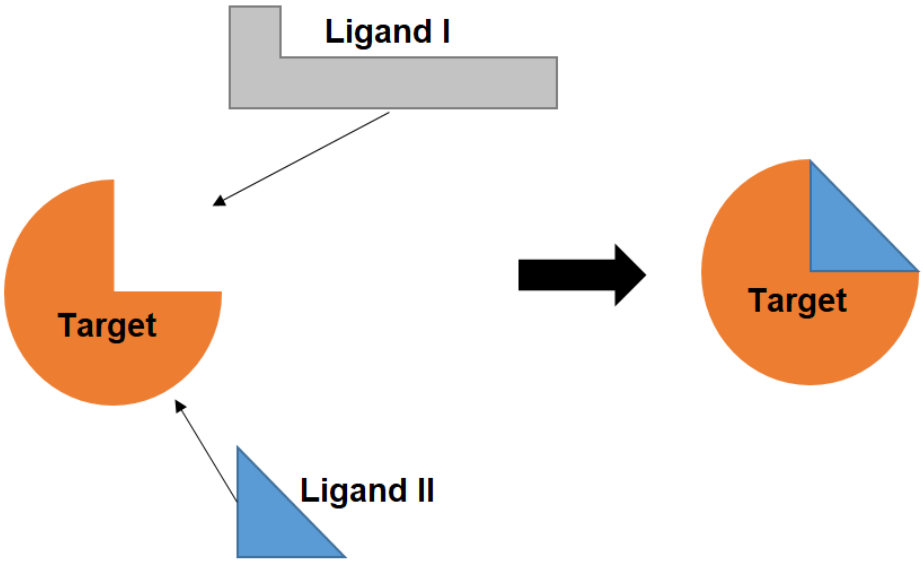


Figure 2 – Identification of a ligand candidate by a typical scoring function

There are basically three important applications of scoring functions in molecular docking. First, they can be used in the determination of the ligand binding site and the conformation between a target and ligand. This approach can be used to search for allosteric sites. Second, they can be used to predict the binding affinity between a protein and ligand. Finally, they can also be used in lead optimization (LI et al., 2013).

Most authors define the scoring functions as three types: force field (FF), empirical and knowledge based (BREDA et al., 2008; FERREIRA et al., 2015; HAGA; ICHIKAWA; DATE, 2016; HUANG; GRINTER; ZOU, 2010). Liu and Wang (LIU; WANG, 2015) define two more types of scoring functions: machine-learning-based and hybrid methods.

The force field scoring functions are based on the intermolecular interactions between the ligand and target atoms, such as the van der Waals, electrostatic and bond stretching/bending/torsional force interactions, obtained from experimental data and in accordance with the principles of quantum mechanics (FERREIRA et al., 2015). Some published force-field scoring functions include the ones described in Li et al (LI et al., 2015), Goldscore (VERDONK et al., 2003), Sybyl/D-Score (ASH et al., 1997) and internal coordinate modeling (ABAGYAN; TOTROV; KUZNETSOV, 1994).

Empirical scoring functions estimate the binding free energy based on weighted structural parameters by adjusting the scoring functions to experimentally determine the binding constants of a set of complexes (BREDA et al., 2008). To create an empirical scoring function, a set of data from protein-binding complexes whose affinities are known is initially used as training. A linear regression is then performed as a way of predicting te values of some variables (HUANG; GRINTER; ZOU, 2010). The weight constants generated by the empirical function are used as coefficients to adjust the equation terms. Each term of the function describes a type of physical event involved in the formation of the ligand-receptor complex. Thus, hydrogen bonding, ionic, nonpolar interactions, desolvation and entropic effects are considered. Some popular empirical, scoring functions include Glide-Score (FRIESNER et al., 2004), Sybyl-X/F-score (CERTARA, 2016) and DOCK 6 empirical force field (ALLEN et al., 2015).

In the knowledge-based scoring functions, the binding affinity is calculated by summing the binding interactions of the atoms of a protein and the molecular target (BREDA et al., 2008). These functions consider statistical observations performed on large databases such as the Protein Data Bank (FERREIRA et al., 2015). The method uses pairwise energy potentials extracted from known ligand-receptor complexes to obtain a general scoring function. These methods assume that intermolecular interactions occurring near certain types of atoms or functional groups occurring more

frequently are more likely to contribute favorably to the binding affinity. The final score is given as a sum of the score of all individual interactions. One example of software that uses knowledge-based scoring function is Paradocks (MEIER et al., 2010).

In addition, machine-learning-based methods (LIU; WANG, 2015) have been considered as a fourth type of scoring function. These techniques have been used in quantitative structure-activity relationship (QSAR) analysis to predict various physical-chemical, biological, and pharmaceutical properties of small molecule compounds. In these types of scoring functions, modern QSAR analysis can be applied to derive statistical models that calculate protein-ligand binding scores. Some scoring functions of this type are NNScore 2.0 (DURRANT; MCCAMMON, 2011), RF-Score-VS (WÓJCIKOWSKI; BALLESTER; SIEDLECKI, 2017), SFCscoreRF (ZILIAN; SOTRIFFER, 2013), SVR-KB (LI; WANG; MEROUEH, 2011) and SVR-EP (LI; WANG; MEROUEH, 2011), ID-Score (LI et al., 2013) and CScore (OUYANG; HANDOKO; KWOH, 2011).

Finally, there are also hybrid scoring functions that combine two or more of the previously defined scoring functions into one. A few recently published examples of this type of scoring function include the hybrid scoring function developed by Tanchuk et al (TANCHUK et al., 2016), SMoG2016 (GENG et al., 2019), GalaxyDock BP2 (BAEK et al., 2017) and iScore (GENG et al., 2018).

2.5 Consensus Docking

In the last decade, a new technique of VS called consensus docking (CD) has been used in some studies (ALIEBRAHIMI et al., 2017; CHERMAK et al., 2016; HOUSTON; WALKINSHAW, 2013; KUKOL, 2011; PARK; EOM; KIM, 2014; POLI; MARTINELLI; TUCCINARDI, 2016; TUCCINARDI et al., 2014) to increase the accuracy of VS studies and reduce the false positives obtained in VS experiments (ALIEBRAHIMI et al., 2017).

This technique is a combination of two different approaches, which the resultant combination is better than a single approach alone (Figure 3). However, Poli, Martinelli and Tuccinardi (POLI; MARTINELLI; TUCCINARDI, 2016) reported that there are few

studies that evaluate the possibility of combining the results of different VS methods to achieve higher success rates in VS studies.

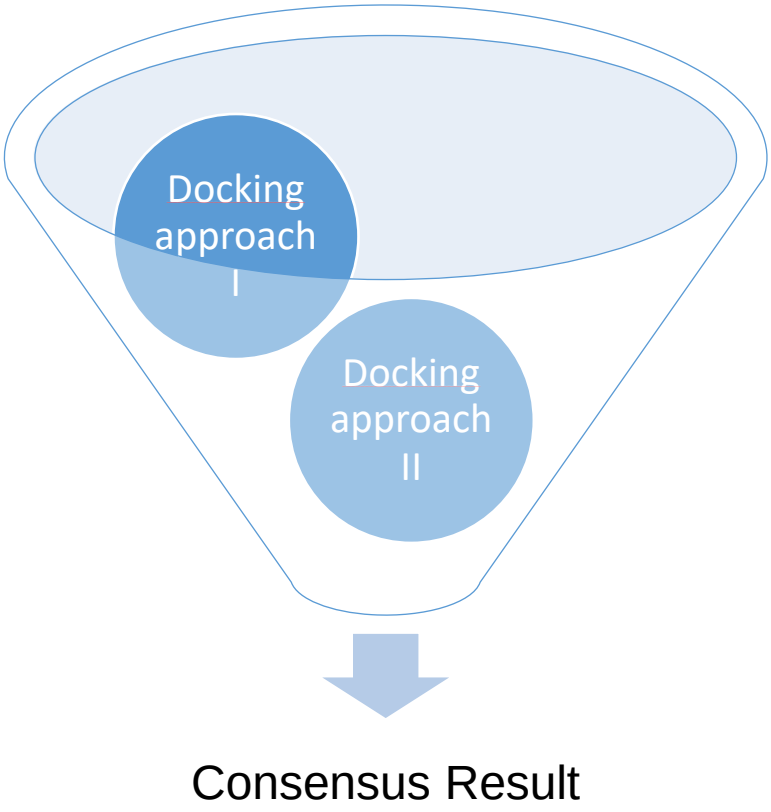


Figure 3 - Consensus Docking approach

Houston and Walkinshaw (HOUSTON; WALKINSHAW, 2013) described the main reason for using this combination the individual program may present incorrect results and these errors are mostly random. Therefore, even when two programs present different results, the combination of these results may, in principle, be much closer to the correct answer than even the best program alone. Houston and Walkinshaw also suggest that CD approaches using two different docking programs improve the precision of the predicted binding mode for any VS study. In this study, which was performed on three different protein targets, the use of the CD approach better predicted the ligand-binding pose than the use of a single approach alone. The same study also verified that a greater level of consensus in a given pose indicates a greater reliability in this result. Finally, the results presented by the authors suggested

that the CD approach works as well as the best VS approaches available in the literature.

Park, Eom and Kim (PARK; EOM; KIM, 2014) use an approach in which they combine the use of the Autodock 4.2 (MORRIS et al., 2009) and Flexx (RAREY et al., 1996) programs. These programs were chosen because both use different types of score functions (force field in Autodock and empirical in Flexx). In this study, they achieved superior performance with the application of consensus docking than with the use of each of the programs alone.

Alternatively, when using two different VS programs, there is extra time to run the two different tools and combine the results. However, Houston and Walkinshaw (HOUSTON; WALKINSHAW, 2013) showed that the increased runtime may be advantageous; using Autodock Vina (TROTT; OLSON, 2009) in a VS approach along with Autodock4 (MORRIS et al., 2009) increased the final runtime by approximately 10%. This combination is interesting given the potential gains from its use.

Therefore, the use of consensus docking is a recent technique, and although there are few papers in the literature on the subject, it seems to be a promising approach for further VS study.

2.6 Virtual Databases

An indispensable condition in performing VS is the availability of the 3D structure of the target protein (CAVASOTTO, 2011) and ligands to be docked. Some databases were created to store 3D structures of molecules. Among them, we highlight some free databases and some databases commercially available. Below we are going to present a brief explanation of them:

- Protein Data Bank (PDB) (BERMAN et al., 2013): PDB is the public database where three-dimensional structures of proteins, nucleic acids and complex molecules have been deposited since 1971. The worldwide PDB organization ensures that PDB files are publicly available to the global community. It is widely used by the academic community and has grown consistently in recent years. In the last 10 years, the number of 3D structures of the PDB increased from 48,169 at the end of 2008 to 147,604 in the end of 2018, an increase of nearly

207%. This implies that in the last 10 years, almost 9,943 new structures have been added to the PDB every year, just over 27 structures per day, on average. The pace of this growth has increased. At the beginning of this decade approximately 25 new entries were added per day on average. In 2018, over 31 new structures were added per day, an average daily growth of 24% compared to 2010. In 2019, until September 30, 32.6 new structures were added per day, on average.

- PubChem (KIM et al., 2016): PubChem is a public database, aggregating information from smaller, more specific databases. It has more than 97 million compounds available.
- ChEMBL (BENTO et al., 2014): ChEMBL is a database of bioactive molecules with medicinal properties maintained by the European Institute of Bioinformatics (EBI) of the European Molecular Biology Laboratory (EMBL). Currently, it has almost 2.3 million compounds and 15.2 million known biological activities.
- ZINC (STERLING; IRWIN, 2015): ZINC is a free database of commercially available compounds for VS. ZINC has more than 230 million commercially available compounds in the 3D format. ZINC is maintained by Irwin and Shoichet Laboratories of the Department of Pharmaceutical Chemistry at the University of California, San Francisco (UCSF).
- NatProDB (UNIVERSIDADE ESTADUAL DE FEIRA DE SANTANA, 2006): The State University of Feira de Santana has made NatProDB available. This database stores 3D structures of the semiarid biome. The pharmacological profile of compounds from the semiarid flora have not yet been studied, which has motivated our research group to deepen the research by their molecular targets (TARANTO et al., 2015).
- Our Own Molecular Target (OOMT) (CARREGAL; JR; TARANTO., 2013): OOMT is a special molecular target database because it has the biological assay for all its molecular targets, and includes specific targets for cancer, dengue, and malaria. OOMT was created by a group of researchers from Federal University of São João del-Rei (UFSJ).
- Brazilian Malaria Molecular Targets (BraMMT) (NUNES et al., 2019): The BRAMMT database comprises thirty-five molecular targets for *Plasmodium*

falciparum retrieved from the PDB database. This database allows *in silico* virtual high throughput screening (vHTS) experiments against a pool of *P. falciparum* molecular targets.

- Drugbank (WISHART et al., 2018): DrugBank is a database that contains comprehensive molecular information about drugs, their mechanisms, their interactions, and their targets. The database contains more than 11,900 drug entries, including nearly 2,538 FDA-approved small molecule drugs, 1,670 biotechnology (protein / peptide) drugs approved by the FDA, 129 nutraceuticals and nearly 6,000 investigational drugs.

Commercially available Databases:

- MDL Drug Data Report (MDDR) (SCI TEGIC ACCELRYS INC., 2019): MDDR is a commercial database built from patent databases, publications and congresses. It has more than 260,000 biologically relevant compounds and approximately 10,000 compounds are added every year.
- ChemSpider (PENCE; WILLIAMS, 2010): ChemSpider is a database of chemical substances owned by the Royal Society of Chemistry. It has more than 71 million chemical structures from over 250 data sources. ChemSpider allows downloading up to 1000 structures per day. Previous contact is needed for the download of more structures, and ChemSpider is therefore not a totally free database.

2.7 Development of Models for Drug Design by Homology Modeling

Although the availability of 3D structures of the molecular target in good resolution is essential for performing Virtual Screening, sometimes it is not possible to obtain it experimentally. In addition, even though the structure is available, the atomic coordinate of high flexible loops could be poorly described by experimental methods. As a result, gaps regions are formed, and it can be near of the binding site. Also, there are some cases where the researchers have the amino acid sequence of the target protein, but do not have their 3D structure. In order for a target structure to be reliably used in the VS process, the 3D structure must be determined as completely as possible, avoiding gaps regions. Noteworthy, the cavity of binding site should be well

described complexed with a ligand and with a correct description of the main forces for molecular recognition. In other words, the molecular target must be in holo form (ligand-bound state) for successful VS campaign (SEELIGER; DE GROOT, 2010).

The prediction of the 3D structure of a protein from its amino acid sequence can be carried out through comparative homology modeling methodology. Homology modeling is a methodology for predicting protein structure based on the general observation that proteins with similar sequences have similar structures (CAVASOTTO; PHATAK, 2009), since the structural conformation of a protein is more highly conserved than its amino acid sequence. Consequently, minor changes in sequence will result in few changes in the 3D structure (HILLISCH; PINEDA; HILGENFELD, 2004). The evaluation of prediction techniques showed that homology modeling is the only technique capable of producing reliable and quality models over a wide range of size while, on the other hand, *de novo* methods are limited to small proteins and peptides (BIASINI et al., 2014).

Homology modeling has gained importance in recent years by filling the gap between the sequence and the structural space of proteins. Thus, this methodology allows researchers to construct structural models which are infeasible by experimental methods, such as X-ray crystallography and Nuclear Magnetic Resonance (NMR) spectroscopy (RAMACHANDRAN; DOKHOLYAN, 2012). IUPAC defines Homology Model as a model of a protein, whose three-dimensional structure is unknown, built from the coordinate data of similar proteins or using alignment techniques and homology arguments (VAN DE WATERBEEMD et al., 1997).

The quality of a homology model depends mainly on the similarity between the structure that will be used as a template and the target protein to be modeled. In general, sequences with higher identities lead to more precise models, since the evolutionary structural divergence will be smaller and the alignment errors will occur with less probability (BIASINI et al., 2014). Hillisch et al. (HILLISCH; PINEDA; HILGENFELD, 2004) point out that if the identity between the template and protein sequence is greater than 50%, the generated model is usually of sufficient quality to be used in the Target Based Virtual Screening. Otherwise, Cavasotto and Phatak (CAVASOTTO; PHATAK, 2009) explain that an template with identity between 25 and 50% can be used to assess target druggability and design mutagenesis experiments.

Finally, Cavasotto and Phatak said those proteins whose identity is between 10 and 25% are speculative at best.

Generally, the homology modeling process involves four steps (MARTI-RENOM et al., 2000):

i. identification and proteins selection;
ii. sequence alignment;
iii. model construction;
iv. model refinement.

Various software may be used in the development of drugs by homology. The most used free software, in academic research, in the process of homology modeling are MODELLER (ESWAR et al., 2007; FISER; ŠALI, 2003) and SWISS-MODEL (BENKERT; KÜNZLI; SCHWEDE, 2009; BIASINI et al., 2014). This software allows the alignment of a sequence to be modeled with related known structures, automatically calculating a new model containing all heavy atoms.

2.7.1 SWISS-MODEL

SWISS-MODEL (BIASINI et al., 2014; GUEX; PEITSCH, 1997; GUEX; PEITSCH; SCHWEDE, 2009) is a web service used in homology modeling. It has three basic types of modeling options:

I. The automated mode;
II. The alignment mode;
III. The project mode.

The automated mode is used when identity between the protein to be modeled and the template used is high enough to allow the whole process to be done in an automated way. In this case, the model is calculated by the SWISS-MODEL homology pipeline. However, for more difficult modeling cases, where the evolutionary distance between the target and the model is large, it is not advisable to use the automatic mode. In this case, the Alignment mode of the server is recommended, where a multiple-sequence alignment of the target, template and other members of the protein family can be submitted to calculate the structural model (BIASINI et al., 2014). A

modeling that uses the assembly of rigid fragments, in addition to some heuristics, are used to improve the insertions and deletions efficiency based on the structural context.

In the project mode, visual inspection and manual alignment manipulation are used because the correct alignment between the generated model and the template cannot be clearly determined by sequence-based methods. In this case, the user will examine and manipulate the alignment between the target and the template using the DeepView structural and visualization analysis tool (Swiss-PDB Viewer) (GUEX; PEITSCH; SCHWEDE, 2009). Then, the SWISS-MODEL will construct the coordinates of the model according to the alignment between target and the template informed by the user.

Finally, the model quality generated by SWISS-MODEL is validated using model quality estimation tools (MQE), such as ANOLEA (MELO; FEYTMANS, 1998) or QMEAN (BENKERT; KÜNZLI; SCHWEDE, 2009).

2.7.2 MODELLER

MODELLER (ESWAR et al., 2007; FISER; ŠALI, 2003; ŠALI, 2013) is a homology modeling software that integrates a large set of features. It is written in Fortran language and uses Python for its control language. All scripts used by MODELLER to perform homology modeling are Python scripts. Thus, Python knowledge is needed to execute even the simplest MODELLER activities.

In the simplest case, user provides an initial alignment between the sequence to be modeled and the template to be used. Next, MODELLER is used to calculate a model containing all non-hydrogen atoms. It implements comparative modeling for satisfying spatial constraints. In addition to being used in the construction of a model, MODELLER can also be used to perform a large set of other auxiliary activities such as: alignment between two proteins, multiple alignment of sequences and structures, calculation of phylogenetic trees and modeling protein loops. To perform these activities optimally, MODELLER uses techniques of Conjugated Gradient and Molecular Dynamics with Simulated Annealing. Once the models are generated by MODELLER, it allows the visualization of the Discrete Optimized Protein Energy (DOPE) (SALI; SALI, 2006) for each generated model. DOPE allows to compare

spatial features of the atoms of the generated model to determine the violations of the molecular restraints. DOPE score explicitly depends on the sizes of the native structures and has an arbitrary scale. Consequently, scores from different proteins cannot be compared directly (SALI; SALI, 2006). However, the DOPE value allows the comparison between the models generated by MODELLER for a specific protein, because in this case, a lower DOPE value indicates a possibly better model.

2.8 Methods of evaluating the quality of a model and simulation

To verify the quality of a homology modeling process or a docking approach, some methods are used to evaluate generated complexes and to verify if the protein generated by the homology modeling or docking is able to reproduce the experimental data results of the ligand-receptor complex. The most common evaluation methods are root mean square deviation (RMSD) (HAWKINS et al., 2008), receiver operating characteristic (ROC), area under the curve ROC (AUC-ROC) (FLACH; WU, 2005; TROTT; OLSON, 2009) and Ramachandran plot (LASKOWSKI et al., 1993).

2.8.1 Root Mean Square Deviation (RMSD)

One of the aspects evaluated in docking programs is the accuracy of the generated geometry (JAIN, 2008). Docking programs attempt to reproduce the conformation of the ligand-receptor complex in the crystallographic structure.

RMSD is a statistical measure that evaluates the quality of the docking process or homology modeling approach performed by a program (DING et al., 2016). The RMSD between two structures can be calculated according to the equation (SARGSYAN; GRAUFFEL; LIM, 2017):

$$RMSD(A,B) = \frac{1}{N}\sum_{i=1}^{n} d_i^2$$

where d is the distance between atom i in the two structures and N is the total number of equivalent atoms. Since the calculation of RMSD requires the same number of atoms in both structures, it is often used in the calculation of only the heavy atoms or backbone of each amino acid residue.

Using the RMSD calculation, it is possible to evaluate if a program was able to reliably reproduce a known crystallographic conformation, as well as their respective intramolecular interactions. To verify if a given program can accomplish this task, complex ligand-targets are subjected to a redocking process. After redocking, the overlap of the crystallographic ligand with the conformation of the ligand obtained with the docking program is performed. Then, the RMSD calculation is used to check the average distance between the corresponding atoms (usually the backbone atoms).

Generally, RMSD values lower than 2 Å indicate that the model generated by docking programs was able to reproduce the intermolecular interactions of the crystallographic complex (COLE et al., 2005; GOWTHAMAN; LYSKOV; KARANICOLAS, 2015; JAIN, 2008; MEIER et al., 2010). Therefore, in this case, the docking is considered very accurate. In the case of binding a large ligand, some authors generally relax this criterion (MÉNDEZ et al., 2003; VERSCHUEREN et al., 2013). For a model generated by homology modeling, evaluating the RMSD value is important, although visual inspection of the generated model is essential.

Figure 4 shows the visualization of the 2YND protein (Plasmodium vivax N-myristoyltransferase in complex with a pyrazole sulfonamide inhibitor) and a model of that same protein generated by homology modeling and using as template 3IU1 (crystal structure of human type-I N-myristoyltransferase with bound myristoyl-CoA) and 4C68 (Plasmodium vivax N-myristoyltransferase in complex with a peptidomimetic inhibitor) proteins. The RMSD between the original 2YND protein and the model generated in MODELLER is 0.25 Å. In Figure 4, the red represents the original 2YND protein and the green represents the protein generated by MODELLER.

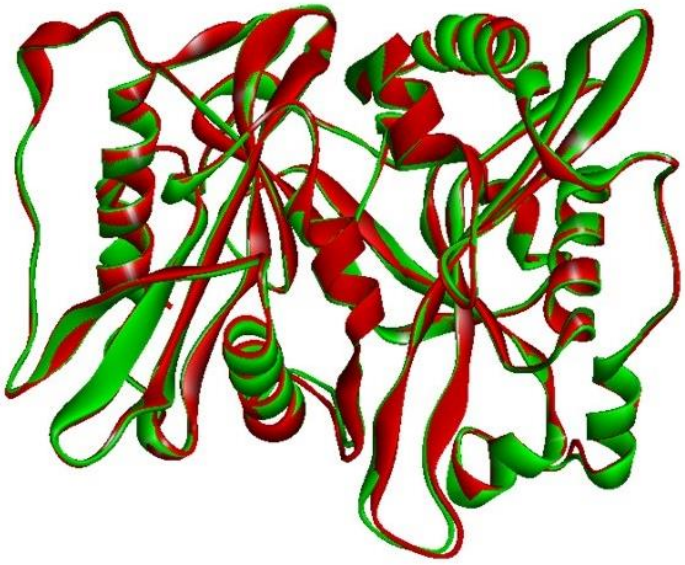


Figure 4 - RMSD between the 2YND protein and a model generated by homology modeling using 4C68 and 3IU1 proteins as template

2.8.2 ROC Curve and AUC

One of the great challenges of VS methods is the ability to differentiate active compounds against the target from inactive ones (false positives) (AWUNI; MU, 2015). Thus, it is important that VS tools have ways to assist their users to distinguish false positives from true positives. The ROC curve and the area under the ROC curve (AUC-ROC) (LÄTTI; NIINIVEHMAS; PENTIKÄINEN, 2016) are widely used methodologies for this purpose.

Active compounds and decoys are used to create the ROC curve and AUC-ROC. Active compounds are those having known biological activity for the molecular target of interest. Some databases, such as ChEMBL (BENTO et al., 2014; GAULTON et al., 2012), allows to search for these compounds. Alternatively, decoys are compounds that, although possessing physical properties similar to an active compound (such as molecular mass, number of rotatable bonds, and logP), have different chemical structures that make them inactive. They are generated from random molecular modifications in the structure of an active compound (HUANG; SHOICHET; IRWIN, 2006). Some databases, such as DUD-E (MYSINGER et al.,

2012) and ZINC (STERLING; IRWIN, 2015), provide decoys for compounds of interest. A small number (greater than 2) of known active compounds have to be used to calculate the AUC-ROC (LÄTTI; NIINIVEHMAS; PENTIKÄINEN, 2016).

After generation of the decoys, a VS process is performed using known active compounds and decoys against a target of interest (YURIEV; RAMSLAND, 2013). For each ligand-target complex, a binding energy is then calculated. The active compounds are expected to have lower binding energy than inactive compounds. The ROC curve plots the distribution of true and false results on a graph, while AUC-ROC allows the evaluation of the probability of a result is false. Hence, AUC-ROC reflects the probability to recover an active compound preferentially to inactive compounds (TRIBALLEAU et al., 2005; ZHAO et al., 2009), allowing verification of the sensitivity of a VS experiment in relation to its specificity. The larger the area under the curve, the better the ability to select active compounds and have fewer false positives.

The AUC value can vary between 0 and 1. Hamza (HAMZA; WEI; ZHAN, 2012) showed a practical way of interpreting the AUC values:

- AUC between 0.90 and 1.00: Excellent
- AUC between 0.80 and 0.90: Good
- AUC between 0.70 and 0.80: Fair
- AUC between 0.60 and 0.70: Poor
- AUC between 0.50 and 0.60: Failure

Therefore, the closer the AUC is to 1, the greater the ability of the VS tool to separate between active and inactive ligands. AUC-ROC values close to 0.5 indicates a random process (OGRIZEK et al., 2015). Acceptable values should be greater than 0.7.

Figure 5 shows an example of an ROC curve generated in a VS performed with the 4O1Z (cyclooxygenase-1 complexed with meloxicam) protein using five active ligands and 250 decoys. The VS tool was able to distinguish well between active and inactive ligands with the generated ROC curve and its respective AUC, which value was 0.86.

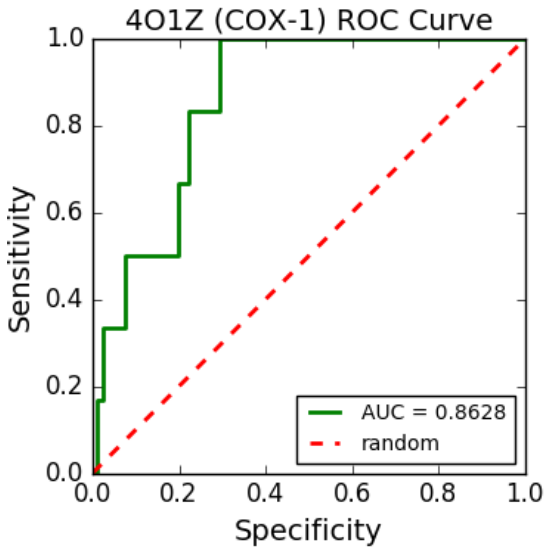
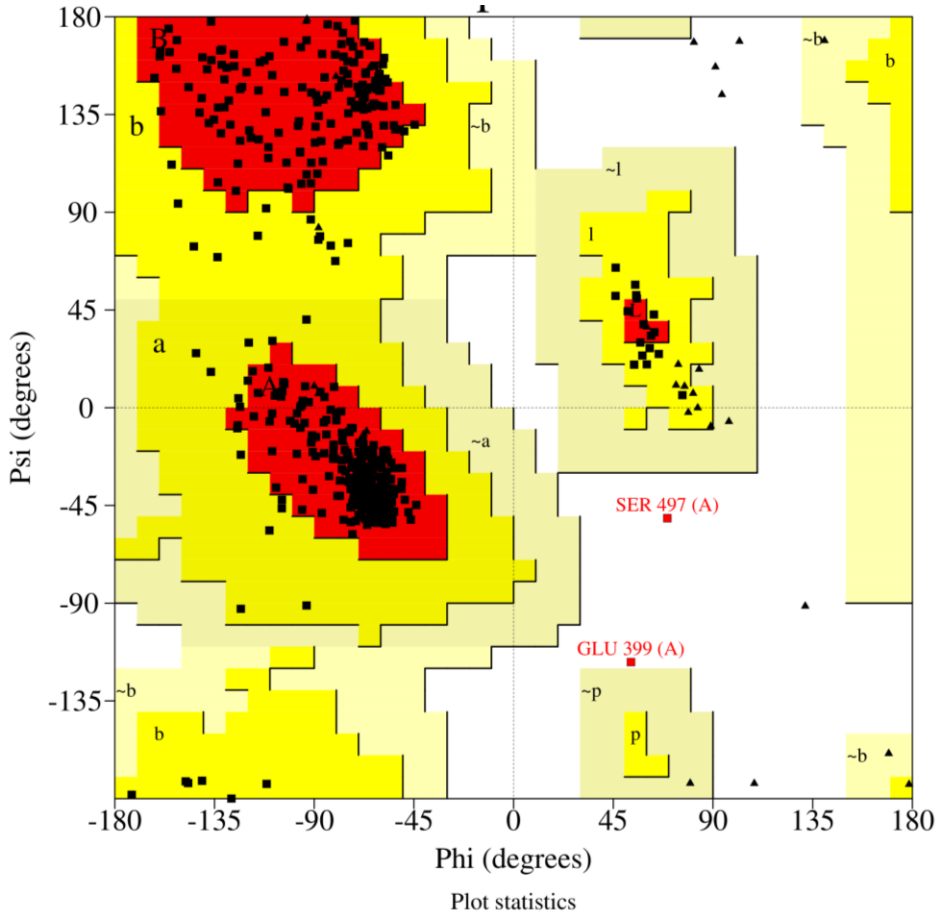


Figure 5 - ROC Curve example of the 4O1Z protein using 5 active ligands and 250 decoys

2.8.3 Ramachandran Plot

The protein model constructed using homology modelling approach should be validated to identify possible errors at different organizational levels of the protein structure and to estimate the quality of its tertiary structure. Correct stereochemistry of a protein is important, since it is critical for its interpretation, especially if the model is subjected to a molecular dynamics simulation (SCHREINER et al., 2011). The stereochemical quality of the model is evaluated through Ramachandran plot (RAMACHANDRAN; RAMAKRISHNAN; SASISEKHARAN, 1963). It evaluates bond lengths, plane angles, planarity of side chain rings, chirality, side chain conformations, planarity of peptide bonds, torsion angles for the backbone and the side chain, the steric clashes between non-bonded atoms. Ramachandran plot assists in the identification of the residues that are in the most favorable and unfavorable regions, orienting the evaluation of the quality of theoretical or experimental models of proteins.

A good model generated through homology modeling must have at least 90% of amino acid residues in favorable region of the Ramachandran plot (LASKOWSKI et al., 1993). Figure 6 shows an example of a Ramachandran plot of the 4PH9 (Ibuprofen bound to cyclooxygenase-2) protein generated by Procheck (LASKOWSKI et al., 1993). In the Figure 6, almost 90% of the amino acid residues are in a favorable region.



Plot statistics

Residues in most favoured regions [A,B,L]	426	89.5%
Residues in additional allowed regions [a,b,l,p]	48	10.1%
Residues in generously allowed regions [~a,~b,~l,~p]	0	0.0%
Residues in disallowed regions	2	0.4%
Number of non-glycine and non-proline residues	476	100.0%
Number of end-residues (excl. Gly and Pro)	2	
Number of glycine residues (shown as triangles)	37	
Number of proline residues	36	
Total number of residues	551	

Figure 6 - Ramachandran plot of the protein 4PH9. In this figure, 2 amino acids are in disallowed regions and so they are highlighted in red.

2.9 Protein Protonation State

The position of the hydrogens atoms cannot be determined experimentally by X-ray crystallography (TEN BRINK; EXNER, 2009). Almost all ligand-receptor interactions are pH-dependent and, to be correctly modeled, the protonation states of the molecules must be appropriately assigned (PETUKH; STEFL; ALEXOV, 2013).

Pólgar et al. (POLGÁR et al., 2007) compared the performance and dependency of the protonation states of the programs Surflex (SPITZER; JAIN, 2012), eHiTS (ZSOLDOS et al., 2007), GOLD (VERDONK et al., 2003) and FlexX-Pharm (HINDLE et al., 2002). It has been found that FlexX-Pharm and GOLD are dependent of the ligand protonation whereas Surflex does not depend. The tests carried out showed that eHITS does not depend on adjustments of the protonation state of the ligand, but this independence of previous knowledge did not result in better results, since eHITS presented the worst outcomes when compared to the other tools.

Brink and Exner (TEN BRINK; EXNER, 2010) show that proteins around the active site may influence the local pH and, therefore, the docking would be more effective if it took into consideration the different protonation state of the ligand and the molecular target. However, by the results of their experiments, the authors verified that approximately 80% of the correctly docked ligands had the best docking result when they were prepared at pH 7.4, which explains the good performance of tools that do not consider local protonation changes. Onufriev and Alexov (ONUFRIEV; ALEXOV, 2013) suggest that there is a tendency for changes in the protonation state upon binding to be minimal at physiologically relevant pH for each complex, suggesting that native receptor-ligand interactions evolved to reduce the energy cost associated with ionization. The authors noted that during a VS process, the correct assignment of protonation states is crucial for classification and scoring of lead candidates. Incorrect protonation state assignment may cause a potential best candidate not to be ranked at the top of the list and thus be thrown out.

Therefore, the use of a ligand and target with the incorrect protonation state can greatly influence the outcome of the docking process. Thus, it is important to use theoretical protocols to adjust the protonation state of a molecule prior to performing the Docking. The adjustment of the protonation state of the ligands and the molecular targets can be done manually. However, since current databases have thousands of compounds (both molecular targets and ligands), there must be ways to do this preparation automatically. In this scenario, several tools were developed that allow this adjustment, such as Glide (FRIESNER et al., 2004), Babel (O'BOYLE et al., 2011), SPORES (TEN BRINK; EXNER, 2009), PROPKA (DOLINSKY et al., 2007).

Davies et al. (DAVIES et al., 2006) said that PROPKA (LI; ROBERTSON; JENSEN, 2005) software was the most accurate method for all residues with the exception of Glu and His. Thus, since PROPKA is an efficient open source tool for adjusting the protonation state, we decide to use it in this study.

2.10 VIRTUAL SCREENING ALGORITHMS

In VS, we are targeting proteins in the human body to find novel ligands that will bind to them. VS can be divided into two classes: structure-based and ligand-based. In structure-based virtual screening, a 3D structure of the target protein is known, and the goal is to identify ligands from a database of candidates that will have better affinity with the target 3D structure. VS can be performed using molecular docking, a computational process where ligands are moved in 3D space to find a configuration of target and ligand that maximizes some scoring function. The ligands in the database are ranked according to their maximum score, and the best ones can be investigated further, e.g., by examining the mode and type of interaction that occurs. Additionally, VS techniques can be divided according to the algorithms used, which can be:

- *Machine Learning-based Algorithms*
    - Artificial neural networks (ANNs) (ASHTAWY; MAHAPATRA, 2018; TAYARANI et al., 2016);
    - Support vector machines (SENGUPTA; BANDYOPADHYAY, 2014; SILVA et al., 2016);
    - Bayesian techniques (ABDO et al., 2010);
    - Decision tree (HO, 1998), (LAVECCHIA, 2015);
    - k-nearest neighbors (kNN) (PETERSON et al., 2009);
    - Kohonen's SOMs and counter propagation ANNs (SCHNEIDER; TANRIKULU; SCHNEIDER, 2009);
    - Ensemble methods using machine learning (KORKMAZ; ZARARSIZ; GOKSULUK, 2015; MELVILLE; BURKE; HIRST, 2009);
- Evolutionary Algorithms
    - Genetic algorithms (ROHILLA; KHARE; TYAGI, 2017; XIA et al., 2017);

    - Differential evolution (FRIESNER et al., 2004; HUI-FANG et al., 2010; SPITZER; JAIN, 2012; THOMSEN; CHRISTENSEN, 2006; VERDONK et al., 2003)(THOMSEN; CHRISTENSEN, 2006);
    - Ant colony optimization (KORB; STUTZLE; EXNER, 2009; REDDY et al., 2017);
    - Tabu Search (BAXTER et al., 1998);
    - Particle swarm optimization (GOWTHAMAN; LYSKOV; KARANICOLAS, 2015; LIU; LI; MA, 2012; NG; FONG; SIU, 2015);
- Local Search used by Autodock Vina (TROTT; OLSON, 2009), SwissDock/EADock (GROSDIDIER; ZOETE; MICHIELIN, 2011) and GlamDock (TIETZE; APOSTOLAKIS, 2007);
- Exhaustive Search used by eHiTS (ZSOLDOS et al., 2007);
- Linear Programming Methods such as Simplex Method (RUIZ-CARMONA et al., 2014);
- Systematic Methods such as Incremental Construction used by FlexX (RAREY et al., 1996), Surflex (SPITZER; JAIN, 2012) and Sybyl-X (CERTARA, 2016);
- Statistical Methods
    - Monte Carlo (HARRISON, 2010);
    - Simulated annealing (SA) (DOUCET; PELLETIER, 2007; HATMAL; TAHA, 2017);
    - Conformational space annealing (CSA) (SHIN et al., 2011);
- *Similarity-based algorithms*
    - Based on sub structures (JONES; WILLETT; GLEN, 1995; RICHMOND et al., 2006; TRESADERN; BEMPORAD; HOWE, 2009);
    - Pharmacochemical (BAJORATH, 2002; CRUZ-MONTEAGUDO et al., 2014; KRISTENSEN; NIELSEN; PEDERSEN, 2013; SOARES, 2011);
    - Overlapping volumes (LEACH et al., 2010; MALDONADO et al., 2006);
    - Molecular interaction fields (MIFs) (WILLETT, 2006);
    - Hybrid approach (BREDA et al., 2008; ELOKELY; DOERKSEN, 2013; FERREIRA et al., 2015; HAGA; ICHIKAWA; DATE, 2016; HUANG; GRINTER; ZOU, 2010; LIU; WANG, 2015);

2.11 VS SOFTWARE PROGRAMS

There are several VS software programs using different docking algorithms that make a VS process easier for the researchers to execute by avoiding the need to have advanced knowledge of computer science and on how to implement the algorithms used in this task. In this regard, VS software can act as a possible cost reducer, since they function as a filter that selects from a database with thousands of molecules that are more likely to present biological activity against a target of interest. VS programs measure the binding affinity of a small molecule (ligand) to a molecular target of interest to determine the interaction energy of the resulting complex (CARREGAL et al., 2017).

Molecular docking software proved to be very promising technique to support the development of new drugs since the 1990s. However, according Santos (DOS SANTOS, 2015), after some years of use, it was identified that the scoring functions did not accurately predict the free energy of binding. Thus, these tools were ordering the list of compounds according to affinity in an incorrect form. Recently, due to the advance of the docking techniques used by the software, the focus has again turned to its use (ROGNAN, 2010).

Proteins are dynamic entities, and they possess an inherent flexibility that allows them to function through molecular interactions within the cell, among cells and even between organisms. Appreciation of the nonstatic nature of proteins is emerging, but to describe and incorporate this into an intuitive perception of protein function is challenging. Flexibility is of overwhelming importance for protein function, and the changes in protein structure during interactions with binding partners can be dramatic.

Table 1 summarizes the main characteristics of software used in VS. The first column contains the software used and its reference. The second contains the type of software license: Free for academic use, Freeware, Open Source or commercial. The Free for academic use license indicates that the software in question can be used for teaching and research in the academic world without a fee. However, it implies that the software has restrictions for commercial use. A freeware license indicates that the software is free. Thus, users can use it without charge and all the functions of the program are available to be used without any restriction. An open source license (OSS) indicates that the software source code is accessible so users can study, change, and

distribute the software to anyone and for any purpose. Software developed under a commercial license indicates that it is designed and developed for a commercial purpose. Thus, in general, it is necessary to pay some licensing fee for its use. The third column indicates on which platforms the software can be used (Windows, Linux or MAC). The next column indicates whether or not the software may consider protein flexibility during anchoring. The docking algorithm column lists the algorithms used by the software to perform the docking. The sixth column, called the scoring function, indicates which scoring functions are used by the software.

Table 1 – Virtual Screening Software programs

Software	License	Platform	Protein Flexibility	Docking Algorithm	Scoring Function
Autodock4 (MORRIS et al., 2009)	Free for Academic Use	Windows, Linux, Mac	Yes	Genetic algorithm Simulated Annealing	Hybrid (Force-field and Empirical)
Autodock Vina (TROTT; OLSON, 2009)	Open Source	Windows, Linux, Mac	Yes	Genetic Algorithm Simulated Annealing Local Search Particle swarm optimization	Hybrid (Empirical and Knowledge-based)
DOCK 6 (ALLEN et al., 2015)	Free for academic use	Windows, Linux, Mac	Yes	Shape fitting (sphere sets) Lowest Energy Binding	Force-Field Empirical
SwissDock / EADock DSS (GROSDIDIER; ZOETE; MICHIELIN, 2011)	Free for academic use	Web	No	Stochastic (Tabu search based) Local Search Combination of broad and local search of the conformational space	Force-Field

Software	License	Platform	Protein Flexibility	Docking Algorithm	Scoring Function
eHiTS (ZSOLDOS et al., 2007)	Freeware for academic Use	Unix	No	Exhaustive search	Hybrid (Empirical and Knowledge Based)
FITTED (CORBEIL et al., 2008; CORBEIL; ENGLEBIENNE; MOITESSIER, 2007)	Commercial	Windows,Linux, Mac	Yes	Genetic Algorithm	Force-Field
FlexX (RAREY et al., 1996)	Commercial	Windows, Linux	No	Incremental Construction	Empirical
FLIPDock (ZHAO; SANNER, 2007)	Freeware for academic Use	Linux e Windows	Yes	Genetic Algorithm	Force-field
Fred (MCGANN, 2011)	Free for academic use	Windows, Linux, Mac	No	Exhaustive search algorithm	Hybrid
GalaxyDock2 (SHIN et al., 2013)	Freeware	Linux	Yes	Conformational Analysis Genetic Algorithm	Force-Field
GeauxDock (FANG et al., 2016)	Open Source	Linux	Yes	Monte Carlo	Hybrid (Empirical and Knowledge based)

Software	License	Platform	Protein Flexibility	Docking Algorithm	Scoring Function
GlamDock (TIETZE; APOSTOLAKIS, 2007)	Freeware	Windows, Linux, Mac	No	Monte Carlo Simulated Annealing Local Search Conformational Analysis	Empirical
Glide (FRIESNER et al., 2004)	Commercial	Windows, Linux	Yes	Conformational Analysis Monte Carlo Sampling	Empirical
GOLD (VERDONK et al., 2003)	Commercial	Linux, Windows	Yes	Genetic Algorithm	Force-Field
ICM (ABAGYAN; TOTROV; KUZNETSOV, 1994)	Commercial	Windows, Linux, Mac	Yes	Monte Carlo minimization	Force-Field
iGEMDOCK / GEMDOCK (HSU et al., 2011)	Freeware	Windows, Linux	Yes	Genetic Algorithm	Empirical
LigandFit (MONTES; MITEVA; VILLOUTREIX, 2007)	Commercial	Linux	Yes	Monte Carlo	Force-Field

Software	License	Platform	Protein Flexibility	Docking Algorithm	Scoring Function
LigDockCSA (SHIN et al., 2011)	-	-	Yes	Conformational space annealing Global optimization	Hybrid (Empirical and Force-Field)
MOE (CHEMICAL COMPUTING GROUP, 2004)	Commercial	Windows, Linux, Mac	Yes	Conformational Analysis	Empirical, Force-Field
ParaDocks (MEIER et al., 2010)	Freeware	Linux	No	Genetic Algorithm	Hybrid (Knowledge-based and Empirical)
rDOCK (RUIZ-CARMONA et al., 2014)	Open Source	Linux	Yes	Genetic Algorithm, Monte Carlo, Simplex minimization	Hybrid (Empirical and Force-field)
SLIDE (SCHNECKE; KUHN, 2000)	Free for academic use	Linux	Yes	Conformational Analysis	Empirical
Surflex (SPITZER; JAIN, 2012)	Commercial	Windows, Linux, Mac	Yes	Incremental Construction	Empirical
Sybyl-X (CERTARA, 2016)	Commercial	Windows	Yes	Incremental Construction	Force Field Based

Software	License	Platform	Protein Flexibility	Docking Algorithm	Scoring Function
vLifeDock (CHOPADE, 2015)	Commercial	Windows, Linux, Mac	Yes	Genetic algorithm	PLP score, XCscore

3. Justifications

The development of a new workflow for VS that integrates the whole process of VS is justified by the lack of automatic integration between free software that need to be used throughout the VS process. In addition, the free tools available have difficult to use interfaces (often they have no graphical interfaces) and require advanced computer skills such as programming languages. Also, the developed platform will allow the reduction of human errors that may occur in the rational drug discovery process, because MolAr integrates several tools used in the process and prepares the files generated by the result of a tool to be used in the following steps. Thus, by automating the whole process of rational development of a new drug, it is easier to repeat an experiment, reducing the human errors that may arise in this process. The interfaces of MolAr platform are intuitive and easy to understand, which assists users in their use. So, it will be possible to produce more reliable results in a lower cost. Finally, as researchers do not have to spend a huge time learning how a program works because, since it will be easy to use, they can dedicate themselves to what is most important in this process, which is to find out possible targets and ligands, the execution of the VS and then carry out in vitro tests.

4. Objectives

4.1 Overall Objective

This thesis proposes a new workflow to enable the execution of the entire VS process in an integrated, automated and effective manner, from protein preparation (homology modeling and protonation state) to virtual screening. In addition, the developed workflow proposes a new way to perform virtual consensus screening between AutoDock Vina and DOCK 6. In this thesis, DOCK 6 utilizes conformations already optimized by AutoDock Vina to improve the VS.

4.2 Specific Objectives

Specific objectives of this thesis are:

- Carry out a broad literature review about the techniques, software and algorithms involved in the VS process, as well as the existing aspects in relation to existing VS systems;
- Develop an integrated workflow considering its performance and ease of use;
- Integrate and optimize existing techniques and tools decreasing the time needed to execute a VS simulation and improving its results;
- Develop and implement a workflow to support the whole VS process, executing simulations and presenting the results obtained in an easy and intuitive form;
- Develop the Virtual Screening features with Autodock Vina, DOCK 6.
- Implementation of consensus virtual screening so that dock 6 uses the best ligand conformations defined by Autodock Vina.
- Register MolAr workflow under Brazilian law;
- Publish papers rated at least B2 by CAPES.

5. Methodology

MolAr is a workflow designed to make it easy to perform the entire virtual screening process using free tools in an integrated manner. To achieve this goal, several steps was performed:

- The proposed workflow has been automated to be used effectively and easily using bash scripts and high-level languages such as Java and Python. The graphical interface was developed in the JAVA language, which was chosen because it is multiplatform and object oriented. There are also some scripts in bash and some scripts developed in the Python language. This was necessary because some of the tools used, such as MODELLER, have their Python-based operation.
- MolAr is available for free download at http://www.drugdiscovery.com.br/. Two versions of the software were developed: a full-featured version that works on Linux and a Windows version that has only the Homology modeling features with MODELLER and virtual screening with Autodock Vina. The full version for Windows is not yet available because not all software and libraries used by Linux workflow have a version for Windows; therefore, some workarounds will need to be used for everything to run on Windows. In homology modeling, for example, which has already been ported to Windows, the Ramachandran chart is generated using http://servicesn.mbi.ucla.edu/PROCHECK/ instead of using PROCHECK software. Autodock Vina has a downloadable version that works on Windows and it has been ported to be used on Windows without major changes.
- MolAr also attempts to simplify the entire installation process for the user. Thus, libraries and software used by MolAr Workflow are automatically installed during MolAr installation.
- Table 2 and Table 3 show the software and libraries automatically installed by MolAr in Linux and why they are needed.
- Table 4 shows the software programs automatically installed on Windows version.

Table 2 - Programs installed by MolAr on Linux and why they are needed

Software	Necessity
Open JDK	Package containing the necessary infrastructure for the development of Java applications.
Python	Used by MODELLER.
Mopac	Refinement of ligands.
MODELLER	Homology Modeling.
Pip	Python package manager on Linux.
Procheck	Ramachandran Plot.
Pdfunite	Join pdf files.
Pymol	View a 3D molecule and calculate RMSD.
Jmol	View a 3D Molecule.
Pdb2pqr	Python package that contains PROPKA, which is used in the protonation state adjustment.
MPI	It is used to carry out DOCK 6 in parallel.
DOCK 6	VS with DOCK 6.
Sphgen	It is used in DOCK6 to generate sets of overlapping spheres to describe the shape of a molecular surface.
Autodock Vina 1.1.2	Virtual screening with Autodock Vina 1.1.2.
DMS	It is used by DOCK 6 to compute the molecular surface of a molecule.
Yad	It is a tool used by MolAr to create graphical dialogs from bash scripts.
Autodocktools	It is used in Autodock Vina.
Chimera	It is used to configure the ligand for docking.
Ambertools	Used to run DOCK 6 with amber force field
Evince	It is used to view pdf files.
Openbabel	It is used to convert one chemical file format to a different one.
mgltools	It is installed to allow use of the adt and Autodock Vina.

Table 3 - Libraries installed by MolAr on Linux and why they are needed

Library	Necessity
biopython	Python scripts
python-dev	Python scripts
python-matplotlib	Python scripts to generate ROC curve
python-sklearn	Python scripts to generate ROC curve
libgfortran3	Procheck
numpy 1.8.2	Autodocktools
libgl1-mesa-dev	Autodocktools
mesa-common-dev	Autodocktools
libxext-dev	Amber Tools
libstdc++5:i386	DS Visualizer
libstdc++5	DS Visualizer

Table 4 – Program installed by MolAr on Windows and why they are needed

Library	Necessity
Open JDK	Package containing the necessary infrastructure for the development of Java applications.
Python	Used by MODELLER.
Mopac	Refinement of ligands.
MODELLER	Homology Modeling.
Autodock Vina 1.1.2	Virtual screening with Autodock Vina 1.1.2.

- After the development of MolAr, it was necessary to evaluate whether it reaches its goal. Thus, it was verified if it could help to carry out a VS and if it would be easy to use. So, MolAr was tested and validated in real uses, seeking to analyze what can be improved, until arriving at a suitable final version, that will be made available to other researchers. In the first stage of the tests, only researchers from the Pharmaceutical Chemistry Laboratory used the program. This contributed to the execution being carried out in a more controlled environment. Thus, the problems that appeared during the execution were solved and served as an opportunity to improve the software. Therefore, the software was used in three researches developed in the Pharmaceutical Chemistry Laboratory of the Federal

University of São João del-Rei. In these studies, Homology Modeling, VS, protein protonation state adjustment and redocking were performed. Then, MolAr build loop regions using MODELLER and performs Virtual Screening using Autodock Vina, DOCK 6 and a consensus between them. Section 6.3 describes the case studies performed.

- Finally, a tutorial on using MolAr has been produced. It teaches, step by step, how to perform all the features of MolAr.

Figure 7 summarizes the methodology used by MolAr to perform the entire Virtual Screening process.

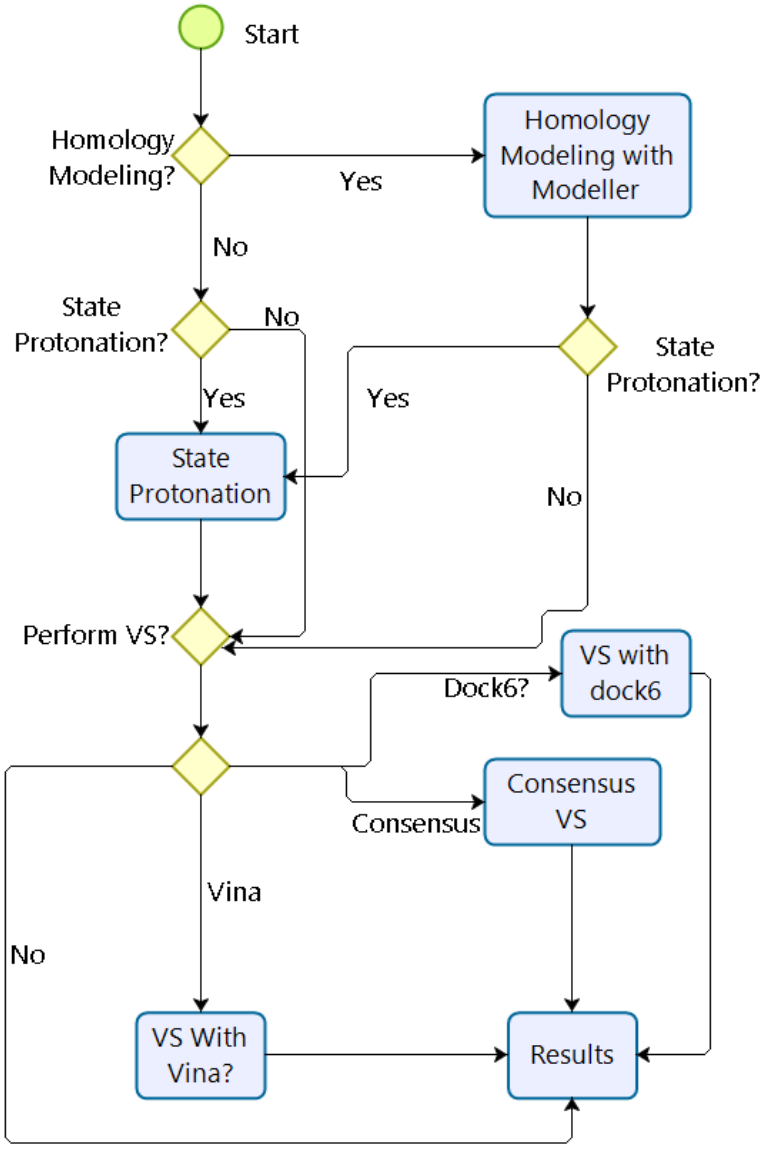


Figure 7 - MolAr methodology workflow

Prior to initiating a Virtual Screening, homology modeling may be required if the target has no 3D structure or if it has GAP regions in its structure (such as loop regions whose 3D structure is generally not defined by x-ray crystallography). It may also be necessary to adjust the protonation state of the target protein. After, Virtual Screening could be performed. When deciding to perform virtual screening, user must define whether VS will be performed with Autodock Vina, DOCK 6, or a Consensus between the 2 tools. In this case, the two tools will be executed sequentially and then a consensus between the 2 results will be made. Finally, MolAr will display the virtual screening result. After the result is displayed, it can be saved, and the ROC curve can be generated.

Below, we will explain in detail de workflow used in MolAr to perform Homology Modeling, VS with Autodock Vina, VS with DOCK 6 and Consensus Virtual Screening.

5.1 Homology Modeling

Figure 8 summarizes the Homology modeling performed by MolAr.

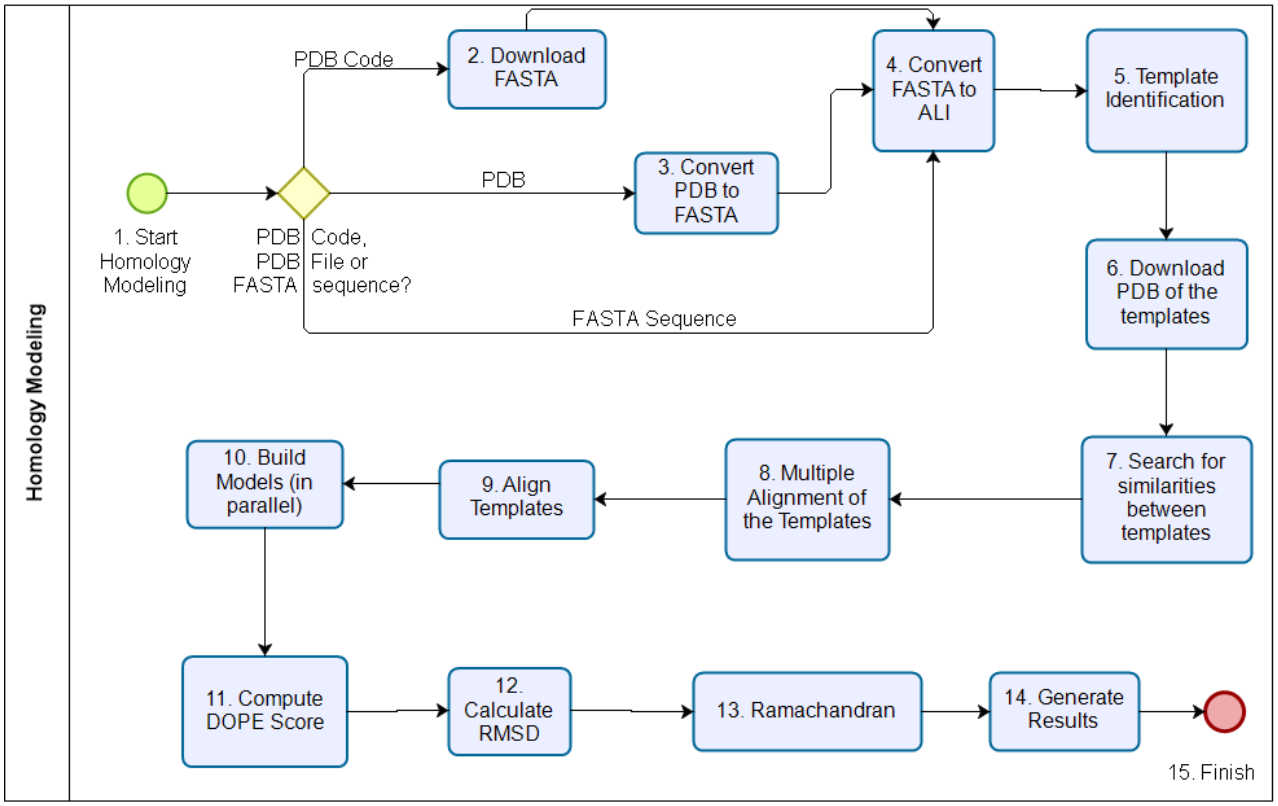


Figure 8 – Homology Modeling Workflow performed by MolAr

1. The first step is to start the Homology Modeling process. MolAr allows simple and easy to use homology modeling using MODELLER;
2. If user entered the PDB code of the target, MolAr will download the FASTA file corresponding to that code;
3. If user entered the PDB file, MolAr will convert the PDB file to a FASTA file;
4. If system has performed steps 2 or 3, or if user entered the FASTA sequence, MolAr will convert the FASTA to a file in the alignment format used by MODELLER;
5. Then, MolAr will identify templates to be used in the Homology Modeling process;
6. So, after selecting the templates that will be used, MolAr downloads the PDB file corresponding to these templates;
7. Then, MolAr will identify similarities between the templates;
8. The next step is the multiple alignment between the selected templates;
9. Next, protein is aligned to the multiple templates selected;
10. Thereon, models will be built by MODELLER in parallel;
11. Then, the DOPE Score will be calculated for each generated model;
12. Next the RMSD between each generated model and the template used is calculated;
13. In order for the user to evaluate the quality of the generated model, the system generates the Ramachandran plot for each model using Procheck(LASKOWSKI et al., 1993);
14. Finally, after carrying out the homology model process, MolAr display the results of the Homology Modeling.

In some cases, the target PDB file exists, but there are gap regions such as loop regions. These residues are in the PDB file, but during X-ray crystallography it was not possible to determine its atomic coordinates. MolAr addresses this issue by Homology Modeling through option Missing Residues. Figure 9 below shows the Missing residues functionality workflow.

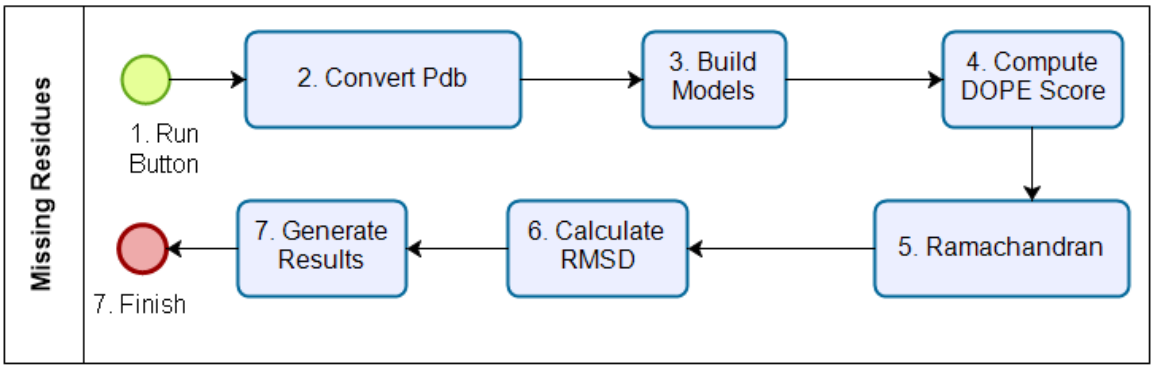


Figure 9 - Missing residues workflow performed by MolAr

Below is a brief explanation of the workflow described in Figure 9:

1. Firstly, user start the Missing residues Functionality;
2. Next, the amino acid sequence of the protein will be read from the PDB file;
3. Then, gaps will be filled;
4. The DOPE Score of each generated model will be calculated;
5. The Ramachandran plot for each model will be generated using Procheck;
6. The RMSD for each generated model will be calculated;
7. Finally, user can see the results.

Finally, the PDB file can have more than one chain and contain gap regions in them. In this case, user should use the multi chain modeling function, whose workflow is defined in Figure 10 and explained below.

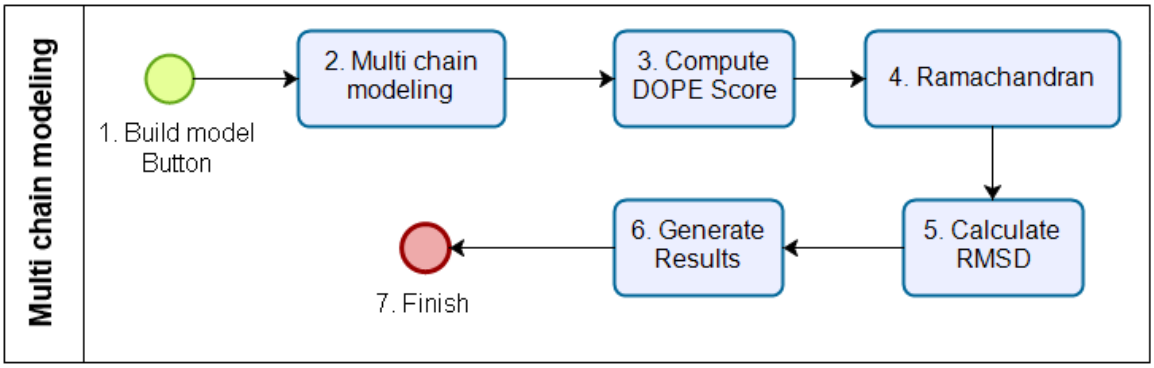


Figure 10 – Multi chain modeling performed by MolAr

The workflow in Figure 10 can be explained as follows:

1. Firstly, user starts the multi chain modeling;
2. Then, the multi chain modeling using MODELLER will be carry out;
3. Next, the DOPE Score will be calculated;

4. After, the Ramachandran plot for each generated model will be created using Procheck;
5. The RMSD will be calculated;
6. Finally, a window with the results will be shown to the user.

5.2 VS with Autodock Vina

VS using Autodock Vina is performed through the Octopus Submenu. An overview of the Octopus (MAIA et al., 2017) workflow can be seen in Figure 11. First, directories of ligands and targets are chosen. Ligands must be in the PDB format and the files in targets folder must be in the Autodock Vina format. The target database has a configuration file with X, Y and Z binding site coordinates, a grid box size delimiting the region for molecular docking simulations and the reference binding energy according to the re-docking of crystallographic ligand. These files are explained in detail in section (detailed in section 6.2.2.3). If user choose to refine ligands, MolAr will perform the refinement using the Run MOPAC (STEWART, 2016) software (explained in section 6.2.2.2). Next, ligands are converted from PDB to PDBQT file format while assigning the rotatable bonds, the Gasteiger-Marsili net atomic charges (GASTEIGER; MARSILI, 1980) and only the hydrogens on polar atoms (oxygen and nitrogen) are kept, while other hydrogens atoms are removed. Then, visual inspection of the geometries of the ligands can be performed using PyMOL (DELANO, 2002). In the next step, the docking is performed using the Autodock Vina, which runs until all the ligands have been docked on a set of targets. Finally, the binding energy results for the complex target/ligand are generated. The standard crystallographic values for the binding energies between ligands and targets are also displayed.

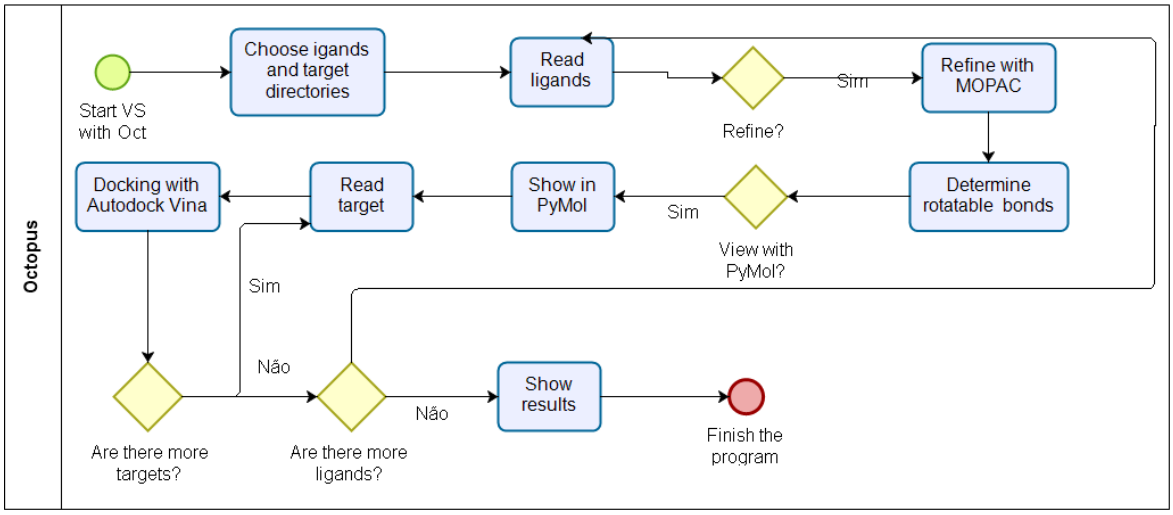


Figure 11 - Octopus Workflow performed by MolAr

5.3 VS with DOCK 6

Figure 12 illustrate the MolAr workflow showing how VS with DOCK 6 is performed.

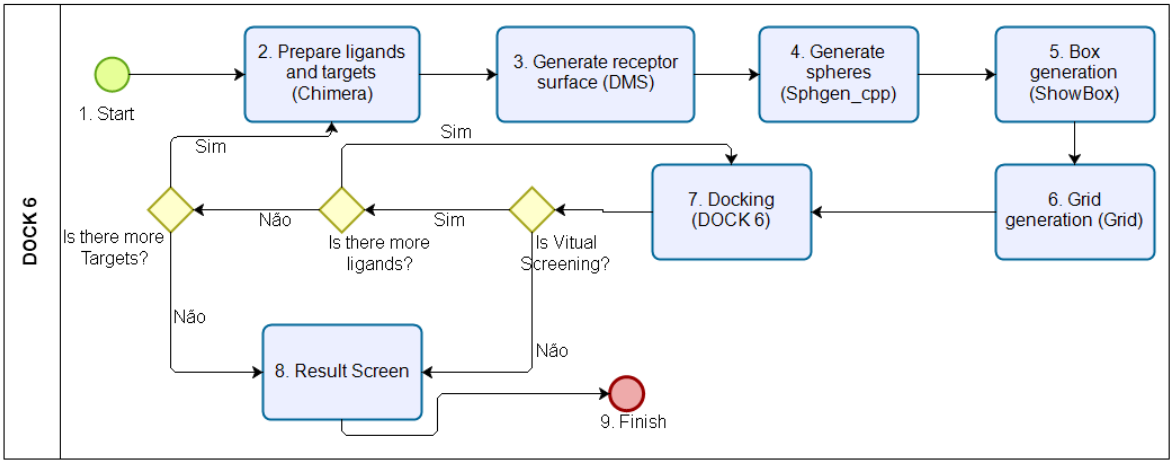


Figure 12 - DOCK 6 Workflow performed by MolAr

The workflow of Figure 12 can be explained as follows:

1. Firstly, user starts redocking or VS with DOCK 6.
2. Initially it is necessary to prepare the molecular target and the ligand. So, a Chimera script is executed, and it performs the following operations:
    a. Adds hydrogen atoms to the system;
    b. Calculate the charge of the structure using AMBER ff99SB protein force field (LINDORFF-LARSEN et al., 2010);

    c. The molecular target and the ligand are detached, and they are saved in different files.
        i. The molecular target is saved without the hydrogen atoms.
3. Next, it is necessary to prepare the molecular target binding site. The active binding site has already been defined a priori, since the program receives as input the target / ligand complex. The first step in preparing the site is the calculation of the target surface to which the solvent will have access. MolAr runs the DMS1 program to do this step;
4. After the molecular surface is created, spheres representing the binding site are created using the sphgen_cpp2 program. The points of that binding site are known as spheres and they are used to define the volume or space within the binding site where the drug will interact. The purpose of the spheres is to generate a grid of sphere centers that reflects the shape of the active site. To do this, MolAr uses grid program. With this, it is the protein / macromolecule and not the drug that defines the shape of the pocket. During docking the sphere centers will be combined with ligand (drug) atoms to generate orientations of the ligand active site. By default, the spheres used will be within a radius of up to 8 Å from the center of the binding site;
5. Next, the box (which is the cubic region where DOCK 6 will perform the docking) is generated. MolAr uses showbox command (DOCK 6 installs this command in the same folder it was installed) to run this step.
6. After the box generation, grid energies are calculated using the grid program (grid is also installed by DOCK 6 in the same folder it was installed).
7. Next, docking can be accomplished using dock6.
8. If user is performing a VS and if there are more ligands, docking will be performed on all ligands. If user is performing a VS and there are not more ligands, but there are more targets, MolAr will carry out all the previous steps to the next target. Next, if it is not a VS, or if there are not more targets, the result screen will be show.
9. Finally, user can save the results and close the DOCK 6 result screen.

[1] DMS program can be downloaded from www.cgl.ucsf.edu/Overview/software.html#dms,
[2] Sphgen_cpp can be downloaded from http://dock.compbio.ucsf.edu/Contributed_Code/sphgen_cpp.htm

5.4 Consensus Virtual Screening (CVS)

The CVS workflow performed by MolAr is showed in Figure 13. Firstly, a VS using Autodock Vina is performed, as showed in Figure 11. Then, the Autodock Vina output PDBQT file is converted to the DOCK 6 format using babel program. Next, VS is performed using DOCK 6, as described in Figure 12. Before displaying the result, a merge of Autodock Vina and DOCK 6 results is done. Finally, CD is calculated by MolAr and the result is displayed to the user on the screen.

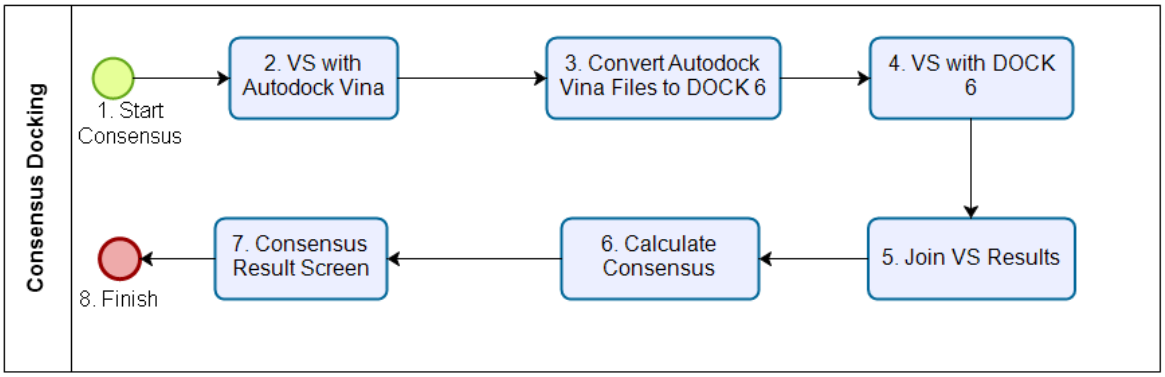


Figure 13 – Consensus Virtual Screening Workflow performed by MolAr

6. Results and Discussion

The accuracy of molecular modeling studies may be affected by the accuracy of the input structures used in the process. Thus, the developing of a software that performs VS require a preparation of the structures that will be used throughout the process. Thus, among the functionalities included in the software developed in this research, there were developed some features to aim in the preparation of the structures used in VS, which involves many tools that aim:

1. add atoms to the protein (building loop regions, for example);
2. check the quality of the model;
3. adjust the protonation state of the protein and ligands;

In this section, we will describe the features developed in the proposed VS software. The first version of the software proposed in this thesis, also called Octopus (MAIA et al., 2017), is available for download and has already been used by researchers from 23 different countries (Appendix F). However, the current version contains substantial improvements over the previous one. The previous version, named Octopus, did not perform the whole VS process. It only performs VS using Autodock Vina. The current version of the developed software assists the execution of the entire VS process, from the preparation of the molecule to the realization of the docking and the VS, besides having tools that allows an analysis of the structures used. The main objective of developing an integrated software is to facilitate the execution of the whole process, making it more automated, agile and less prone to human errors. The name of the developed software was chosen by joining the prefix of two English words, Molecular and Architecture, which thus formed the name MolAr.

MolAr is composed by a set of tools that act in an integrated form to automate VS simulations. Although it has simple and intuitive interfaces, it has many customization options and requires less time to run the entire process. As a differential to other VS software, the proposed one uses only free tools and it has an intuitive interface that does not require any prior programming language knowledge. As a result, it is able to facilitate the execution of the VS experiments by the automation of the whole VS process.

In addition, it is necessary to install several tools and libraries to perform the VS. Many of them are difficult to install. MolAr automates the installation of all tools and libraries needed to perform the VS process.

In the next subsections, we describe in detail the installation procedure of MolAr, as well as some case studies.

6.1 Installation setup

The difficulty of installing free software used in VS is one of the problems faced by users. Often, in addition to installing the program, it is necessary to install libraries used by them and the need for these libraries is often not reported in the program.

In order to overcome this problem, the installation of MolAr automatically installs all the software used with few user interventions. The installation procedure is different on Linux and Windows, but it is quite simple. So, to install MolAr, one just need to:

Linux:

I. First, configure the installation by entering the command ./config.sh, which will install the most current version of JAVA and display a screen that asks for MODELLER (ŠALI, 2013) and MOPAC (STEWART, 2016) license (Figure 14).

II. After, it is necessary to run the command ./install.sh. The installation may take several minutes because a lot of programs and libraries will be downloaded and installed (Table 2 and Table 3). It is necessary to follow the installation and confirm the installation of each program.

III. After executing the steps above, user just must execute the command molar and the interface of Figure 15 will be displayed.

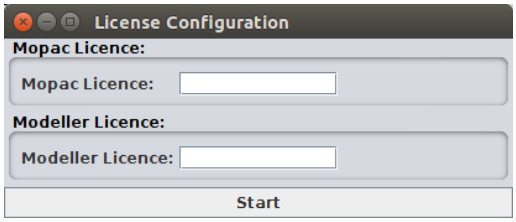


Figure 14 - License setup screen displayed during MolAr installation setup

Windows:

- Run the molar_win.exe command. It will install all the necessary software so that the version available for Windows can be used.

6.2 MolAr

MolAr was developed using bash scripts and high-level languages, such as Java and Python. Most of the code developed is in java, although there are Python and bash scripts developed during the implementation of MolAr. The code developed in the Java language has 25990 lines of code, divided into 121 java classes. As a result, 812 pages would be required to show the contents of these files, making it difficult its inclusion in this work. So, they were

not included in this thesis. However, MolAr source code was registered into Instituto Nacional de Propriedade Intelectual (INPI)3.

Figure 15 shows the main screen of MolAr program. MolAr consists of 4 main menus, which are: Target Builder, Docking, Tools and Help. In the following subsections, MolAr will be described in detail.

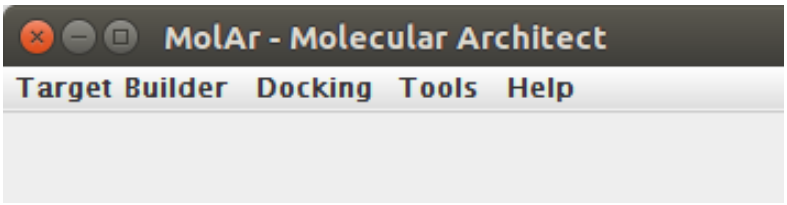


Figure 15 - MolAr main screen

6.2.1 Target Builder Menu

The Target Builder menu contains the features related to the Homology Modeling process, also known as Comparative Modeling (described in subsection 2.7).

The prediction of the 3D structure of a protein from its amino acid sequence can often be made by Homology Modeling. Homology Modeling is a methodology for predicting the protein 3D structure of a protein based on the general observation that proteins with similar sequences have similar structures (CAVASOTTO; PHATAK, 2009). Homology Modeling can also be used to add atoms to the protein (building of loops, regions whose 3D structure is not represented).

The quality of a homology model depends mainly on the similarity between the structure that will be used as a model and the target protein to be modeled. There is several software used in the development of drugs by homology modeling. One of the most used free software is MODELLER (ESWAR et al., 2007). This software allows the alignment of a sequence to be modeled with related known structures, automatically calculating the new model containing all the atoms except the hydrogens. Although the tool is quite complete, it is not easy to use by many users, because it is necessary for the user to have knowledge of the Python programming language to perform the modeling, as well as knowledge of how MODELLER works. So, the goal of building Target Builder functionality in MolAr is to make MODELLER easy to use and eliminate the need for the knowledge of the Python programming language. Figure 16 shows the Target Builder menu expanded in MolAr.

[3] http://www.inpi.gov.br/

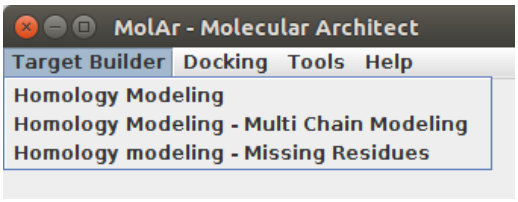


Figure 16 - Target Builder menu

As can be observed, this menu covers a set of tools used in homology modeling. Target Builder menu encompasses the features having interface with MODELLER. The great advantage provided by MolAr in implementing this integration is the user-friendly interface with MODELLER and allowing modeling customizations to be made by the user. This Menu has three options: Homology Modeling, Homology Modeling – Multi chain modeling and Homology Modeling - Missing Residues.

Next, we will describe each of these options in detail.

6.2.1.1 Item Homology Modeling

Target Builder performs homology modeling process in 15 steps (Figure 8). At each step, multiple scripts are executed.

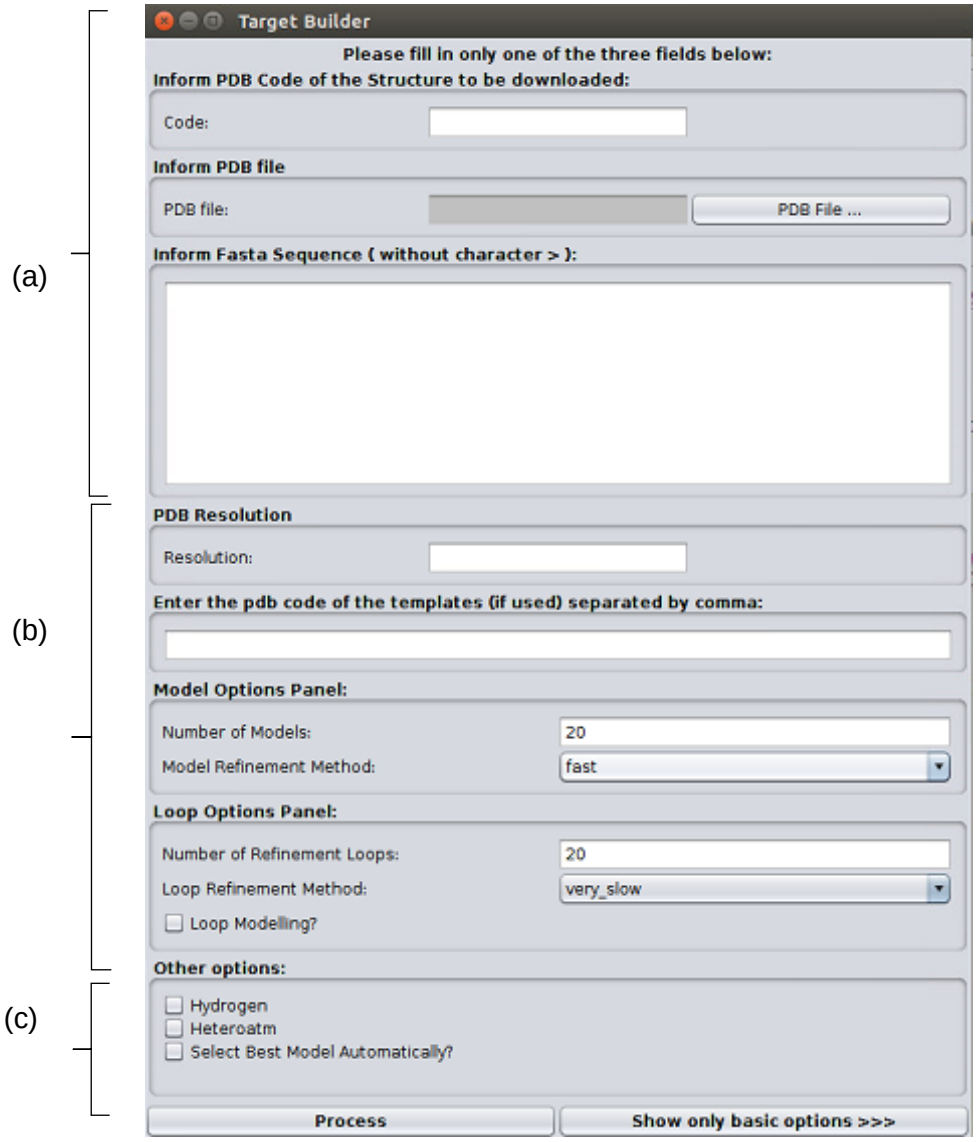


Figure 17: MolAr homology modeling screen (a) Sequence Data; (b) Simulation Parameters; (c) Other Options

The first option of the Figure 16 corresponds to the Homology Modeling functionality. It is the main feature of Target Builder. It works as follows:

- This feature allows simple and easy to use homology modeling using MODELLER. Figure 17 shows the main interface of homology modeling Target Builder feature. It is required to the user to fill in one of the following fields: PDB code, PDB File or FASTA sequence of the structure (Figure 17a). In Figure 17b user can inform templates to be used, the resolution of the PDB file (used to generate Ramachandran plot), the number of models that will be generated and the number

of refinement loops to be made during the process. See modeler online manual4 for details. It is noteworthy that the use of several models generally decreases the quality of the generated model (NIKOLAEV et al., 2018). Moreover, user must select witch modeling method will be used. The possible methods are, in order of optimization: very_fast, fast, slow, very_slow, slow_large. The name of the methods indicates that the more we want to optimize, the slower the modeling will be. It is possible to do the loop refinement in each model. So, in this case, user must inform the amount of refinement loops as well as the refinement method, which follows the same nomenclature and meaning of the modeling methods explained above. Finally, user can also indicate if they want to generate the new model with heteroatoms and hydrogen (Figure 17c). If user select the option “Select Best Model Automatically”, MolAr will select the best model based only in DOPE energy. If user does not select this option, after Target Builder performs the modeling, it will ask to the user to select the best model based on a list containing the following data: DOPE energy, RMSD and Ramachandran plot. After user presses the Process button, MolAr will check its modeling parameters to verify whether the user has entered the PDB code, a PDB file, or a FASTA sequence.

- Then, if user does not specify specific templates to be used in the Target Builderl main page (Figure 17b), MolAr will search in the MODELLER database and list the proteins having an identity greater than 25% (Figure 18). Following, user will have to select a template protein among the possible ones presented by MolAr. After selection, the modeling continues;

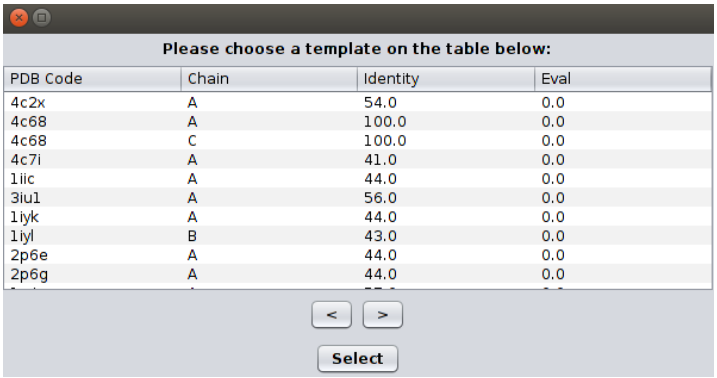


Figure 18 – Select Templates Screen displayed by MolAr during the homology modeling process

[4] https://salilab.org/modeller/manual/

- Next, MolAr will download the PDB files corresponding to the selected templates.
- Finally, after carrying out the homology model process defined in Figure 8, a window with the results will be shown to the user by MolAr. It allows user to choose the best model based on DOPE energy, Ramachandran plot and RMSD (Figure 19). Based on these parameters, users can make an initial evaluation of the quality of the model. However, if the "Select best model automatically" option is chosen in Figure 17c, Target Builder will select automatically, at the end of the modeling, the best option based on the smaller DOPE Energy. So, in this case, Figure 19 will not be displayed.

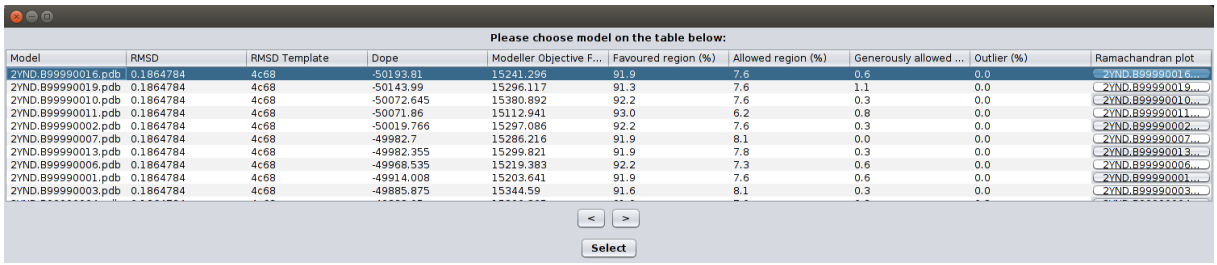


Figure 19 – Homology modeling result screen

During homology modeling process, it is important to know if the model developed has a good quality. Thus, for each model produced, MolAr displays the value of the RMSD relative to the target template (second column on Figure 19) and allows the visualization of the Ramachandran plot (which is shown in Figure 20, after clicking on button in the last column of the table in Figure 19). So, user can decide which model is the best. The results screen also allows any column to be sorted in ascending or descending order by just clicking on it.

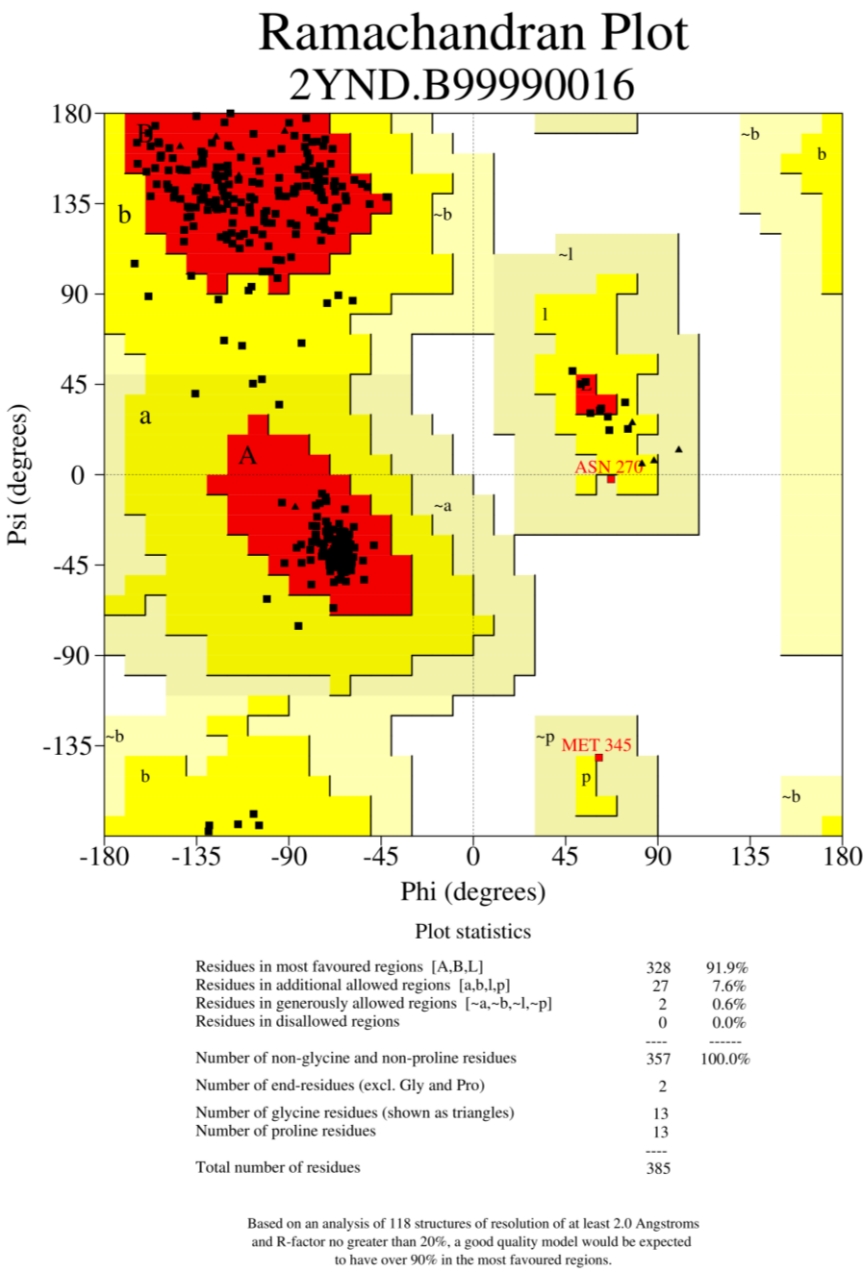


Figure 20 - Ramachandran plot of the model selected on Figure 19

The Python scripts used to run MODELLER were implemented based on the MODELLER tutorial scripts5. Although the scripts provided by the tutorial works well, they need to be modified before each experiment to adjust the parameters to be used (protein and templates used, number of models and loops of refinements, etc.). The modified version was generalized to work for any experiment, considering only the parameters entered by the user in MolAr main interface (Figure 17). In addition, scripts in MolAr allow the generation of models

[5] https://salilab.org/modeller/tutorial/

with heteroatoms. Also, the scripts took a long time to run in some situations. So, they were parallelized, using multiple cores, to be faster. Finally, since scripts do not need to be modified every time they are executed, this will probably reduce the occurrence of human errors.

6.2.1.2 Item Homology Modeling – Multi chain modeling

MolAr also allows user to model a multi chain protein with gaps in its 3D structure (Figure 21). In this case, the PDB file of the target and the template, and the chains to be aligned should be informed. This process is executed after pressing the Build model button, which carry out the multi chain modeling.

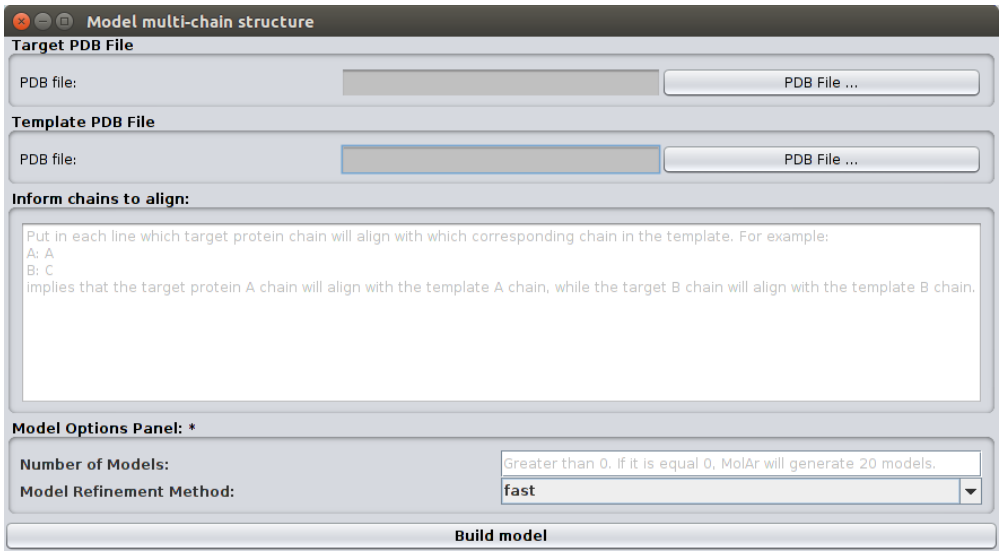


Figure 21 – Multi chain modeling screen

Below, we will explain how it works to the user.

1. Firstly, after filling in the form in Figure 21, user will press the button Build model.
2. Next, the multi chain modeling using MODELLER will be carry out according with the values of the fields Template PDB file, Number of models and Model refinement method.
3. Then, a window with the results will be shown to the user Figure 22.

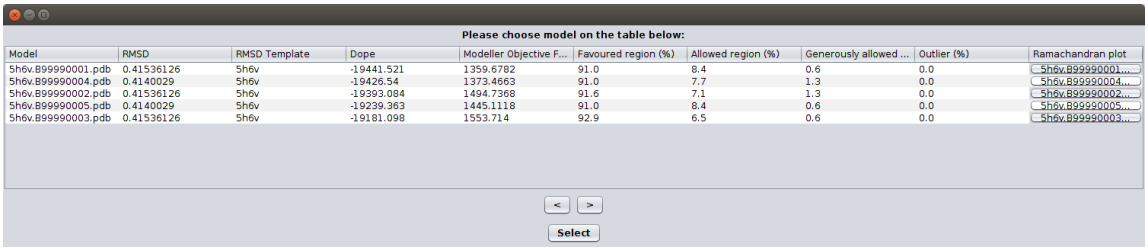


Figure 22 - Multi chain modeling result screen

4. Then, user can save the selected best model and close the window.

6.2.1.3 Item Homology Modeling – Missing Gaps

The downloaded PDB files could have some missing residues. These residues are documented in the PDB file, but during X-ray crystallography it was not possible to determine its atomic coordinates. The Missing Residues option was created to address this problem by Homology Modeling. If there are gaps in the target protein PDB file, MolAr may still attempt to complete these gaps (Figure 23). After filling in the data from the Figure 23 the process of build models with missing residues will began.

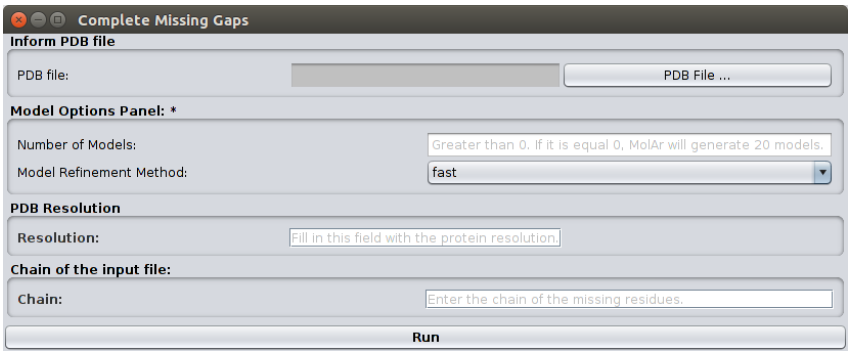


Figure 23 - Missing gaps screen

Missing residues feature works in the following way:

1. Firstly, after filling in the form in Figure 21, user will press the button Run.
2. Next, the workflow defined in Figure 9 will be carried out.
3. Finally, a window with the results will be shown to the user (Figure 24).

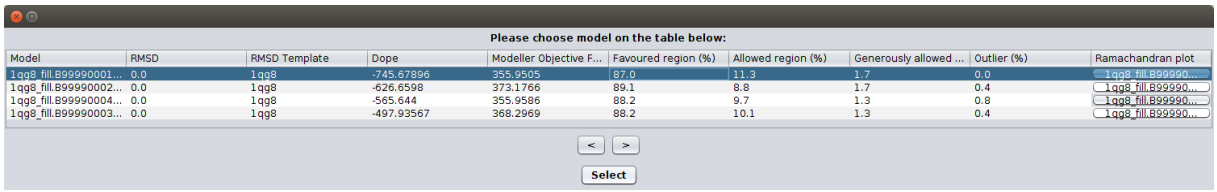


Figure 24 - Missing gaps result screen

4. Then, user can save the selected best model and close the window.

6.2.2 Docking Menu

The docking menu has the features that allow the realization of Molecular Docking and Virtual Screening. The developed software allows the realization of the Virtual Screening through Autodock Vina (TROTT; OLSON, 2009) DOCK 6 (ALLEN et al., 2015) or through a consensus between them.

6.2.2.1 Octopus

Octopus performs VS using Autodock Vina. In relation to our previous work (MAIA et al., 2017), a graphical interface was developed in Java allowing Octopus to be more user-friendly and some bugs were fixed. Octopus allows the execution of four main functionalities (Figure 25), which are: Virtual Screening with Mopac, Virtual Screening without Mopac, Run Mopac and Database Manager.

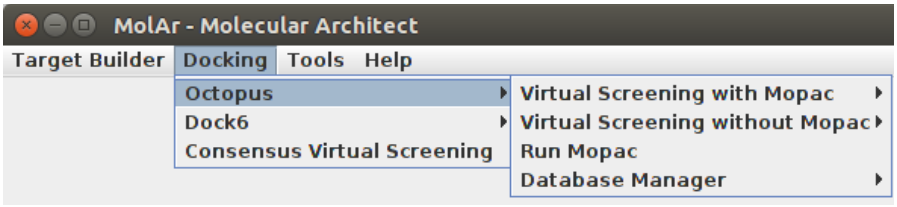


Figure 25 - Octopus submenu

6.2.2.2 RUN MOPAC

RUNMOPAC is a Python software (Appendix A) developed by our group and registered in INPI. This software calculates the net atomic charges for each atom in each molecule avoiding a massive work by the user. Next, the ligands, in PDB file format, are refined through MOPAC2016 (STEWART, 2016) using the Parametric Method 7 (PM7) (STEWART, 2013) and EF routine (STEWART et al., 2012) to search for the structure of local minimum.

The Run Mopac screen (Figure 26) has only one field. In this screen, user needs to select the folder where the ligands whose geometry should be refined are located. Then, after clicking the Run Mopac button, MolAr will refine the geometry of the ligands in the selected folder.

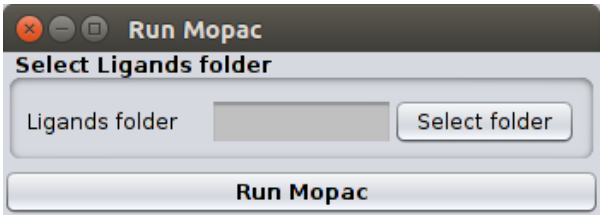


Figure 26 - Run Mopac screen

6.2.2.3 Virtual Screening with Mopac and Without Mopac

MolAr performs the VS process using Autodock Vina (TROTT; OLSON, 2009) as described in the workflow of Figure 11 through Octopus feature. Octopus carry out the VS in 2 different forms, depending on which menu option was chosen (Figure 25), which are: with a

previous execution of Mopac (choosing Virtual Screening with Mopac) or without running MOPAC (choosing Virtual Screening without Mopac). For both the screen displayed is the same as in Figure 27. The only difference is whether Mopac will be carry out.

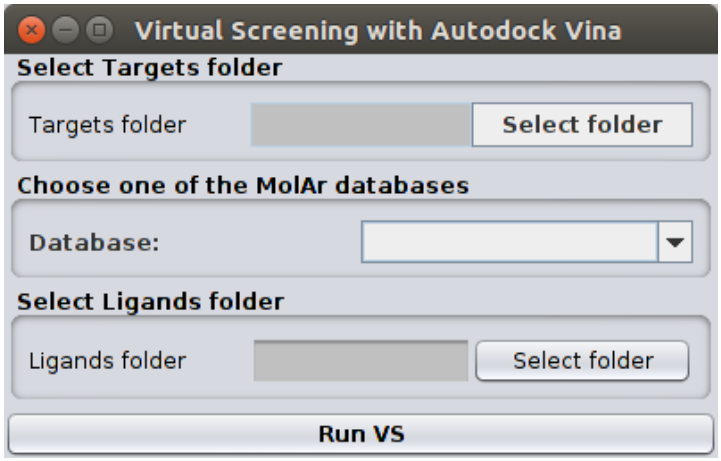


Figure 27 - Octopus main screen

In the main Octopus screen, you can select the targets database where the VS should be performed and the ligands folder. The ligands folder must have all ligands to be used in VS, in the PDB format.

The target database, on the other hand, must be in the PDBQT format, which is used by Autodock Vina. MolAr has two integrated databases that are the Our Own Molecular target (OOMT) (CARREGAL et al., 2017) and the Brazilian Malaria Molecular Target (BRAMMT) (NUNES et al., 2019). The OOMT database comprises various receptors from the Protein Data Bank (PDB), and it includes specific targets for cancer, dengue, and malaria. The BRAMMT database comprise receptors for Plasmodium falciparum. If any of the predefined databases are selected (OOMT or BRAMMT), the Select Folder button is disabled (as in Figure 27, where the BRAMMT database is selected). Otherwise, user can select another database. To illustrate how the target database should be, we will use the OOMT database structure as an example. Other databases can be automatically added to MolAr. It can be done by adding the database, inside the folder where MolAr is installed, in the Octopus / database subfolder.

The OOMT database has 42 targets and each one is placed in a specific folder, identified by the PDB code of the target protein and a file containing the reference binding energy values between each target and its crystallographic ligand (Figure 28).

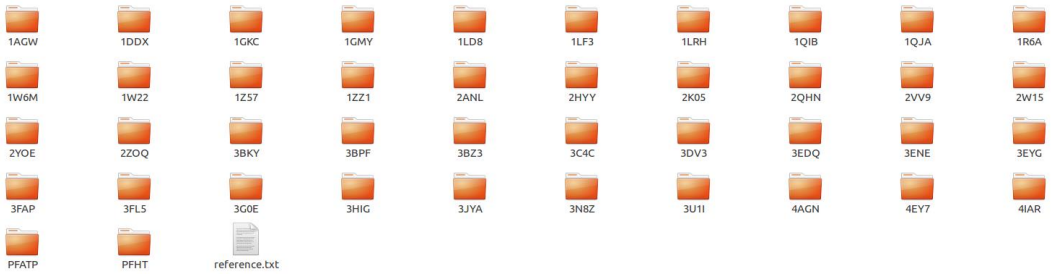


Figure 28 - OOMT database

Within each of the target folders, there should initially be 2 files. The target protein file in the PDBQT format and the conf file. Figure 29 shows an example for the 1AGW target protein (Crystal structure of the tyrosine kinase domain of fibroblast growth factor receptor 1 in complex with su4984 inhibitor). The 1AGW.pdbqt file has the 1AGW protein in the PDQT format.

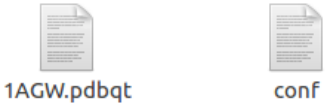


Figure 29 – Contents of the folder 1AGW of OMMT database

The conf file has the information about the binding site. Figure 30 shows the conf file used for the 1AGW protein. It is important to fill in only the coordinates for the center of the binding site (center_x, center_y and center_z), the dimensions of the box to be used (size_x, size_y, and size_z) and the exhaustiveness parameter to change the amount of computational effort used during a docking experiment. The default exhaustiveness value is 8 and increasing it to about 24 provides a better docking result. (FORLI et al., 2016).

```
exhaustiveness = 24

center_x = 9.877
center_y = 3.592
center_z = 23.95

size_x = 20
size_y = 20
size_z = 20
```

Figure 30 - conf file for 1AGW protein

The target database, with the files and folder structure explained above, can be easily created using MolAr's Database Manager feature.

6.2.2.4 Database Manager

The Database Manager feature (Figure 31) allows to manage the Octopus database and avoids VS failure due to database inconsistencies. It verifies if the databases used by Octopus are correct and also allows the user to create and edit new databases. The Database manager has three basic features: (1) creation of a new database (Figure 31(a) and Figure 31(b)), (2) fix problems in an existing database (Figure 31(c)) or (3) edit the data stored in an existing database (Figure 31 (d)). So, if there is any missing or incorrect data in the database it is possible to add it manually.

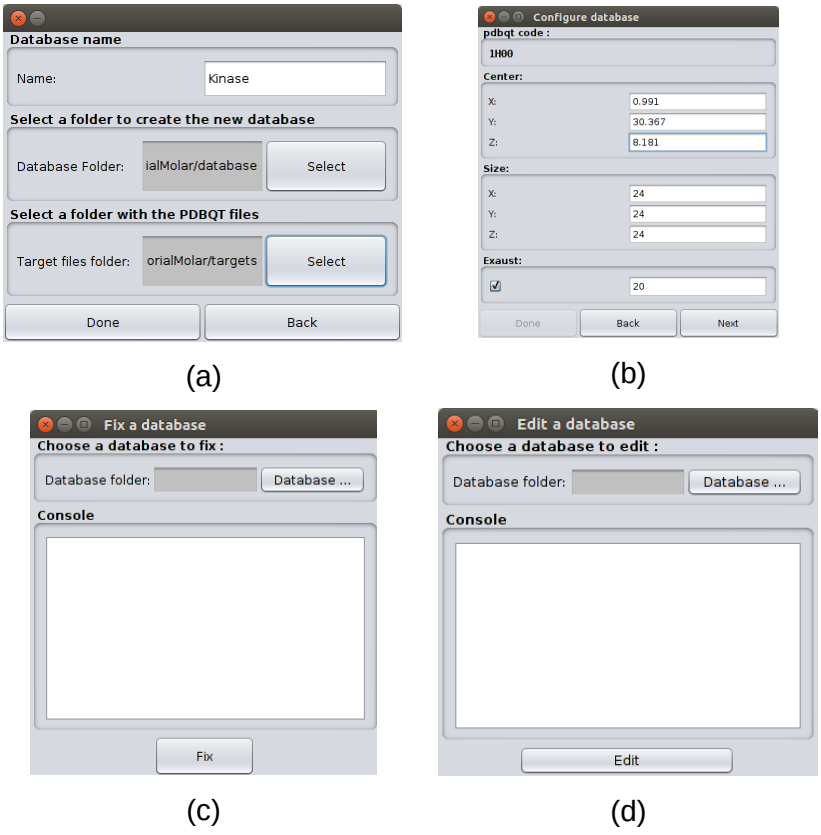


Figure 31 – Database Manager: (a) Create database main screen (b) Configure database screen; (c) Fix database screen; (d) Edit database screen

6.2.2.5 DOCK 6

The DOCK 6 program was the first docking program (ALLEN et al., 2015; KUNTZ et al., 1982) and it was created in the 1980s by Irwin and Kuntz's Group of Pharmaceutical

Chemistry Laboratory of University of California. Even though DOCK 6 is a powerful program with multiple usage options, it does not have a graphical interface.

Thus, a graphical interface was created to access the main features of DOCK 6. Before creating this interface, to do a VS or a simple redocking using DOCK 6, multiple command lines was required and in addition, it was necessary to modify several configuration files. If one of the steps required to perform the docking was forgotten or done incorrectly, the whole docking was compromised. So, this process is prone to many errors, such as an incorrect configuration, which leads to a waste of time to perform virtual screening.

MolAr allows the user to dock using DOCK 6 using a friendly graphical interface and with the minimum effort of the user. So, to do the docking process, user should do fewer interventions in the process. Thus, it is expected that fewer human errors will be committed, besides expecting a gain in the speed of its execution, since everything is done automatically. The graphical interfaces developed allow the use of DOCK 6 to execute a redock or to perform a Virtual Screening. Figure 12 illustrates the workflow of DOCK 6. By default, MolAr uses the parameters set in Appendix G for DOCK 6 input files.

6.2.2.6 Redocking with DOCK 6

Redocking aims to restore the initial position of a ligand in a crystallographic structure of a ligand/target complex. Usually redocking is performed to verify if the docking parameters for the program used are able to predict the structure and interactions of a known complex.

Figure 32 shows the MolAr Redock Screen using DOCK 6. As can be observed, for the default redocking it is necessary to inform the PDB file containing the protein-ligand complex which the redock will be performed and the PDB code of the ligand to be considered during the redocking process.

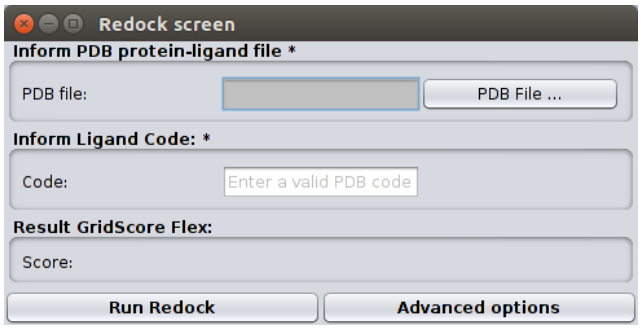


Figure 32 - Redock basic screen

Redocking can be done using the following methods: GridScore Flex; GridScore Flex with Amber, GridScore Rigid, GridScore Rigid with Amber. Amber performs some molecular

dynamics simulations during docking with DOCK 6, but it spends much more time to perform VS. The GridScore Flex method is used by default in redocking. If user want to use a docking method other than the default (GridScore Flex), he must click in the Advanced options button. Thus, the screen in Figure 33 is displayed and the docking method field will allow to change the docking method. DOCK 6 allows many other possible configurations that cannot be made on the MolAr screen. MolAr maintain this possibility, allowing the program to remain flexible by inserting the specific configuration files with the required settings. Thus, you must enter the .in file with the required settings. The files must be in the same format defined by the DOCK 6 manual.

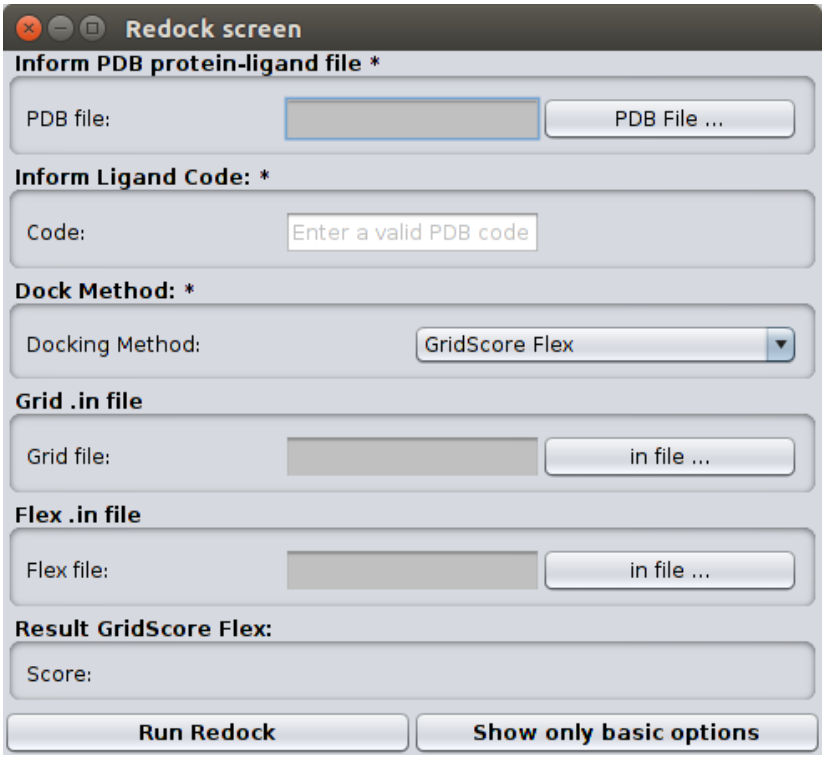


Figure 33 - Redock advanced screen

6.2.2.7 Virtual Screening with DOCK 6

VS can identify the most promising compounds for biological assays and decrease the costs associated with drug development (WESTERMAIER; BARRIL; SCAPOZZA, 2015). MolAr allows a VS to be performed as defined in Figure 12. Figure 34 shows the VS screen using DOCK 6. It is necessary to inform only the target/ligand complex folder and the ligands folder that will be used in the VS process.

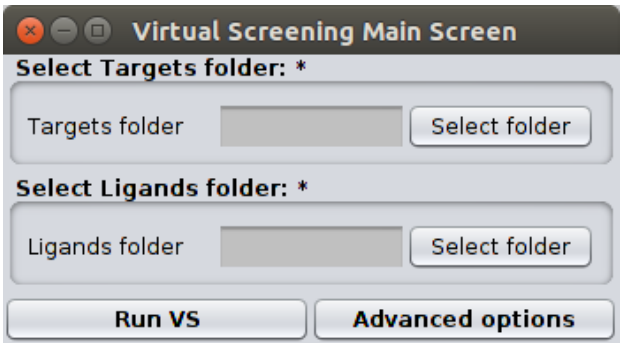


Figure 34 - Virtual screening with DOCK 6 main screen

If user press the Advanced options button, MolAr will allow the docking method to be changed (Figure 35). GridScore flex is the default if user does not change it.

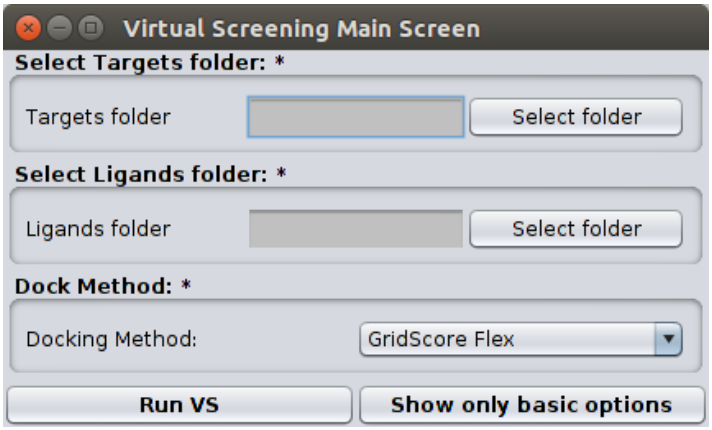


Figure 35 - Virtual Screening with DOCK 6 main screen after pressing the advanced options button

After performing the VS, a result screen is displayed (Figure 36), with the VS result. This screen shows the result of the VS and allows to order the result by PDB code, ligand used or by the Grid Score. The results screen also allows the data to be filtered according to the data of each column (Figure 37).

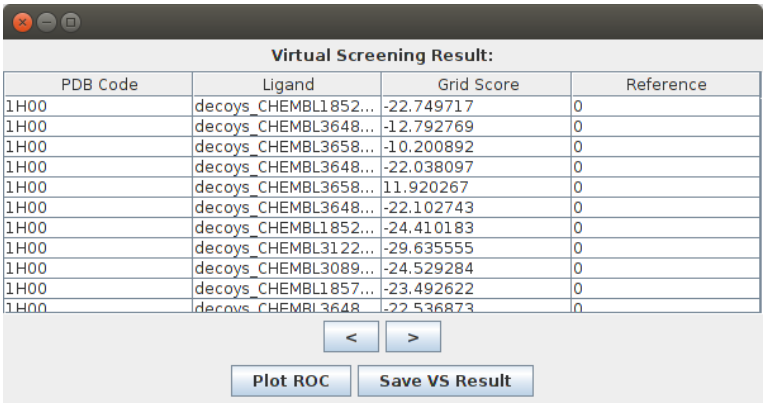


Figure 36 – VS with DOCK 6 result screen

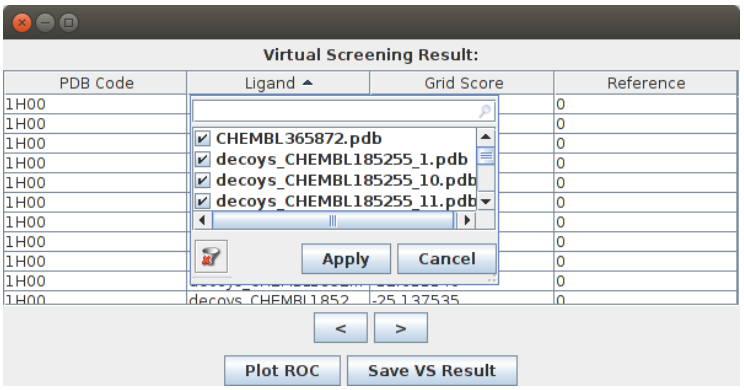


Figure 37 - Filter being performed by MolAr in the ligand column

Finally, when filtering the screen by the PDB code, selecting the active ligands and clicking on the Plot ROC command, the ROC curve is generated and AUC is calculated, as shown in Figure 38.

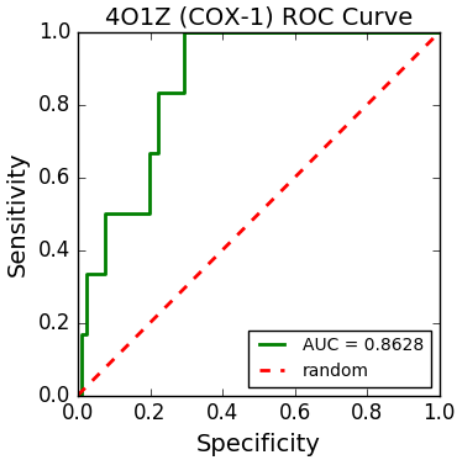


Figure 38 - ROC Curve for 4O1Z VS

6.2.2.8 Consensus Virtual Screening (CVS)

CVS is a relatively recent technique and although it has not yet been widely applied in the literature (POLI; MARTINELLI; TUCCINARDI, 2016), it presents some promising results (ALIEBRAHIMI et al., 2017; CHERMAK et al., 2016; HOUSTON; WALKINSHAW, 2013; PARK; EOM; KIM, 2014). CVS assumes that two distinct VS approaches combined yield a better result than a single approach alone (HOUSTON; WALKINSHAW, 2013).

However, this is a difficult approach to handle, as the use of more than one approach involves handling entries in different formats and using different programs. In addition, these programs are often difficult to use because they do not have a graphical interface. In order to solve this problem, MolAr implements the Virtual Consensus Screening between DOCK 6 and

Autodock Vina. As can be seen in Figure 39, for the realization of the CVS approach, users only need to inform MolAr the folder of the molecular target database, the ligands folder and de docking method. MolAr also allows the use of the BRAMMT or OOMT databases. In this case, the targets folder field is automatically populated. If user does not select the Docking Method, MolAr will select GridScore Flex as default. The other options for the Docking Method field are GridScore Flex with Amber, GridScore Rigid, GridScore Rigid with Amber. After completing all fields, the user only needs to press the command Run Consensus.

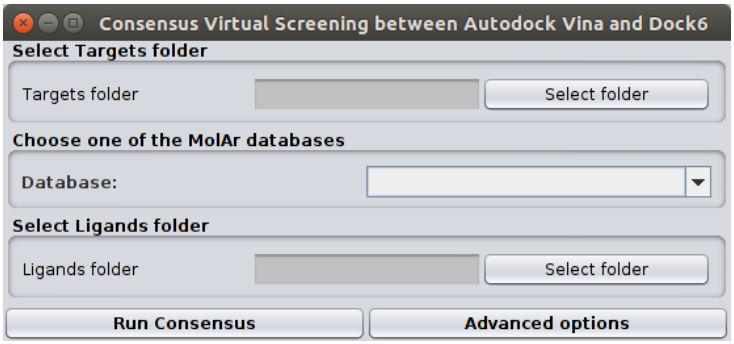


Figure 39 - CVS main screen

After completing all fields, the user only needs to press the command Run Consensus. During the Consensus MolAr will perform all the necessary conversions and it will carry out the VS using both, the Autodock Vina and DOCK 6. At the end of the simulation, MolAr will display DOCK 6 and Autodock Vina binding energy and in a third column, a classification according to the CVS which is performed. The result of the scoring function displayed by Autodock Vina and DOCK 6 is normalized to values between 0 and 10. The Consensus result calculated by MolAr corresponds to the average between these 2 values. Thus, the consensus scoring function is calculated by averaging all normalized DOCK6 and Autodock Vina scoring function values according to equation (1):

$$consensus = \frac{(normalized\ DOCK6\ result)+(normalized\ autodock\ vina\ result)}{2} \quad (1)$$

Then the list is ordered, according with the consensus calculated between Autodock Vina and DOCK 6 scoring functions.

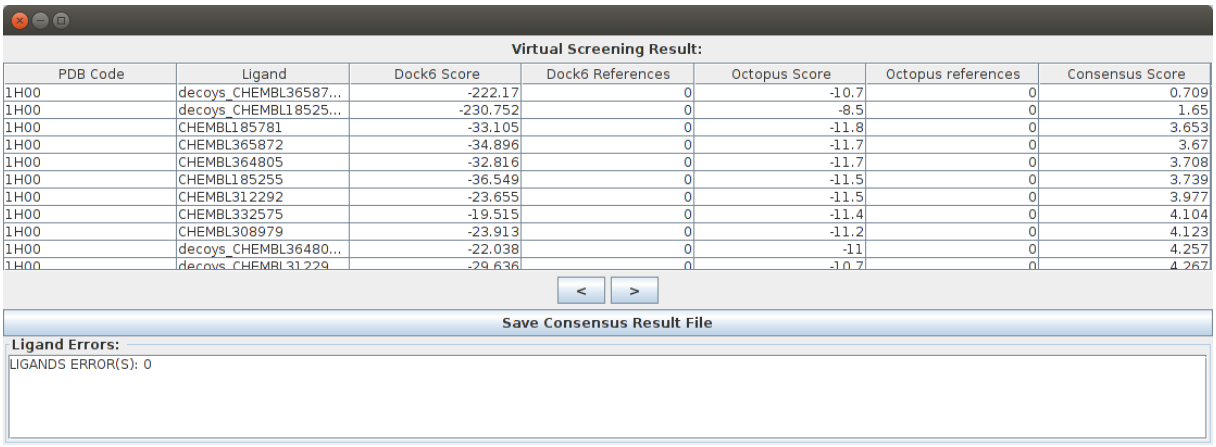


Figure 40 - CVS result screen

6.2.3 Tools Menu

The MolAr has a set of tools to support the realization of the entire VS process. These tools allows, for example: (a) visualize the three-dimensional structure of a molecule; (b) make the quality analysis of a structure; (c) carry out the adjustment of the protonation state; or (d) generate the ROC curve of a previous VS performed in the software. These features have been grouped in the Tools menu and in the following subsections, they will be described in detail.

6.2.3.1 View sub menu

In order to visualize the 3D structures of a PDB file (Figure 41a), Jmol (HERRÁEZ, 2006) and PyMol (DELANO, 2002) tools were integrated into the developed software. Figure 41 shows the screen that allows opening a protein with JMol (Figure 41(a)) or Pymol (Figure 41(b)). It is necessary to inform the PDB file of the protein whose 3D structure is to be visualized or its PDB code. If the PDB file is informed, the protein is displayed. If the PDB code of the protein is reported, the PDB file is downloaded and then displayed in JMol. Figure 41(c) shows the 3D structure of the 2YND protein using JMol and Figure 41(d) shows the same protein using Pymol.

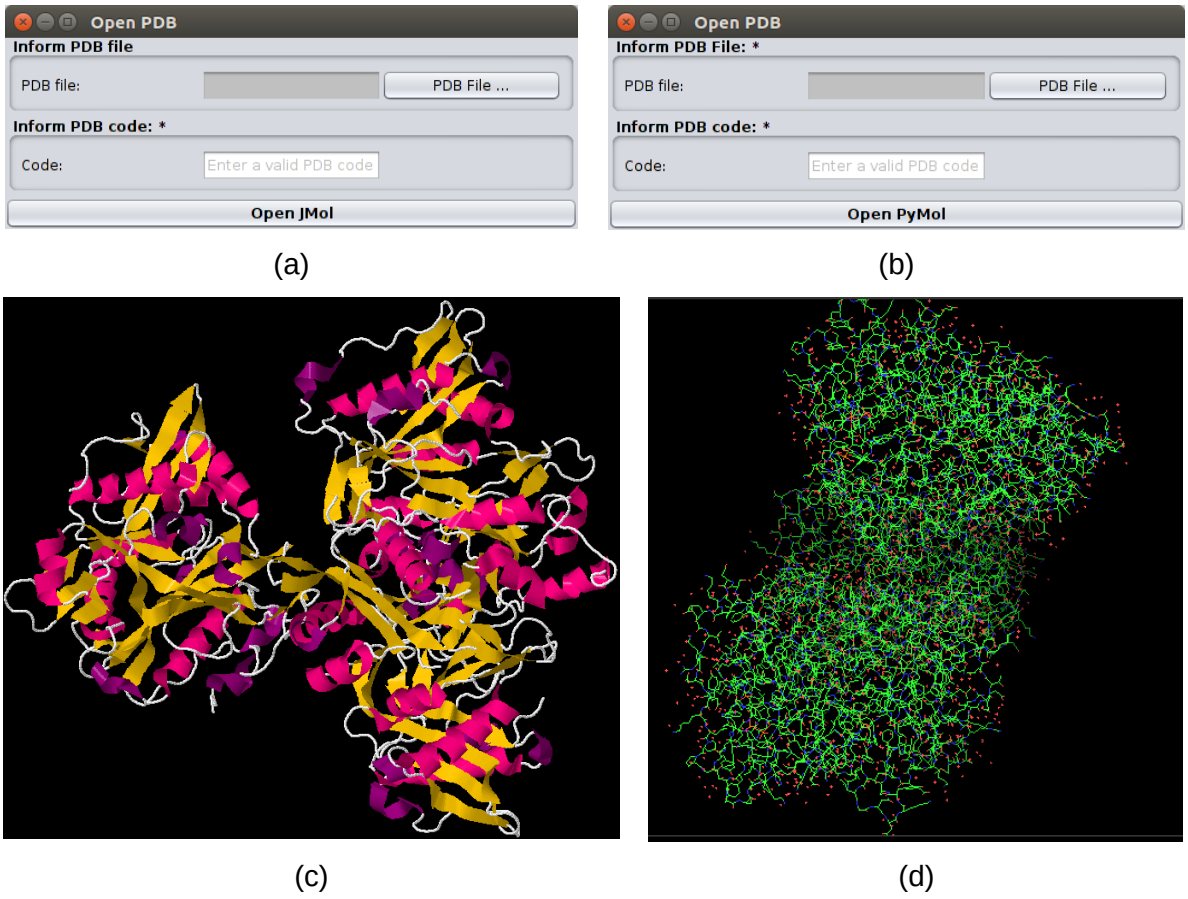


Figure 41 – View sub menu (a) Open PDB file screen with Jmol; (b) Open PDB file screen with PyMol; (c) Visualization of the 3D structure of 2YND protein in JMol; (d) Visualization of the 3D structure of 2YND protein in Pymol

6.2.3.2 Structure analysis submenu

The developed framework allows the analysis of the quality of a structure through the RMSD and the Ramachandran Plot. Figure 42 shows the screen that allows the generation of the Ramachandran Plot. In this screen, it is necessary to inform the PDB file that contains the 3D structure and the resolution of the protein to which the Ramachandran Plot will be generated. Based on this information, MolAr uses Procheck (LASKOWSKI et al., 1993) to generate the Ramachandran Plot. Figure 43 shows an example of the Ramachandran Plot generated by the Software for the protein 4XWK (P-glycoprotein co-crystallized with BDE-100).

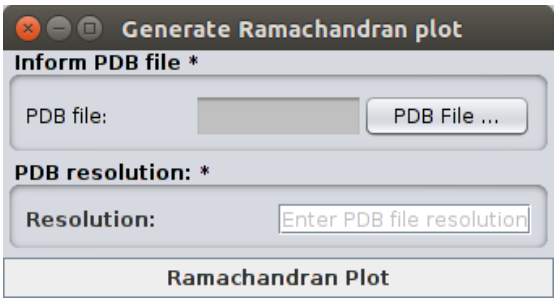
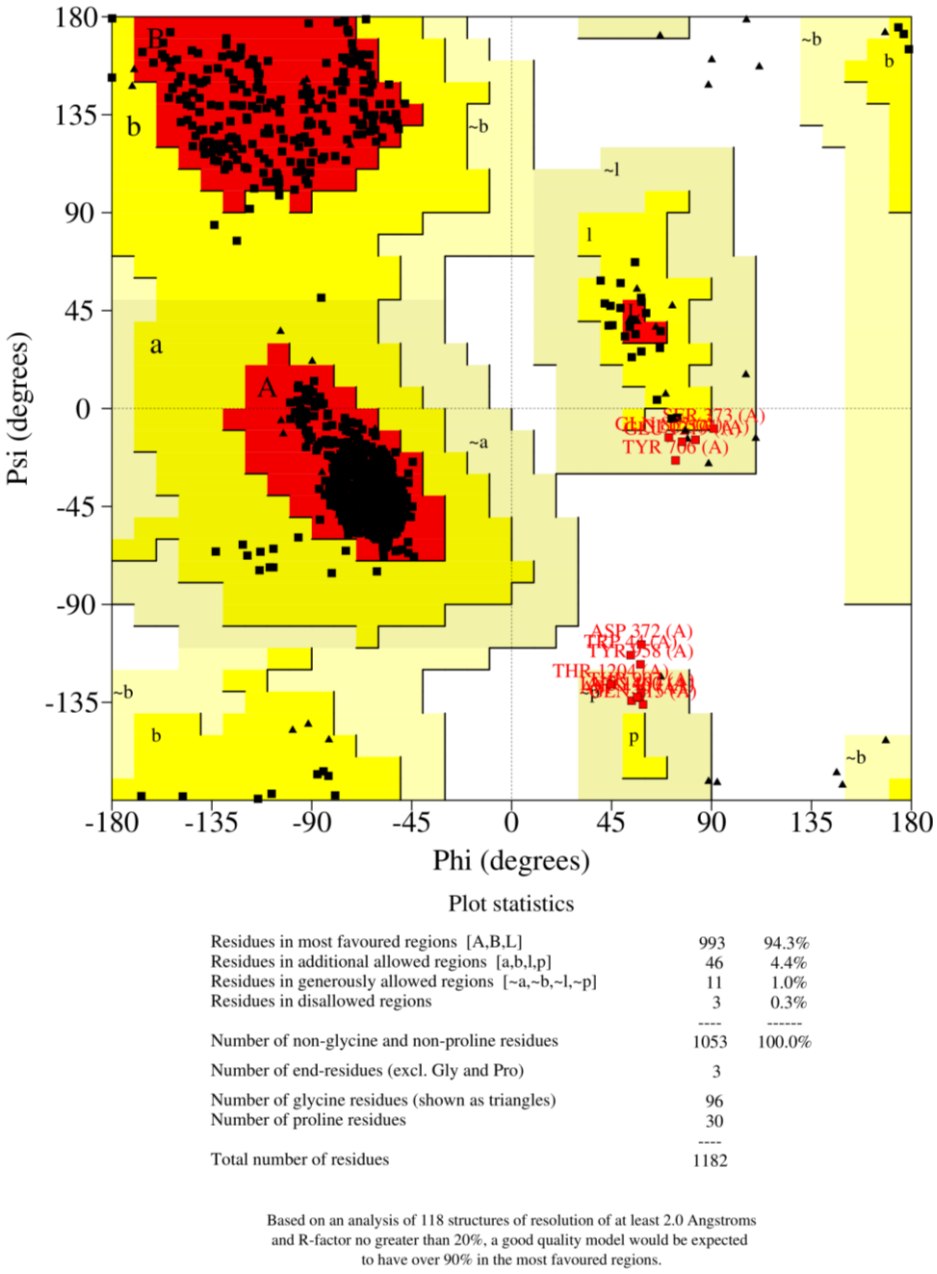


Figure 42 - Ramachandran plot screen



Plot statistics

Residues in most favoured regions [A,B,L]	993	94.3%
Residues in additional allowed regions [a,b,l,p]	46	4.4%
Residues in generously allowed regions [~a,~b,~l,~p]	11	1.0%
Residues in disallowed regions	3	0.3%
Number of non-glycine and non-proline residues	1053	100.0%
Number of end-residues (excl. Gly and Pro)	3	
Number of glycine residues (shown as triangles)	96	
Number of proline residues	30	
Total number of residues	1182	

Based on an analysis of 118 structures of resolution of at least 2.0 Angstroms and R-factor no greater than 20%, a good quality model would be expected to have over 90% in the most favoured regions.

Figure 43 - Ramachandran plot of the 4XWK protein using MolAr

The RMSD's calculation is done through a Python script that uses Pymol library and if user wants to carry out this calculation, user only need to enter the 2 proteins that will be compared. Figure 44 shows an example of a RMSD calculation between the 2YND protein and a model generated in MolAr by homology modeling to this protein.

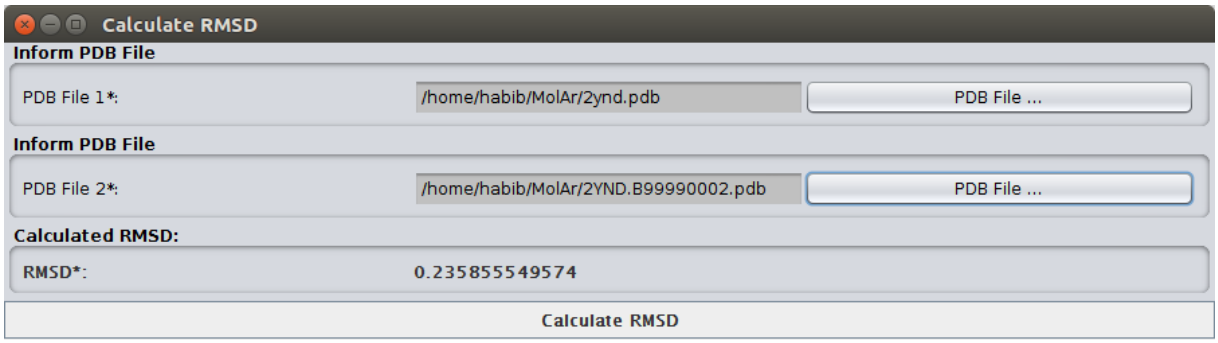


Figure 44 - RMSD calculation between two proteins after homology modeling

6.2.3.3 Protonation Submenu

Before performing Docking and Virtual Screening, it is necessary to adjust the protonation state of the 3D structures involved. It may be necessary to adjust the protonation state of both the proteins involved and the ligands.

Protein Protonation

There are some free software that allow you to adjust the state of protonation of a protein, such as H++ (ANANDAKRISHNAN; AGUILAR; ONUFRIEV, 2012) and PROPKA (DOLINSKY et al., 2007).

PROPKA uses a heuristic method to calculate pKa perturbations due to desolvation, hydrogen bonding, and charge-charge interactions. PROPKA (DOLINSKY et al., 2007) is a tool integrated with PDB2PQR Linux package. It makes it possible to generate the topology file in other formats, depending only on the force field selected during the process of adjusting the protonation state. The available force field options are:

1. Amber (WANG; CIEPLAK; KOLLMAN, 2000);
2. Charmm (MACKERELL et al., 1998);
3. Parse (SITKOFF; SHARP; HONIG, 1994; TANG et al., 2007);
4. Tyl06 (TAN; YANG; LUO, 2006);
5. Peopb (CZODROWSKI et al., 2006);
6. Swanson (SWANSON et al., 2007).

PROPKA requires the execution of a command line in the Linux prompt. It has several options, so, it is prone to user errors. However, it allows a more stable integration with the

developed software because it is installed locally, and it permits the creation of a graphical interface to facilitate tits use. Thus, after accessing Tools -> Protonation -> Protein protonation submenu, MolAr displays Figure 45. It shows a simple interface created for the adjustment of the protein protonation state using PROPKA.

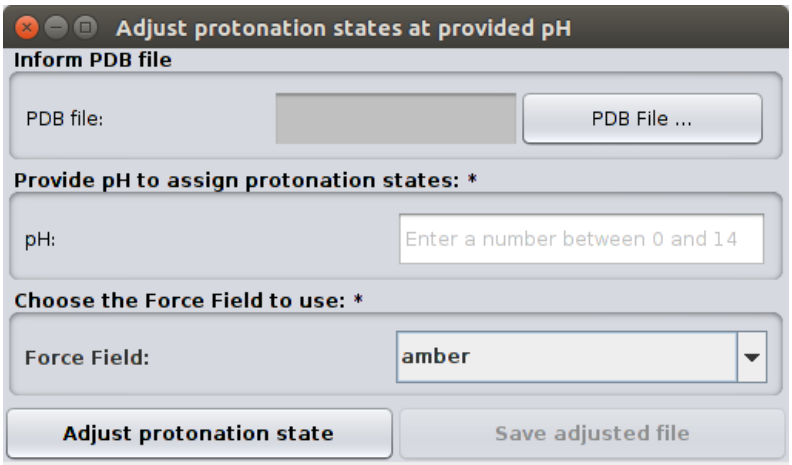


Figure 45 - Adjust the protonation state screen

In the interface of Figure 45, the PDB file, the pH and the force field to be used in this procedure must be reported for the adjustment of the protonation state. *It is important to emphasize that the choice of the force field is very important in this process.* This choice defines, among other things, the nomenclature of residues, which is important in the next stages of molecular modeling studies, and the generation of topology files. The interface of the Figure 45, created to automatically adjust the protonation state, is quite intuitive and therefore does not require prior knowledge of how to use the PDB2PQR program. After filling all fields and pressing the Adjust protonation state button, MolAr performs the protein protonation state adjustment and enables the Save adjusted file button, which allows the adjusted protein to be saved (Figure 46).

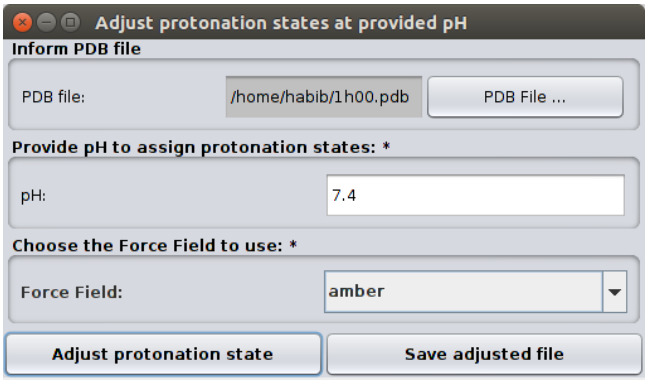


Figure 46 - Adjust protein protonation state

Ligand Protonation

MolAr also allows the protonation state of the ligands to be adjusted using Babel (O'BOYLE et al., 2011).

To do so, simply access the Tools -> Protonation -> Ligand protonation submenu. In this case, MolAr will display Figure 47. Then, after filling in the screen fields (the folder containing the ligands that will have the protonation state adjusted and the pH field), and pressing the "Adjust protonation state" button, MolAr will adjust the protonation state.

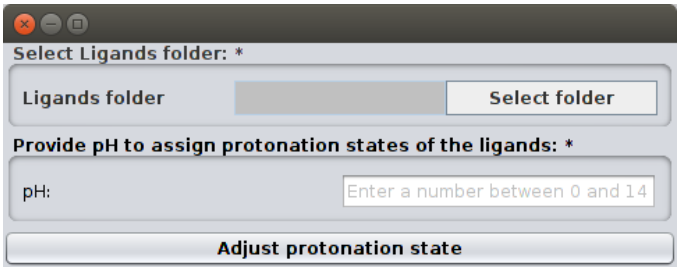


Figure 47 - Adjust ligand protonation state

6.2.3.4 Generate Roc Curve submenu

After performing a virtual screening, one of the most important points to be observed is if the program used was able to separate active compounds from inactive ones. MolAr allows the generation of the ROC curve of a given VS, in order to verify if it can separate these compounds, by just pressing the ROC Curve command from the Tools Menu. So, Molar will show the screen of Figure 48. Then, select the file with the result of the Virtual Screening, generated after its execution, and press the button Plot ROC.

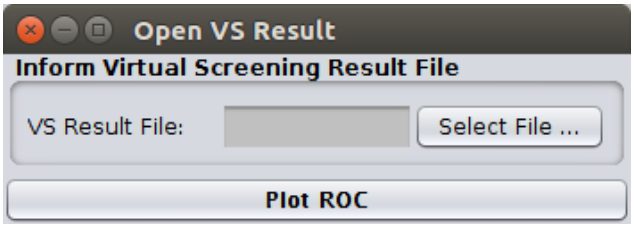


Figure 48 – ROC Curve main screen

After this button is pressed, the ROC curve is generated. Figure 49 shows an example of a ROC curve. It can be verified that the generated ROC Curve also shows the value of the Area Under the Curve (AUC). As discussed previously, if this value is higher than 0.7, we can verify that the program was able to correctly distinguish between active and inactive compounds (HAMZA; WEI; ZHAN, 2012).

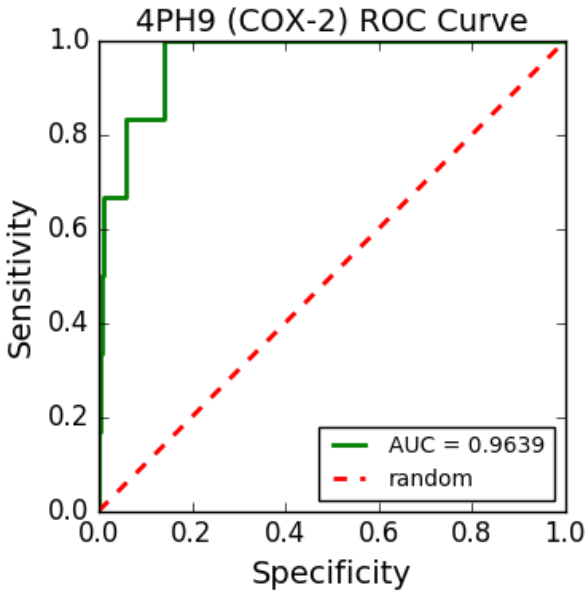


Figure 49 - ROC Curve for the protein 4PH9 generated by the ROC curve screen

6.3 Case studies

The software was used in three case studies developed in the Pharmaceutical Chemistry Laboratory of the Federal University of São João del-Rei:

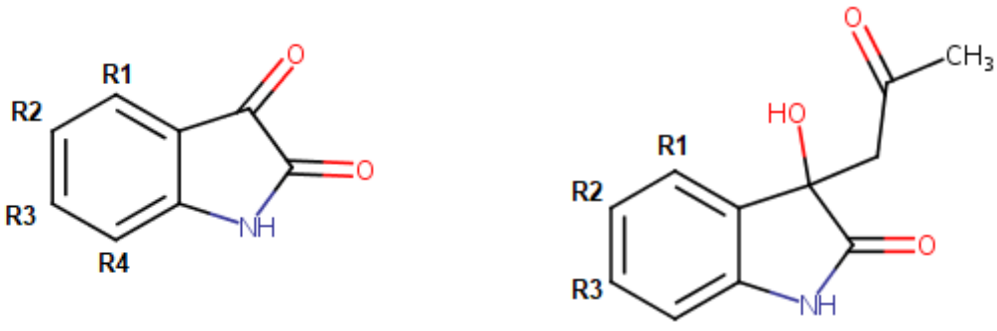
I. Evaluating of Pharmacological effects of Isatine and Convolutamine on cyclooxygenase-1, cyclooxygenase-2 and Mu opioid molecular target
II. Investigation of the best in silico model for DNA-Ligand systems;
III. Evaluation of the consensus virtual screening approach implemented by MolAr

In the following subsections, we will describe the results of the case studies in detail.

6.3.1 Evaluating of Pharmacological effects of Isatine and Convolutamine on cyclooxygenase-1, cyclooxygenase-2 and Mu opioid molecular target

Isatin has been shown antibacterial (LIAN; SUN; ZHU, 2016), antitumor (HOU et al., 2008), antinociceptide (FIGUEIREDO et al., 2013; GIORNO et al., 2016; VENKATESHWARLU et al., 2012) and anti-inflammatory(VENKATESHWARLU et al., 2012) effects.

This case study will perform the Inverse Virtual Screening (IVS) analysis of isatin and convolutamidine, as well as their derivatives to search for the putative receptor among cox-1, cox-2 and Mu opioids, as demonstrated by Matheus (MATHEUS et al., 2007) and Figueiredo (FIGUEIREDO et al., 2013). The general structure of isatin with respectively substitutes and the codes of each isatin derivative defined by Matheus (MATHEUS et al., 2007) and Figueiredo (FIGUEIREDO et al., 2013) are shown in Figure 50, Table 5 and Table 6.



(a) (b)

Figure 50 - General Structure of (a) Isatin (b) Convolutamydine

Isatin (R1, R2, R3, R4 = H)
5-Fluoroisatin (R1=H, R2=F, R3=H, R4=H)
5-Chloroisatin (R1=H, R2=Cl, R3=H, R4=H)
6-Chloroisatin (R1=H, R2=H, R3=Cl, R4=H)
7-Chloroisatin (R1=H, R2=H, R3=H, R4=Cl)
4-Bromoisatin (R1=Br, R2=H, R3=H, R4=H)
5-Iodoisatin (R2=I)
5-Methylisatin (R2=CH3)

Table 5 – Isatin and Isatin derivatives

Convolutamydine A (R1=Br, R2=H, R3=Br)
Convolutamydine analogue 1 obtained from Isatin (R1=R2=R3=H)
Convolutamydine analogue 2 obtained from Isatin (R1=H, R2=Br, R3=H)

Table 6 - Convolutamydine and synthetic analogues obtained from isatins

Among the opioid receptors associated in analgesia, we chose the Mu receptor, because it is the most involved in this process. The other opioid receptors (Delta, Kappa and Sigma) have a lower involvement with the pain process. IVS uses compound library as well as VS, but the focus is the ligand (CARREGAL et al., 2012). Consequently, the goal is to determine the possible biological activities that the chemical compound may possess, besides being useful for the identification of molecular targets responsible for undesired effects.

Firstly, the targets for COX (COX-1 and COX-2) and Opioid (Mu) inhibitors were selected from Protein Data Bank (PDB) (BERMAN, 2000). The targets were selected after searching for the following terms: COX-1, COX-2, Mu opioid receptor. We choose targets with the best resolution and complexed with some drug.

The five active ligands for each target obtained from ChEMBL and whose 3D structure was downloaded from ZINC (GAULTON et al., 2012) were choose for decoy generation. The

selected targets and their PDB code, followed by their respective ligands and their ZINC codes are:

COX

- COX-1 (4O1Z - Cyclooxygenase-1 Complexed with Meloxicam): Naproxen (ZINC105216), Ketorolac (ZINC11012), Ibuprofen (ZINC38141758), Flurbiprofen (ZINC323), Ketoprofen (ZINC5560), Suprofen (ZINC57503)
- COX-2 (4PH9 - Cyclooxygenase-2 Complexed with Ibuprofen): Celecoxib (ZINC2570895), Etoricoxib (ZINC579472), Lumiracoxib (ZINC7563), Rofecoxib (ZINC7455), Valdecoxib (ZINC6694)

Opioids

- Mu (5C1M - mu-opioid receptor bound to the agonist BU72): Morphine (ZINC2041330), Fentanyl (ZINC2522669), Codeine (ZINC3806721), Etonitazene (ZINC4215955), Sufentanil (ZINC538386)

Then, using babel, the mol2 files of the ligands downloaded from ZINC were converted to PDB. Then, homology modeling was performed using MolAr.

Target Builder

The higher number of templates generated during homology modeling, the greater the probability of creating a good model. In this research, fifty models were generated for each target using MolAr Target Builder menu. The lowest DOPE energy model was selected. As we used MolAr to construct only the loop regions, the RMSD of the generated models was very low (less than 0.1) and, therefore, it was not an important metric to be considered in the generated model. For the same reason, the Ramachandran plot changed little in the generated model, when we compare it with the original target.

Virtual Screening

After building Loop regions of the selected targets using Target Builder, the next step of this experiment was to prepare them for docking, creating a database with them, following the Autodock Vina format.

First, the file downloaded from PDB was opened using Chimera. Once opened, the crystallographic ligands of the target were removed. The ligand complexed to the target was saved in a separated PDB file. Next, hydrogens were added to the crystallographic ligand using Chimera. Then, ADT was used to determine the coordinates of the target binding site and the size of the box used was 24. Finally, we did a redocking between each target and its crystallographic ligand using MolAr, to verify whether the MolAr was able to dock it correctly.

DUD-E (MYSINGER et al., 2012) was used to select decoys for each of the selected active ligands. For each ligand, DUD-E creates fifty decoys from ZINC so that their physicochemical properties coincide with the physicochemical properties of the active ligands. However, it selects the fifty most dissimilar compounds, by topology, when compared to the active ligands. Thus, although they have similar physicochemical characteristics, they are not active. The realization of VS experiments using decoys selected by DUD-E allows to verify the ability of the tool to distinguish between active and inactive ligands for a specific target. To verify this ability, we use ROC curve analysis.

After decoys generation, we perform each test in triplicate. First, to verify the reliability of the tests to each target, we perform a separate VS for each one. Therefore, each VS experiment was performed with the following configuration:

1. A single target.
2. Ligands selected for that target.
3. The decoys generated in DUD-E for the ligands of that target.

Afterwards, results were evaluated, and ROC curve was generated. Finally, tests were performed on all Targets and in the active ligands proposed by Matheus (MATHEUS et al., 2007) and Figueiredo (FIGUEIREDO et al., 2013).

Results and discussion

It was performed two main groups of experiments. Firstly, we generate a ROC curve for each target. Then, tests were performed on all targets, active selected ligands and in the proposed ones.

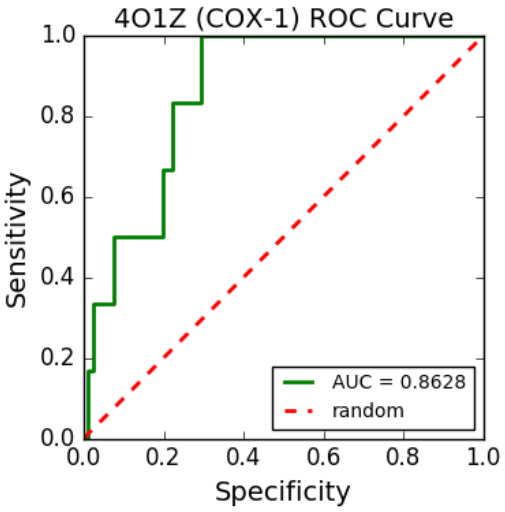
ROC curve

The ability of the tool to distinguish between active and inactive ligands was evaluated in terms of the area under the ROC curve (AUC). ROC Curves allows to verify the sensitivity of a VS experiment in relation to its specificity. The larger the area under the curve, the better the ability to select active components and the smaller the number of false positives.

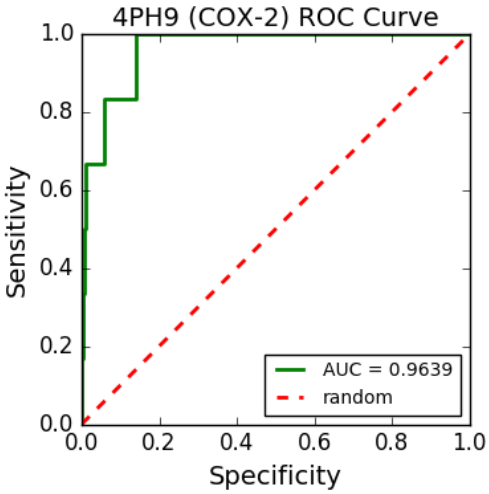
The AUC value will vary between 0 and 1 (or 0 and 100%, respectively). The closer the AUC is to 1 (or 100%), the greater the ability of the VS tool to separate between active and inactive ligands.

In Figure 51, we show the ROC curve generated for the VS experiments for each target and its AUC, considering the active ligands and the inactive ones generated by DUD-E.

COX-1



COX-2



Mu Opioid

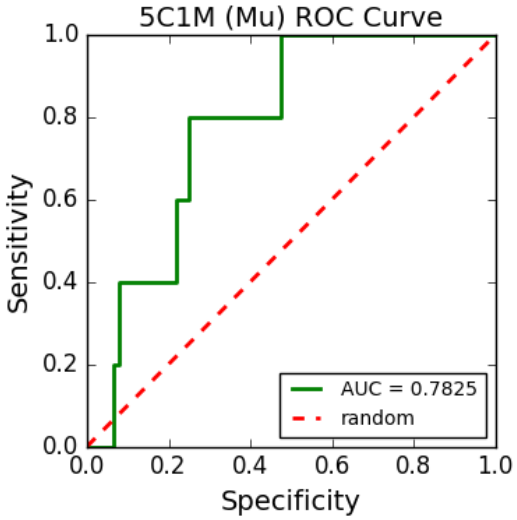


Figure 51 - ROC Curves for COX-1, COX-2 and Mu opioid targets

The ROC curves on Figure 51 shows that MolAr was able to distinguish active ligands from inactive ones appropriately for 4O1Z (COX-1), 4PH9 (COX-2) and 5C1M (Mu opioid receptor). In the experiments, for 4O1Z, AUC ranged from 0.8317 to 0.8628 while in the 4PH9 target AUC ranged from 0.9636 to 0.9658. For the Mu opioid receptor (5C1M) AUC varies from 0.7418 to 0.7825. Hamza, Wei and Zhan (HAMZA; WEI; ZHAN, 2012) showed that the closer the AUC is to 1, the greater the ability of the VS tool to separate between active and inactive ligands, but values above 0.7 are acceptable, meaning that the VS tool has successfully separated active ligands from inactive ones.

The AUC value can vary between 0 and 1. Hamza (HAMZA; WEI; ZHAN, 2012) showed a practical way of interpreting the AUC values:

- AUC between 0.90 and 1.00: Excellent
- AUC between 0.80 and 0.90: Good
- AUC between 0.70 and 0.80: Fair
- AUC between 0.60 and 0.70: Poor
- AUC between 0.50 and 0.60: Failure

Therefore, the closer the AUC is to 1, the greater the ability of the VS tool to separate between active and inactive ligands. AUC-ROC values close to 0.5 indicates a random process (OGRIZEK et al., 2015). Acceptable values should be greater than 0.7.

Inverse Virtual Screening

After the generation of the ROC curve to all targets, a VS was performed using all ligands and targets. Table 7 shows the VS results for COX-1 (4O1Z on Table 7) and COX-2 (4PH9 on Table 7). Except for 5-lisa_1 compound when bound to COX-1, all evaluated compounds have a binding energy close to the ibuprofen´s. So, they will probably bind to COX-1 and COX-2.

Table 7 - Virtual Screening results for COX-1 and COX-2

(Binding Energy in Kcal/mol)

ID	Ligands	4PH9	4O1Z
1	4PH9_zinc_2570895_CELECOXIB.pdbqt	-10.4	-
2	4PH9_zinc_7455_ROFECOXIB.pdbqt	-9.3	-
3	4PH9_zinc_6694_VALDECOXIB.pdbqt	-9.1	-
4	4PH9_zinc_7563_LUMIRACOXIB.pdbqt	-9.0	-
5	4PH9_zinc_579472_ETORICOXIB.pdbqt	-8.6	-
6	4O1Z_KETOROLAC_zinc_11012.pdbqt	-	-8.8
7	4O1Z_FLURBIPROFEN_zinc_323.pdbqt	-	-8.6
8	4O1Z_KETOPROFEN_zinc_5560.pdbqt	-	-8.3
9	4O1Z_SUPROFEN_zinc_57503.pdbqt	-	-7.8
10	4O1Z_NAPROXEN_zinc_105216.pdbqt	-	-7.7
11	5-Fisa_1.pdbqt	-7.5	-6.8
12	5-Clisa_1.pdbqt	-7.2	-6.7
13	5-Meisa_2.pdbqt	-7.2	-6.8

ID	Ligands	4PH9	4O1Z
14	5-Methylisatin.pdbqt	-7.1	-7.0
15	5-Fisa_2.pdbqt	-7.1	-7.1
16	IBUPROFEN_zinc_2647	-7.1	-6.8
17	5-Fluoroisatin.pdbqt	-7.0	-6.8
18	7-Chloroisatin.pdbqt	-7.0	-6.5
19	convolutamydeA_1.pdbqt	-7.0	-6.3
20	convolutamydeA_2.pdbqt	-7.0	-6.3
21	convolutamydeA_composto1_1.pdbqt	-7.0	-6.4
22	convolutamydeA_composto1_2.pdbqt	-7.0	-6.3
23	5-Chloroisatin.pdbqt	-6.9	-6.8
24	4-Bromoisatin.pdbqt	-6.8	-7.0
25	6-Chloroisatin.pdbqt	-6.8	-6.7
26	convolutamydeA_composto2_1.pdbqt	-6.8	-6.5
27	5-Clisa_2.pdbqt	-6.8	-7.1
28	5-lisa_1.pdbqt	-6.8	-5.7
29	5-lisa_2.pdbqt	-6.8	-7.0
30	5-Meisa_1.pdbqt	-6.8	-7.2
31	convolutamydeA_composto2_2.pdbqt	-6.7	-6.9
32	Isatin.pdbqt	-6.7	-6.5
33	5-Iodoisatin.pdbqt	-6.6	-6.5

The VS experiments results for Mu opioid receptor (Table 8), shows there is no isatin derivative with lower docking energy than the greater energy of the active ligands (Sufentanil). The ROC Curve for Mu Opioid receptor (Figure 51), demonstrated that the results for this target is reliable. Thus, one can conclude that the probability of these compounds binding to the Mu opioid target is low. So, isatin derivatives probably does not have any opioid effect against Mu opioid target.

Table 8 - Virtual Screening results for Mu Opioid receptor.

(Binding Energy in Kcal/mol)

ID	TARGETS	5C1M
1	5C1M_zinc_2041330_morphine.pdbqt	-8.9
2	5C1M_zinc_2522669_Fentanyl.pdbqt	-8.8
3	5C1M_4N6H_zinc_3806721_codeina.pdbqt	-8.4
4	5C1M_zinc_4215955_Etonitazene.pdbqt	-8.1
5	5C1M_zinc_538386_Sufentanil.pdbqt	-7.8
6	5-Fluoroisatin.pdbqt	-6.1
7	5-Meisa_2.pdbqt	-7.2
8	Isatin.pdbqt	-5.9
9	5-Fisa_1.pdbqt	-7.1
10	5-Fisa_2.pdbqt	-6.8
11	5-Meisa_1.pdbqt	-6.8
12	convolutamydeA_composto2_1.pdbqt	-6.7
13	5-Clisa_1.pdbqt	-6.8
14	convolutamydeA_composto2_2.pdbqt	-6.8
15	5-Clisa_2.pdbqt	-6.7
16	5-lisa_1.pdbqt	-6.6
17	5-lisa_2.pdbqt	-6.6
18	6-Chloroisatin.pdbqt	-6.1
19	convolutamydeA_1.pdbqt	-6.7
20	convolutamydeA_composto1_1.pdbqt	-6.7
21	5-Methylisatin.pdbqt	-6.2
22	7-Chloroisatin.pdbqt	-6.2
23	4-Bromoisatin.pdbqt	-6.0
24	5-Chloroisatin.pdbqt	-6.0
25	5-Iodoisatin.pdbqt	-5.7
26	convolutamydeA_2.pdbqt	-7.0
27	convolutamydeA_composto1_2.pdbqt	-7.0

So, after carry out the IVS against COX-1, COX-2 and Mu opioid receptor, the results presented shows that isatin derivatives proposed by Figueiredo et al. (FIGUEIREDO et al., 2013), Giorno et al. (GIORNO et al., 2015) and Matheus et al. (MATHEUS et al., 2007) could

have antinociceptive action on COX-1 and COX-2 receptors. It is also possible to conclude that the probability of the Mu opioid receptor being inhibited by isatin compounds is low, since there are no isatin or convolutamidine derivatives with binding energy lower than the lowest binding energy of known ligands.

6.3.2 Investigation of the best in silico model for DNA-Ligand systems

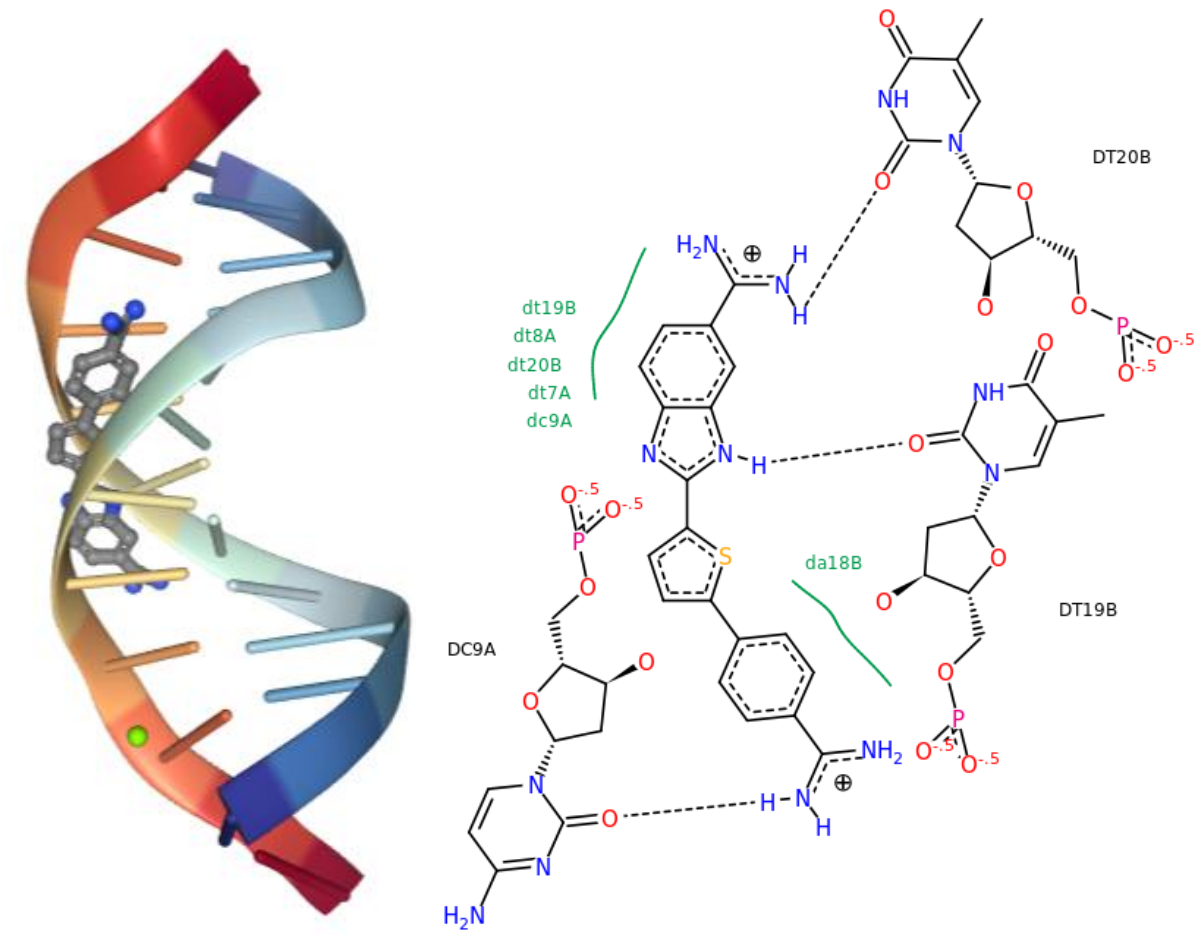
DNA is a common main target in the treatment of several genetic diseases, most notably cancer, due to its importance in the cell cycle (GILAD; SENDEROWITZ, 2014).

However, drugs that interact with these receptors usually have high toxicity due to low selectivity between the DNA of normal and abnormal cells, so motivating the development of new drugs targeting this receptor (SILVERMAN, 2004).

Molecular docking (MD) is widely validated for protein-ligand systems and, despite recent works that already use MD to model DNA-ligand complexes, there is not yet a consensus about the best in silico model to nucleic acids because DNA has several unique properties, such as high charge density and high flexibility (GILAD; SENDEROWITZ, 2014).

In order to compare recent in silico models on DNA-ligand systems, MolAr was used to evaluate the different approaches used by Autodock Vina (TROTT; OLSON, 2009) and DOCK6 (BROZELL et al., 2012), as well as the combination of these two (CVS which intends, with the combination of the result of different VS approaches, to improve the reliability of VS results). ROC curve analysis was used to compare and validate these approaches.

The active compounds were selected from Srivastava et al. (SRIVASTAVA et al., 2011). They were used to compare several molecular docking approaches, using 57 crystal structures of DNA-ligand complexes with known minor groove binders as ligands (SRIVASTAVA et al., 2011). In order to perform this case study, the four most active ligands define by Srivastava (SRIVASTAVA et al., 2011) and the DNA model related to them were selected, and decoys from these compounds were obtained using the DUD-E (MYSINGER et al., 2012) service. The target used was 1VZK (A Thiophene Based Diamidine Forms a "Super" AT Binding Minor Groove Agent) which was the same used by Srivastava et al (SRIVASTAVA et al., 2011). Figure 52 shows the 3D structure of the target 1VZK and the Interactions in 2D with its crystallographic ligand (D1B).



(a) (b)

Figure 52 - 1VZK (a) 3D Structure view (b) Interactions in 2D

Then three virtual screening simulations were performed as follows:

1. Minimization of the ligands with the MOPAC program, followed by virtual screening with Autodock Vina;
2. Virtual screening with DOCK 6;
3. Consensus virtual screening between approaches 1 and 2. The main idea in performing a CVS was to check, through the area on the curve, whether the use of the CVS between the DOCK 6 and the Autodock Vina would increase the reliability of the docking.

Results

MolAr was able to differentiate active compounds from inactive ones since all configuration shown excellent AUC values (Figure 53).

The AUC values remained consistent in all ROC curves. Autodock Vina (Figure 53a) has shown better results than DOCK 6 (Figure 53b) probably due to its search algorithm that relies on random changes of conformation. This feature was able to search outside of local sites of energy minima, while DOCK 6 uses an *anchor-and-grow* search algorithm. CVS carry out a VS along with Autodock Vina as the first approach and then carry out a second VS using DOCK 6 as a refinement tool. The resulting ROC curve of the CVS (Figure 53c) showed even better results than the virtual screening of DOCK 6 or Autodock Vina alone.

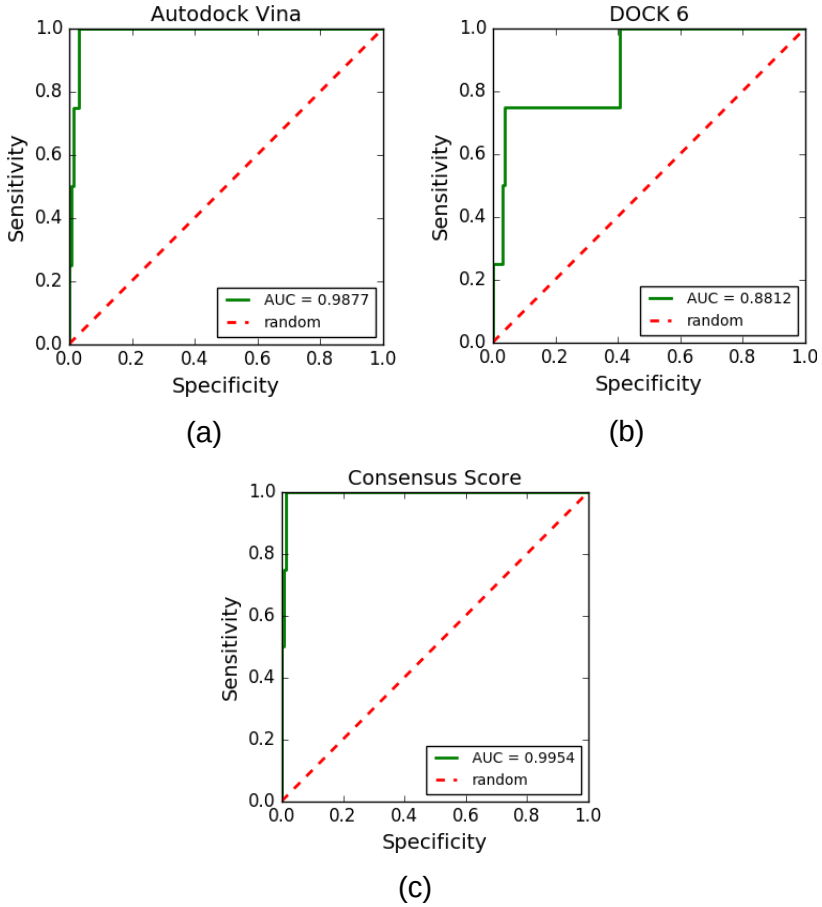


Figure 53 – ROC curves (a) ROC curve obtained after performing VS with Autodock Vina; (b) ROC curve obtained after performing VS with DOCK 6; (c) ROC curve obtained after performing consensus virtual screening.

Finally, it is important to note that even though DOCK 6 had, worse results than Autodock Vina, it contributed to the improvement of the results. It can be checked, for example, the position of actives ligands between all molecules in Virtual Screening. In the Table 9, the active ligand with highest energy according to DOCK 6 (127d) was in position 71th. In Octopus, the active ligand with the highest energy was in position 21th (2dnd). In the consensus between them, the active with highest energy was identified as the 8th between all the molecules. This

was because, for example, the active ligand with the highest energy identified by DOCK 6 (127d) was the one with the lowest energy in Autodock Vina. The same was true for Autodock vina's worst active ligand (2dnd). In DOCK6 it was the 10th lowest energy among all ligands. Thus, when MolAr calculates CVS between DOCK 6 and Autodock Vina, the result was improved.

Table 9 - Position of the active ligands identified by DOCK 6, Autodock Vina and Consensus

Active	DOCK 6		Autodock Vina		Consensus	
	Energy	Position	Energy	Position	Energy	Position
121d	-57.61	1th	-8.9	10th	1.97	2th
1eel	-45.30	8th	-9.7	4th	1.46	1th
2dnd	-42.83	10th	-8.4	21th	2.79	8th
127d	-27.77	71th	-11.5	2th	2.76	7th

The AUC-ROC curve from the CVS suggests that even though with DOCK 6 and Autodock Vina showing different results, the combination of these results may in principle be closer to the correct answer because the reliability of the result was increased and there was a better prediction of the positions of the active ligands.

6.3.3 Evaluation of the consensus virtual screening approach implemented by MolAr

MolAr CVS algorithm uses, the resulting ligand conformation after docking with Autodock Vina as an input to DOCK 6. Thus, it is expected that, starting from a ligand conformation already evaluated and optimized by Autodock Vina previously, DOCK 6 will obtain better results.

It was necessary to verify whether this strategy would not have the opposite effect. Therefore, six targets from the DUD-E database were used and 10 active ligands were selected for each target. DUD-E has in its database a set of 102 targets and 22886 active compounds for these targets, an average of 224 active ligands per target.

Performing virtual screening using all these targets and ligands would be impracticable, since the time spent would be enormous for performing Virtual Screening with the Consensus Screening setting chosen. For the CVS setting chosen, in addition to Autodock Vina VS, a VS was performed using DOCK 6 with Grid Score using amber. This choice promises better results, as it performs some molecular dynamics simulations during docking with DOCK 6, but it leads to a much longer execution time than GridScore flex without amber. In addition, besides

performing the virtual screening of the active ligands, it would be necessary to generate decoys for the chosen ligands to carry out a VS and plot the ROC curve.

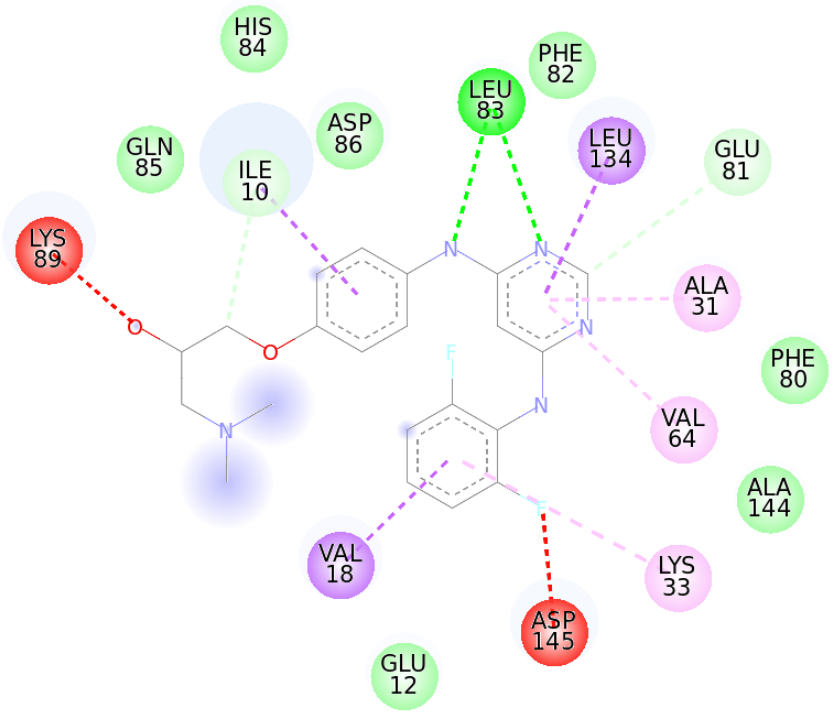
Thus, a subset of the DUD38 targets was used. DUD38 is a subset of DUD that can be subdivided into six target families: Metalloenzyme (4), Nuclear Hormone Receptor (8), Kinase (9), Folate Enzyme (2), Serine Protease (2), and finally a diverse family called Other Enzymes (13). In our tests, we used 1 target from each of the 6 families of DUD38, thus a total of 6 targets. For each target, we selected 10 active ligands and we use DUD to generate 50 decoys for each ligand. The targets were chosen considering mainly their resolution in the PDB. A homology modeling was performed to reconstruct GAP regions (such as loop regions), for all chosen targets. Table 10 lists the chosen targets and

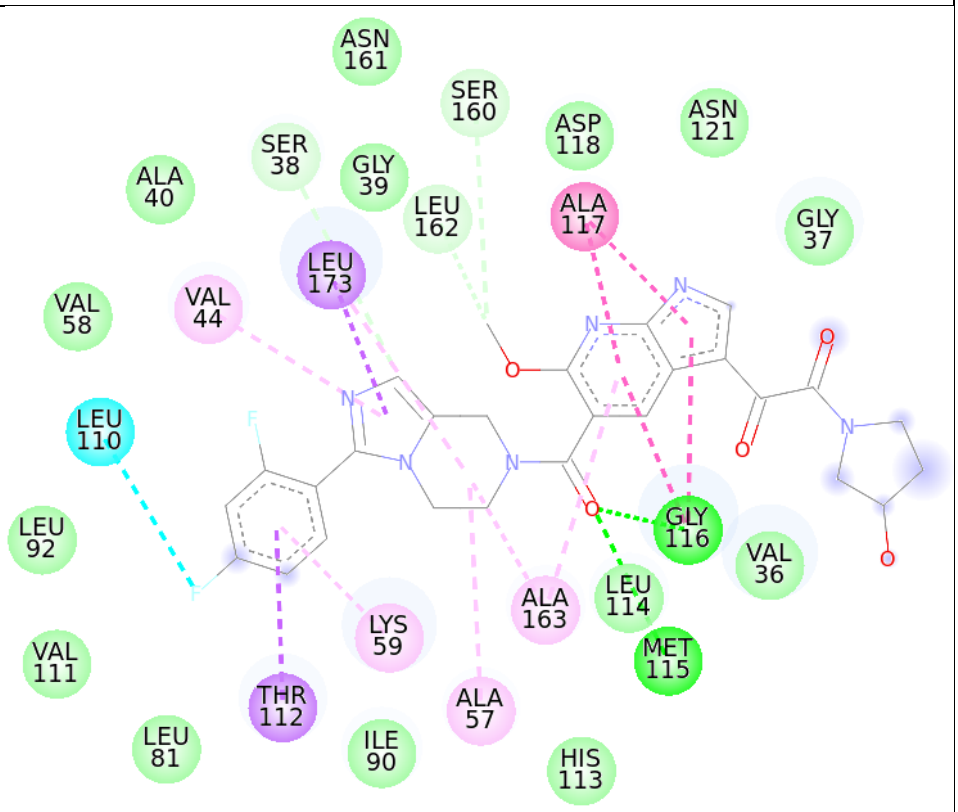
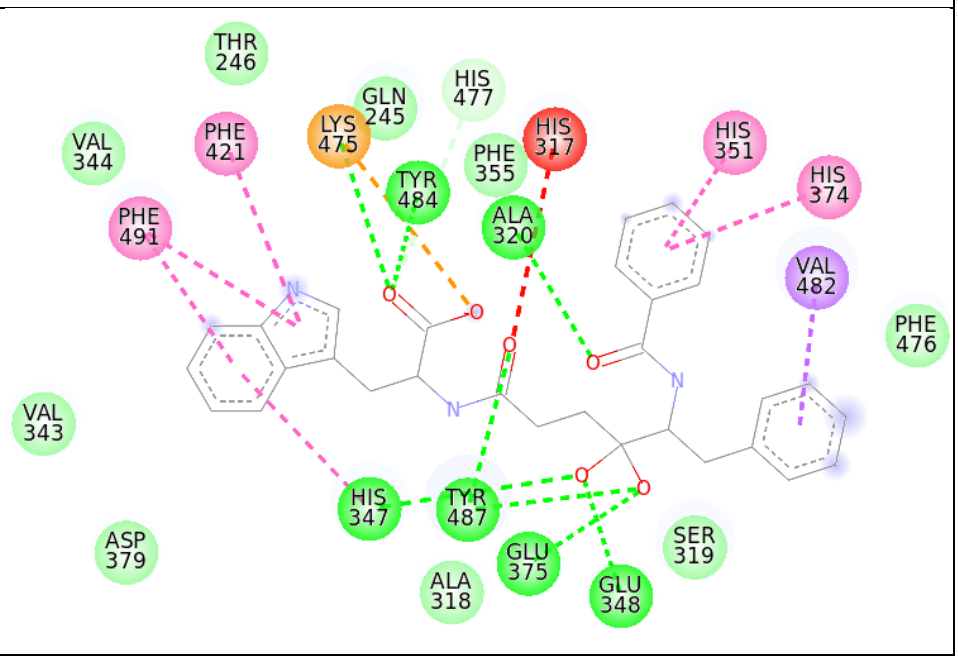
Table 11 shows the interactions of crystallographic ligands with the targets used in this case study.

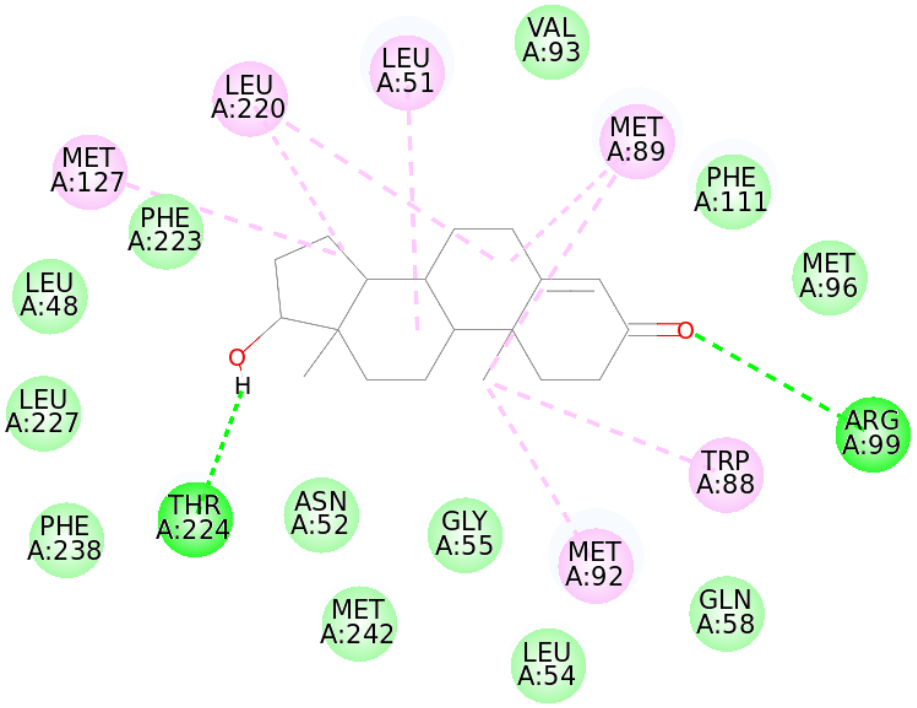
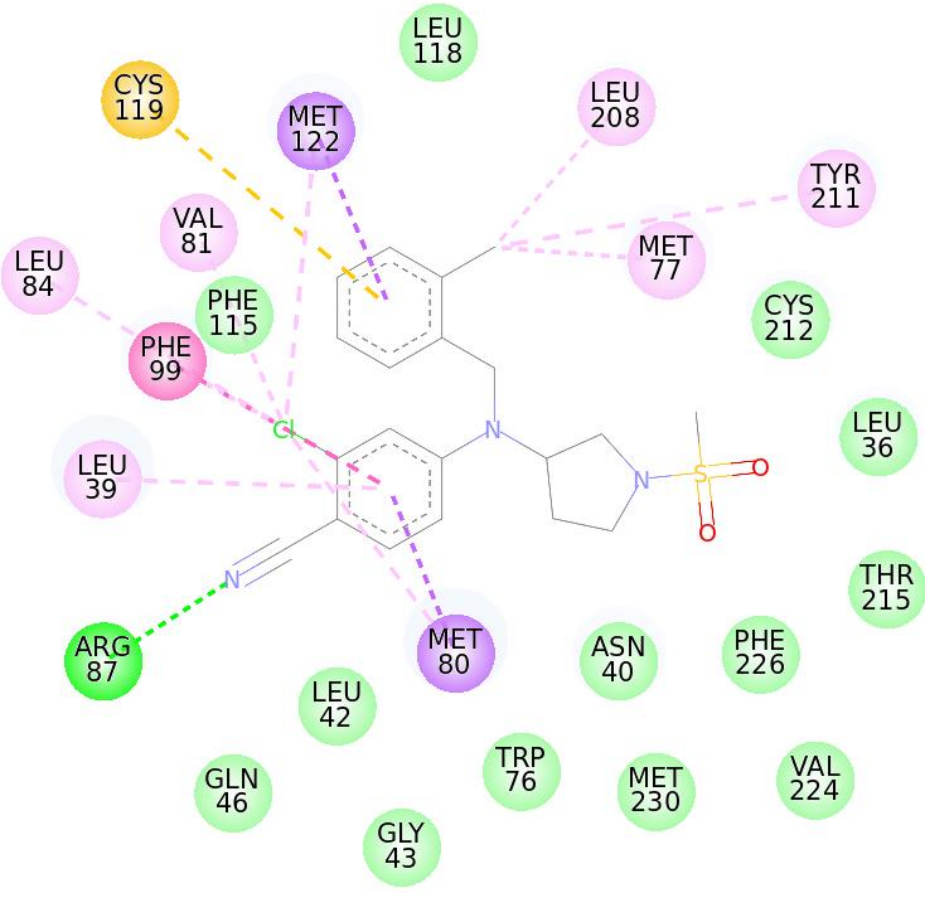
Table 10 - Subset of Targets chosen in DUD38

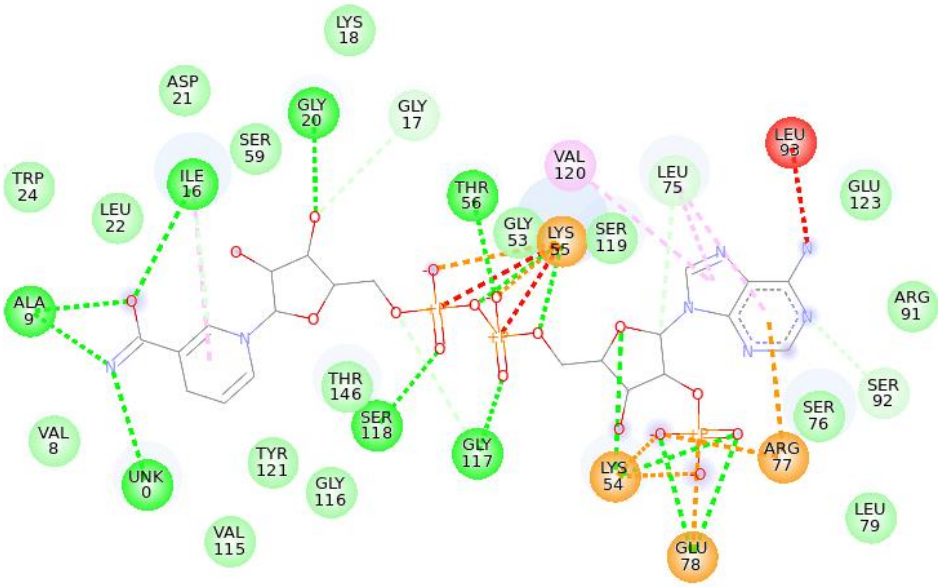
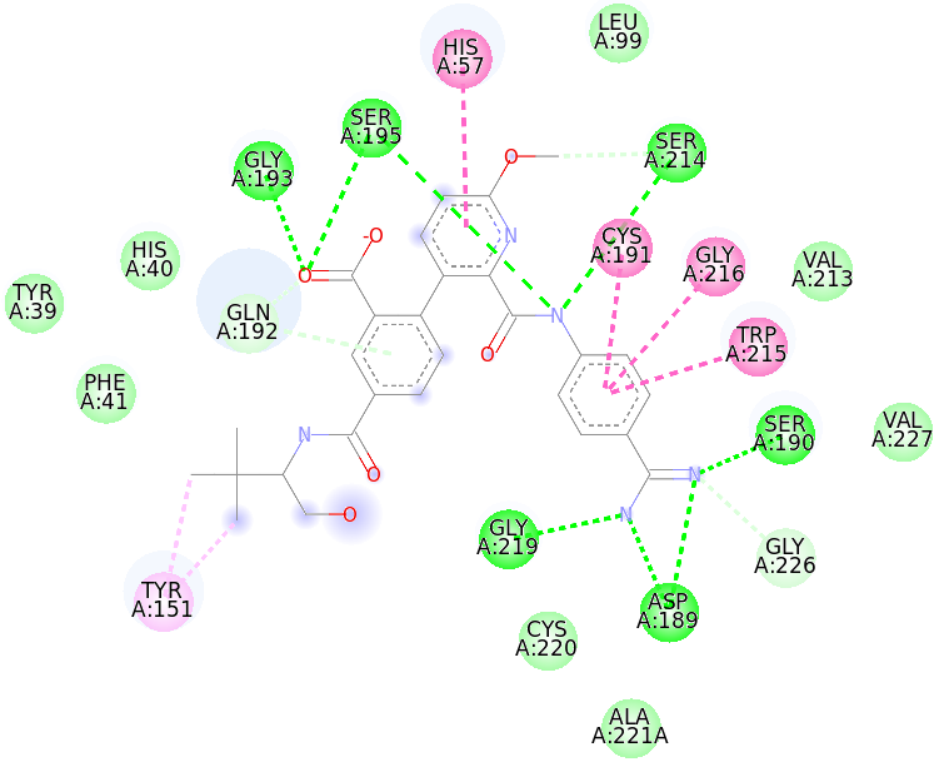
Family	PDB code
Kinase	1H00 (CDK2 in complex with a disubstituted 4, 6-bis anilino pyrimidine CDK4 inhibitor)
	2QD9 (P38 Alpha Map Kinase inhibitor based on heterobicyclic scaffolds)
Metalloenzyme	3BKL (Testis ACE co-crystal structure with ketone ACE inhibitor kAW)
Nuclear Hormone Receptor	2AM9 (Crystal structure of human androgen receptor ligand binding domain in complex with testosterone)
	3KBA (Progesterone receptor bound to sulfonamide pyrrolidine partial agonist)
Folate Enzyme	3NXO (Perferential Selection of Isomer Binding from Chiral Mixtures: Alternate Binding Modes Observed for the E- and Z-isomers of a Series of 5-Substituted 2,4-Diaminofuro[2,3-d]pyrimidines as Ternary Complexes with NADPH and Human Dihydrofolate Reductase)
Serine Protease	2AYW (Solution Structure of Drosophila melanogaster SNF RBD2)
Other Enzymes	1XL2 (HIV-1 Protease in complex with pyrrolidinmethanamine)

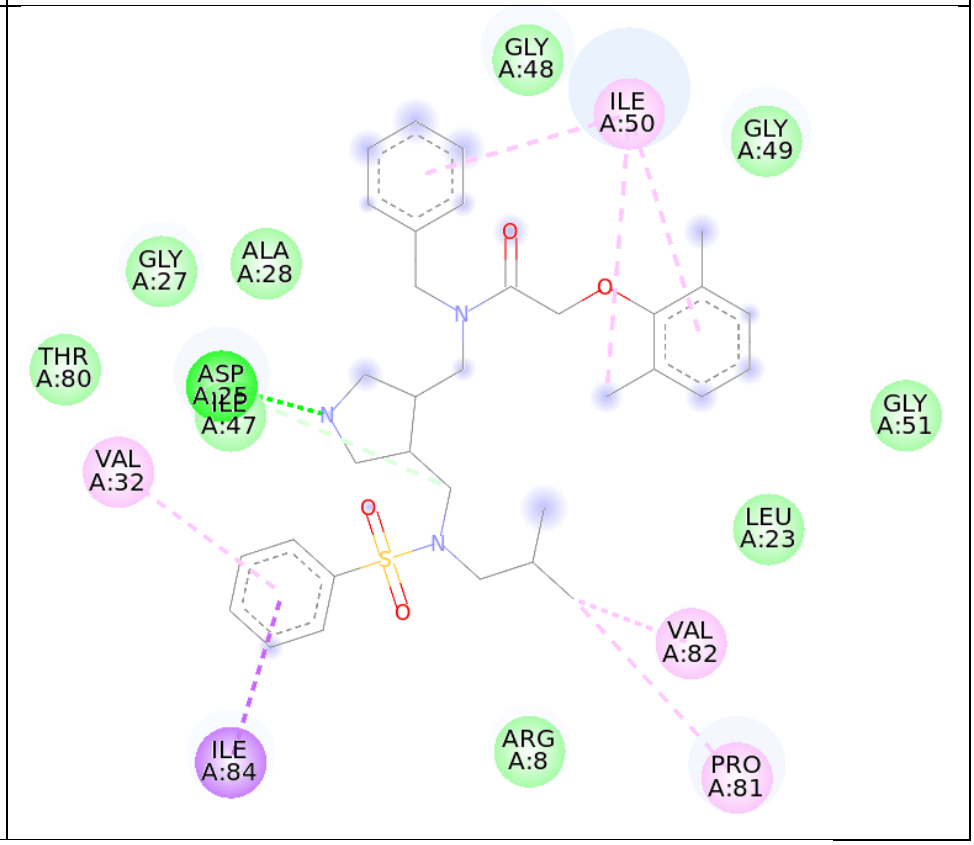
Table 11 - Interactions of crystallographic ligands with the targets used in this case study

Protein/Ligand	Interactions in 2D
1H00/FCP	

Protein/Ligand	Interactions in 2D
2QD9/LGF	
3BKL/KAW	

Protein/Ligand	Interactions in 2D
2AM9/TES	
3KBA	

Protein/Ligand	Interactions in 2D
	
2AYW/ONO	

Protein/Ligand	Interactions in 2D
1XL2/189	

After choosing targets, it was necessary to choose which active ligands would be used for each target. Then, we used MolAr to perform a virtual screening using Autodock Vina in each of the chosen targets and all DUD active ligands for each target. The 10 active ligands with the best energy were chosen, for each target. This pre-selection of the active ligands was necessary, since it would be impracticable to perform a virtual screening in the chosen configuration against all active ligands of each target and their respective decoys in order to generate the ROC curve. Thus, since Autodock Vina has a shorter runtime than DOCK6, it was chosen to perform these tests (in general, for the selected targets, this step spent around 1 day for each one in a computer with the following configurations: Intel(R) Xeon(R) E-2174G CPU @ 3.80GHz 16GB RAM).

Consensus virtual screening was performed for all targets and another VS using only DOCK 6 was carry out to compare the influence of using the Autodock Vina output as input on DOCK 6 with the result obtained by DOCK 6 using the original ligands and its decoys. On

average, due to the configuration chosen for the tests, virtual consensus screening for each target took 2 weeks to be performed. However, there were targets, such as Metalloenzymes, where VS took up to 30 days for a single target.

Figure 54 outlines the experiment performed in this case study:

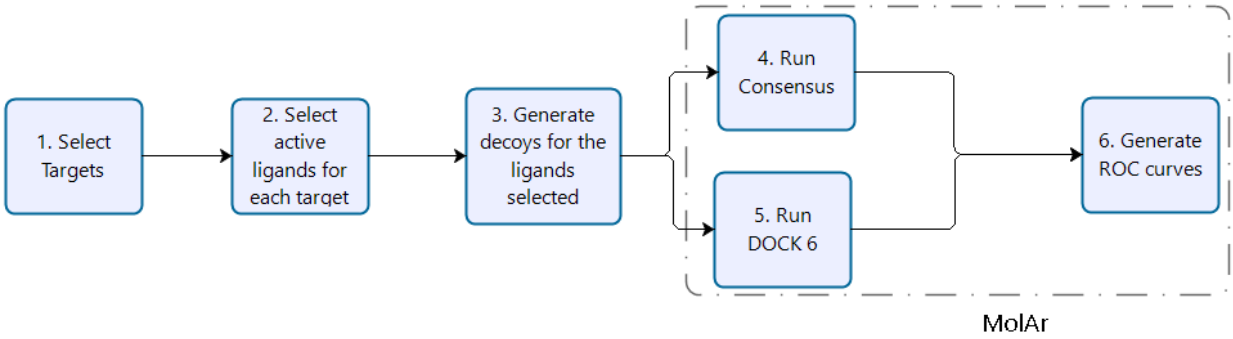


Figure 54 – DUD-E experiments workflow

Results

In this experiment, a virtual consensus screening was performed and then a VS using only DOCK 6 was carried out to compare DOCK 6 performance using the final ligand conformations defined by Autodock Vina and the original conformations.

In the experiments performed there was an improvement in the ROC curve of up to 42% (3BKL protein), which demonstrates that the approach used can lead to significant gains. It is clear from the table above showing the results of the experiments performed that, with the exception of 3KBA protein, in all other proteins, there was an improvement in the ROC curve when executing DOCK 6 based on the final conformation of ligands defined by Autodock Vina, compared to the execution of DOCK 6 using the original ligand conformation. The 3KBA protein showed the same AUC in both scenarios. Table 12 summarizes the results.

Table 12 - Comparison between running DOCK 6 on ligands conformation provided by Autodock Vina or running DOCK 6 on original ligands

Family	PDB code	AUC		
		Dock6	Dock6 after Autodock Vina	ROC curve improvement
Kinase	1H00	0,85	0,96	13%
	2QD9	0,48	0,59	23%
Metalloenzyme	3BKL	0,57	0,81	42%
	2AM9	0,35	0,46	31%

Nuclear Hormone Receptor	3KBA	0,56	0,56	0%
Folate Enzyme	3NX0	0,63	0,80	27%
Serine Protease	2AYW	0.90	0.95	6%
Other Enzymes	1XL2	0,63	0,76	21%

Thus, we can conclude that the CVS can increase the reliability of the tests performed by DOCK 6, because it tends to decrease the false negatives identified by DOCK 6.

7. Final considerations and future work

This thesis presented MolAr, a new software that aims to assist the Virtual Screening process. One of the main contributions of our work is the development of a software composed of several integrated tools and working as a workflow for the execution of the whole VS process. MolAr automates the entire VS, minimizing the need for human interference and thereby reducing the chances of error. MolAr allows the researcher to perform docking without any advanced computing knowledge. A new CVS approach between DOCK 6 and Autodock Vina has also been implemented, allowing the user to easily perform this task. The tools used during software development are free for academic use.

The major functionalities provided by MolAr are:

- Target Builder allowing Homology Modeling design;
- Virtual Screening using Autodock Vina and DOCK 6;
- Virtual Consensus Screening between DOCK 6 and Autodock Vina automating all the necessary process to be performed for the Consensus Screening without any user interference;
- Database Manager, which allows user to easily manage Octopus databases.
- Visualization of the 3D structure of a molecule;
- Analysis of a structure by the Ramachandran Plot and RMSD calculation;
- Adjusting the protonation state of a molecular targets and ligands;
- ROC curve generation.

Finally, three case studies were performed with MolAr. In the first case MolAr was used to evaluate the Pharmacological effects of Isatine and Convolutamine on cyclooxygenase-1, cyclooxygenase-2 and Mu opioid molecular target carrying out a Virtual Screening using Autodock Vina. It was found that isatin derivatives evaluated may have antinociceptive action on the COX-1 and COX-2 receptors. On the other hand, the likelihood of the isatin derivatives evaluated acting to inhibit the Mu opioid receptor is low, since there are no isatin derivatives with lower binding energy than the lowest binding energy of known opioid ligand.

In the second case, MolAr performed a virtual screening in DNA-Ligand systems. It was verified that in the case study evaluated, both the Autodock Vina and the DOCK 6 presented good reliability. However, it was found that although the results found were excellent, the ROC curve analysis suggests that the use of CVS increased the reliability of screening, compared with the execution of the VS tools separately. In addition, the identification of active ligands has been improved because with the consensus approach all active ligands have been

identified in the top 8. In Autodock Vina all active ligands have been identified in the top 22, whereas in DOCK 6 they were identified in the top 71.

Moreover, MolAr performs the whole CVS process automatically. Thus, the user does not have to worry about the various steps required to perform the CVS between the Autodock Vina and the DOCK 6, such as preparation of the ligands, conversion of the output generated by the Autodock Vina to the DOCK 6 format, comparison between the results generated by the two software programs (which can be difficult because they are in different units), etc.

Finally, the last case validates the strategy used in virtual consensus screening. In the strategy implemented by MolAr, the final ligands conformations resulting from the anchoring previously performed by Autodock Vina was used as input to the VS carried out by DOCK 6. In the experiments performed, the improvement in the AUC reached up to 42%. Thus, the use of the final conformation determined by Autodock Vina as input to the virtual screening performed at DOCK 6 seems to be a good strategy for improving the reliability of screening performed by DOCK 6.

So, this thesis demonstrated that MolAr was able to perform *in silico* simulations correctly, with very good results. Finally, MolAr has the advantages of having easy-to-use interface that eliminates the need to use multiple command line programs. In addition, MolAr automates the installation of software programs and libraries required in the Virtual Screening process, contributing to the reduction of time and errors that may occur. The Linux version has all features while the window´s version is limited. In Windows, MolAr only allows the homology modeling and virtual screening with AutoDock Vina. MolAr is freely available to be downloaded at http://www.drugdiscovery.com.br/.

MolAr was made available in October 2019, but it is already being used by researchers from 5 different countries, while Octopus is already being used by researchers from 23 different countries. The MolAr´s paper has been submitted to the Journal of Chemical Information and Modeling, and the number of downloads will increase after its publication.

There are several pending steps to be performed on the software such as:

- Permission to add new features by third parties. This feature would allow other researchers to develop packages with new features that could be easily added to MolAr;
- Addition of a ligand database, such as DrugBank. A database of active ligands would allow more targets to be easily tested against existing drugs on the market. This would make it easier to conduct studies such as drug repositioning, for example.
- Generating a full-featured version that works on Windows;

- Inclusion in MolAr of a feature that would allow ligand-based virtual screening.
- Implementation of artificial intelligence techniques that could allow better ligand ranking in a virtual screening.

References

ABAGYAN, Ruben; TOTROV, Maxim; KUZNETSOV, Dmitry. ICM - A new Method for Protein Modeling and Design: Applications to Docking and Structure Prediction from Distorted Native Conformation. **Journal of computational chemistry**, [s. l.], v. 15, n. 5, p. 488–506, 1994.

ABDO, Ammar et al. Ligand-based virtual screening using bayesian networks. **Journal of Chemical Information and Modeling**, [s. l.], v. 50, n. 6, p. 1012–1020, 2010.

ALIEBRAHIMI, Shima et al. Identification of Phytochemicals Targeting c-Met Kinase Domain using Consensus Docking and Molecular Dynamics Simulation Studies. **Cell Biochemistry and Biophysics**, [s. l.], p. 1–11, 2017. Disponível em: <http://dx.doi.org/10.1007/s12013-017-0821-6>

ALLEN, William J. et al. DOCK 6: Impact of new features and current docking performance. **Journal of Computational Chemistry**, [s. l.], v. 36, n. 15, p. 1132–1156, 2015.

ANANDAKRISHNAN, Ramu; AGUILAR, Boris; ONUFRIEV, Alexey V. H++ 3.0: Automating pK prediction and the preparation of biomolecular structures for atomistic molecular modeling and simulations. **Nucleic Acids Research**, [s. l.], v. 40, n. W1, p. 537–541, 2012.

ARROWSMITH, John. A decade of change. **Nature Reviews Drug Discovery**, [s. l.], v. 11, n. 1, p. 17–18, 2012. Disponível em: <http://dx.doi.org/10.1038/nrd3630>

ASH, Sheila et al. SYBYL Line Notation (SLN): A Versatile Language for Chemical Structure Representation. **Journal of Chemical Information and Modeling**, [s. l.], v. 37, n. 1, p. 71–79, 1997. Disponível em: <http://pubs.acs.org/cgi-bin/doilookup/?10.1021/ci960109j>

ASHTAWY, Hossam M.; MAHAPATRA, Nihar R. Task-Specific Scoring Functions for Predicting Ligand Binding Poses and Affinity and for Screening Enrichment. **Journal of Chemical Information and Modeling**, [s. l.], v. 58, n. 1, p. 119–133, 2018.

AWUNI, Yaw; MU, Yuguang. Reduction of false positives in structure-based virtual screening when receptor plasticity is considered. **Molecules**, [s. l.], v. 20, n. 3, p. 5152–5164, 2015.

BAEK, Minkyung et al. GalaxyDock BP2 score: a hybrid scoring function for accurate protein–ligand docking. **Journal of Computer-Aided Molecular Design**, [s. l.], v. 31, n. 7, p. 653–666, 2017. Disponível em: <https://doi.org/10.1007/s10822-017-0030-9>

BAJORATH, Jürgen. Integration of virtual and high-throughput screening. **Nature Reviews Drug Discovery**, [s. l.], v. 1, n. 11, p. 882–894, 2002. Disponível em: <http://www.nature.com/doifinder/10.1038/nrd941>

BANEGAS-LUNA, Antonio-Jesús; CERÓN-CARRASCO, José P.; PÉREZ-SÁNCHEZ, Horacio. A review of ligand-based virtual screening web tools and screening algorithms in large molecular databases in the age of big data. **Future Medicinal Chemistry**, [s. l.], v. 10, n. 22, p. 2641–2658, 2018. Disponível em: <https://www.future-science.com/doi/10.4155/fmc-2018-0076>

BAXTER, Carol A. et al. Flexible docking using Tabu search and an empirical estimate of binding affinity. **Proteins: Structure, Function and Genetics**, [s. l.], v. 33, n. 3, p. 367–382, 1998.

BENKERT, Pascal; KÜNZLI, Michael; SCHWEDE, Torsten. QMEAN server for protein model quality estimation. **Nucleic Acids Research**, [s. l.], v. 37, n. suppl_2, p. W510–W514, 2009. Disponível em: <http://link.springer.com/10.1007/978-1-61779-588-6>

BENTO, A. Patrícia et al. The ChEMBL bioactivity database: an update. **Nucleic Acids Research**, [s. l.], v. 42, n. D1, p. D1083–D1090, 2014. Disponível em: <http://www.ncbi.nlm.nih.gov/pubmed/24214965>

BERMAN, H. M. The Protein Data Bank. **Nucleic Acids Research**, [s. l.], v. 28, n. 1, p. 235–242, 2000. Disponível em: <https://academic.oup.com/nar/article-lookup/doi/10.1093/nar/28.1.235>

BERMAN, Helen M. et al. The future of the protein data bank. **Biopolymers**, [s. l.], v. 99, n. 3, p. 218–222, 2013. Disponível em: <http://www.pubmedcentral.nih.gov/articlerender.fcgi?artid=3684242&tool=pmcentrez&rendertype=abstract>

BIASINI, Marco et al. SWISS-MODEL: modelling protein tertiary and quaternary structure using evolutionary information. **Nucleic acids research**, [s. l.], v. 42, n. Web Server issue, p. W252-8, 2014. Disponível em: <http://www.ncbi.nlm.nih.gov/pubmed/24782522>

BREDA, Ardala et al. Virtual Screening of Drugs: Score Functions, Docking, and Drug Design. **Current Computer-Aided Drug Design**, [s. l.], v. 4, n. April, p. 265–272, 2008.

BROZELL, Scott R. et al. **Evaluation of DOCK 6 as a pose generation and database enrichment toolJournal of Computer-Aided Molecular Design**, 2012.

CARPENTER, Kristy A. et al. Deep learning and virtual drug screening. **Future Medicinal Chemistry**, [s. l.], v. 10, n. 21, p. 2557–2567, 2018. Disponível em: <https://www.future-science.com/doi/10.4155/fmc-2018-0314>

CARREGAL, A. P.; JR, M. Comar; TARANTO., A. G. Our Own Molecular Targets Data Bank

(OOMT). **Biochemistry and Biotechnology Reports**, [s. l.], v. 2, p. 14–16, 2013.

CARREGAL, Ana Paula et al. Inverse virtual screening studies of selected natural compounds from Cerrado. **International Journal of Quantum Chemistry**, [s. l.], v. 112, n. 20, p. 3333–3340, 2012.

CARREGAL, Ana Paula et al. Docking-based virtual screening of Brazilian natural compounds using the OOMT as the pharmacological target database. **Journal of Molecular Modeling**, [s. l.], v. 23, n. 4, p. 111, 2017. Disponível em: <http://link.springer.com/10.1007/s00894-017-3253-8>

CAVASOTTO, Claudio N. Homology models in docking and high-throughput docking. **Current topics in medicinal chemistry**, [s. l.], v. 11, n. 12, p. 1528–1534, 2011.

CAVASOTTO, Claudio N.; PHATAK, Sharangdhar S. Homology modeling in drug discovery: current trends and applications. **Drug Discovery Today**, [s. l.], v. 14, n. 13–14, p. 676–683, 2009. Disponível em: <http://linkinghub.elsevier.com/retrieve/pii/S1359644609001469>

CERTARA. **SYBYL-X Suite 2.1**CertaraCertara USA, Inc. 100 Overlook Center, Suite 101, Princeton, NJ 08540 USA, , 2016. Disponível em: <https://support.certara.com/software/molecular-modeling-and-simulation/sybyl-x/>. Acesso em: 23 ago. 2017.

CHAN, Margaret. **Ten years in public health 2007-2017**. [s.l: s.n.].

CHEMICAL COMPUTING GROUP. **Molecular Operating Environment (MOE)**. 2004.

CHERMAK, Edrisse et al. Introducing a clustering step in a consensus approach for the scoring of protein-protein docking models. **PLoS ONE**, [s. l.], v. 11, n. 11, p. 1–15, 2016.

CHOPADE, Atul R. Molecular Docking Studies of Phytocompounds From Phyllanthus Species as Potential Chronic Pain Modulators. **Scientia Pharmaceutica**, [s. l.], v. 83, n. 2, p. 243–267, 2015. Disponível em: <http://www.mdpi.com/2218-0532/83/2/243>

COLE, Jason C. et al. Comparing protein-ligand docking programs is difficult. **Proteins: Structure, Function, and Bioinformatics**, [s. l.], v. 60, n. 3, p. 325–332, 2005. Disponível em: <http://doi.wiley.com/10.1002/prot.20497>

CORBEIL, Christopher R. et al. Docking Ligands into Flexible and Solvated Macromolecules . 2 . Development and Application of F ITTED 1 . 5 to the Virtual Screening of Potential HCV Polymerase Inhibitors. **Journal of Chemical Information and Modeling**, [s. l.], p. 902–909, 2008.

CORBEIL, Christopher R.; ENGLEBIENNE, Pablo; MOITESSIER, Nicolas. Docking ligands into flexible and solvated macromolecules. 1. Development and validation of FITTED 1.0. **Journal of Chemical Information and Modeling**, [s. l.], v. 47, n. 2, p. 435–449, 2007.

CRUZ-MONTEAGUDO, Maykel et al. Activity cliffs in drug discovery: Dr Jekyll or Mr Hyde? **Drug Discovery Today**, [s. l.], v. 19, n. 8, p. 1069–1080, 2014.

CUMMINGS, Jeffrey; REIBER, Carl; KUMAR, Parvesh. The price of progress: Funding and financing Alzheimer's disease drug development. **Alzheimer's and Dementia: Translational Research and Clinical Interventions**, [s. l.], v. 4, p. 330–343, 2018. Disponível em: <https://doi.org/10.1016/j.trci.2018.04.008>

CZODROWSKI, Paul et al. Development, validation, and application of adapted PEOE charges to estimate pKa values of functional groups in protein-ligand complexes. **Proteins: Structure, Function, and Bioinformatics**, [s. l.], v. 65, n. 2, p. 424–437, 2006. Disponível em: <http://doi.wiley.com/10.1002/prot.21110>

DAS, Pratik Swarup; SAHA, Puja. A REVIEW ON COMPUTER AIDED DRUG DESIGN IN DRUG DISCOVERY. **World Journal of Pharmacy and Pharmaceutical Sciences**, [s. l.], v. 6, n. 7, p. 279–291, 2017. Disponível em: <http://wjpps.com/wjpps_controller/abstract_id/7294>

DAVIES, Matthew N. et al. Benchmarking pKa prediction. **BMC Biochemistry**, [s. l.], v. 7, p. 1–12, 2006.

DELANO, W. L. The PyMOL Molecular Graphics System, Version 1.8. **Schrödinger LLC**, [s. l.], p. http://www.pymol.org, 2002.

DENG, Zhi Luo et al. Exploring the biologically relevant chemical space for drug discovery. **Journal of Chemical Information and Modeling**, [s. l.], v. 53, n. 11, p. 2820–2828, 2013.

DEVI, R. Vasundhara; SATHYA, S. Siva; COUMAR, Mohane Selvaraj. Evolutionary algorithms for de novo drug design - A survey. **Applied Soft Computing Journal**, [s. l.], v. 27, p. 543–552, 2015. Disponível em: <http://dx.doi.org/10.1016/j.asoc.2014.09.042>

DING, Yun et al. Assessing the similarity of ligand binding conformations with the Contact Mode Score. **Computational Biology and Chemistry**, [s. l.], v. 64, p. 403–413, 2016. Disponível em: <http://linkinghub.elsevier.com/retrieve/pii/S1476927116302493>

DOLINSKY, Todd J. et al. PDB2PQR: Expanding and upgrading automated preparation of biomolecular structures for molecular simulations. **Nucleic Acids Research**, [s. l.], v. 35, n. SUPPL.2, p. 522–525, 2007.

DOS SANTOS, F. M. **Uso de fingerprints de farmacóforos potenciais para comparação de sítios protéicos e de ligantes ativos.** 2015. Universidade federal de Minas Gerais, [s. l.], 2015.

DOUCET, Nicolas; PELLETIER, Joelle N. Simulated annealing exploration of an active-site tyrosine in TEM-1β-lactamase suggests the existence of alternate conformations. **Proteins: Structure, Function, and Bioinformatics**, [s. l.], v. 69, n. 2, p. 340–348, 2007. Disponível em: <http://doi.wiley.com/10.1002/prot.21485>

DUFFY, Bryan C. et al. Early phase drug discovery: Cheminformatics and computational techniques in identifying lead series. **Bioorganic and Medicinal Chemistry**, [s. l.], v. 20, n. 18, p. 5324–5342, 2012. Disponível em: <http://dx.doi.org/10.1016/j.bmc.2012.04.062>

DURRANT, Jacob D.; MCCAMMON, J. Andrew. NNScore 2.0: A neural-network receptor-ligand scoring function. **Journal of Chemical Information and Modeling**, [s. l.], v. 51, n. 11, p. 2897–2903, 2011.

DUTKIEWICZ, Zbigniew; MIKSTACKA, Renata. Structure-Based Drug Design for Cytochrome P450 Family 1 Inhibitors. **Bioinorganic Chemistry and Applications**, [s. l.], v. 2018, p. 1–21, 2018. Disponível em: <https://www.hindawi.com/journals/bca/2018/3924608/>

ELOKELY, Khaled M.; DOERKSEN, Robert J. Docking Challenge: Protein Sampling and Molecular Docking Performance. **Journal of Chemical Information and Modeling**, [s. l.], v. 53, n. 8, p. 1934–1945, 2013. Disponível em: <http://www.pubmedcentral.nih.gov/articlerender.fcgi?artid=3755091&tool=pmcentrez&rendertype=abstract>

ESWAR, Narayanan et al. Comparative Protein Structure Modeling Using MODELLER. **Current Protocols in Protein Science**, Hoboken, NJ, USA, v. 50, n. 1, p. 2.9.1-2.9.31, 2007. Disponível em: <http://doi.wiley.com/10.1002/0471140864.ps0209s50>

FANG, Ye et al. Geauxdock: Accelerating Structure-Based Virtual Screening With Heterogeneous Computing. **PloS one**, [s. l.], v. 11, n. 7, p. e0158898, 2016.

FERREIRA, Leonardo et al. Molecular Docking and Structure-Based Drug Design Strategies. **Molecules**, [s. l.], v. 20, n. 7, p. 13384–13421, 2015. Disponível em: <http://www.mdpi.com/1420-3049/20/7/13384/>

FERREIRA, Rafaela S.; GLAUCIUS, Oliva; ANDRICOPULO, Adriano D. Integração das técnicas de triagem virtual e triagem biológica automatizada em alta escala: oportunidades e desafios em P&D de fármacos. **Química Nova**, [s. l.], v. 34, n. 10, p. 1770–1778, 2011.

Disponível em: <http://www.scielo.br/scielo.php?script=sci_arttext&pid=S0100-40422011001000010&lng=pt&nrm=iso&tlng=en>

FIGUEIREDO, Gabriela S. M. et al. Convolutamydine A and synthetic analogues have antinociceptive properties in mice. **Pharmacology Biochemistry and Behavior**, [s. l.], v. 103, n. 3, p. 431–439, 2013. Disponível em: <http://dx.doi.org/10.1016/j.pbb.2012.09.023>

FISER, András; ŠALI, Andrej. MODELLER: Generation and Refinement of Homology-Based Protein Structure Models. **Methods in Enzymology**, [s. l.], v. 374, p. 461–491, 2003.

FLACH, Peter a.; WU, Shaomin. Repairing concavities in ROC curves. In: IJCAI INTERNATIONAL JOINT CONFERENCE ON ARTIFICIAL INTELLIGENCE 2005, **Anais**... [s.l: s.n.]

FORLI, Stefano et al. Computational protein-ligand docking and virtual drug screening with the AutoDock suite. **Nature protocols**, [s. l.], v. 5, n. 11, p. 905–919, 2016.

FRIESNER, Richard A. et al. Glide: A New Approach for Rapid, Accurate Docking and Scoring. 1. Method and Assessment of Docking Accuracy. **Journal of Medicinal Chemistry**, [s. l.], v. 47, n. 7, p. 1739–1749, 2004. Disponível em: <http://pubs.acs.org/doi/abs/10.1021/jm0306430>

GASTEIGER, Johann; MARSILI, Mario. Iterative partial equalization of orbital electronegativity-a rapid access to atomic charges. **Tetrahedron**, [s. l.], v. 36, n. 22, p. 3219–3228, 1980.

GAULTON, Anna et al. ChEMBL: a large-scale bioactivity database for drug discovery. **Nucleic Acids Research**, [s. l.], v. 40, n. D1, p. D1100–D1107, 2012. Disponível em: <http://nar.oxfordjournals.org/lookup/doi/10.1093/nar/gkr777>

GENG, Cunliang et al. iScore: A novel graph kernel-based function for scoring protein-protein docking models. [s. l.], p. 1–26, 2018.

GENG, Cunliang et al. iScore: A novel graph kernel-based function for scoring protein-protein docking models. **Bioinformatics**, [s. l.], p. acs.jcim.6b00610, 2019. Disponível em: <http://pubs.acs.org/doi/abs/10.1021/acs.jcim.6b00610>

GILAD, Yocheved; SENDEROWITZ, Hanoch. Docking studies on DNA intercalators. **Journal of Chemical Information and Modeling**, [s. l.], v. 54, n. 1, p. 96–107, 2014.

GIORNO, Thais Biondino Sardella et al. Central and peripheral antinociceptive activity of 3-(2-oxopropyl)-3-hydroxy-2-oxindoles. **Pharmacology Biochemistry and Behavior**, [s. l.], v. 135,

p. 13–19, 2015. Disponível em: <http://dx.doi.org/10.1016/j.pbb.2015.05.004>

GIORNO, Thais Biondino Sardella et al. Antinociceptive effect and mechanism of action of isatin, N-methyl isatin and oxopropyl isatin in mice. **Life Sciences**, [s. l.], v. 151, p. 189–198, 2016. Disponível em: <http://dx.doi.org/10.1016/j.lfs.2016.02.052>

GOWTHAMAN, Ragul; LYSKOV, Sergey; KARANICOLAS, John. DARC 2.0: Improved docking and virtual screening at protein interaction sites. **PLoS ONE**, [s. l.], v. 10, n. 7, p. 1–24, 2015.

GROSDIDIER, Aurélien; ZOETE, Vincent; MICHIELIN, Olivier. SwissDock, a protein-small molecule docking web service based on EADock DSS. **Nucleic Acids Research**, [s. l.], v. 39, p. 270–277, 2011.

GUEX, Nicolas; PEITSCH, Manuel C. SWISS-MODEL and the Swiss-PdbViewer: An environment for comparative protein modeling. **Electrophoresis**, [s. l.], v. 18, n. 15, p. 2714–2723, 1997.

GUEX, Nicolas; PEITSCH, Manuel C.; SCHWEDE, Torsten. Automated comparative protein structure modeling with SWISS-MODEL and Swiss-PdbViewer: A historical perspective. **ELECTROPHORESIS**, [s. l.], v. 30, n. S1, p. S162–S173, 2009. Disponível em: <http://doi.wiley.com/10.1002/elps.200900140>

HAGA, Jason H.; ICHIKAWA, Kohei; DATE, Susumu. Virtual Screening Techniques and Current Computational Infrastructures. **Current pharmaceutical design**, [s. l.], v. 22, n. 23, p. 3576–84, 2016. Disponível em: <http://www.ncbi.nlm.nih.gov/pubmed/27075580>

HAMZA, Adel; WEI, Ning-Ning; ZHAN, Chang-Guo. Ligand-Based Virtual Screening Approach Using a New Scoring Function. **Journal of Chemical Information and Modeling**, [s. l.], v. 52, n. 4, p. 963–974, 2012. Disponível em: <http://pubs.acs.org/doi/10.1021/ci200617d>

HARRISON, Robert L. Introduction to Monte Carlo Simulation. **AIP Conf. Proc.**, [s. l.], v. 1204, p. 17–21, 2010.

HATMAL, Ma'mon M.; TAHA, Mutasem O. Simulated annealing molecular dynamics and ligand–receptor contacts analysis for pharmacophore modeling. **Future Medicinal Chemistry**, [s. l.], v. 9, n. 11, p. 1141–1159, 2017. Disponível em: <https://linkinghub.elsevier.com/retrieve/pii/S0021999107002422>

HAWKINS, P. C. D. et al. How to do an evaluation: Pitfalls and traps. **Journal of Computer-Aided Molecular Design**, [s. l.], v. 22, n. 3–4, p. 179–190, 2008.

HAWKINS, Paul C. D.; SKILLMAN, A Geoffrey; NICHOLLS, Anthony. Comparison of Shape-Matching and Docking as Virtual Screening Tools Comparison of Shape-Matching and Docking as Virtual Screening Tools. [s. l.], v. 50, n. December 2006, p. 74–82, 2007.

HERRÁEZ, Angel. Biomolecules in the computer: Jmol to the rescue. **Biochemistry and Molecular Biology Education**, [s. l.], v. 34, n. 4, p. 255–261, 2006. Disponível em: <http://doi.wiley.com/10.1002/bmb.2006.494034042644>

HILLISCH, Alexander; PINEDA, Luis Felipe; HILGENFELD, Rolf. Utility of homology models in the drug discovery process. **Drug Discovery Today**, [s. l.], v. 9, n. 15, p. 659–669, 2004.

HINDLE, Sally A. et al. Flexible docking under pharmacophore type constraints. **Journal of Computer-Aided Molecular Design**, [s. l.], v. 16, n. 2, p. 129–149, 2002.

HO, Tin Kam. The random subspace method for constructing decision forests. **IEEE Transactions on Pattern Analysis and Machine Intelligence**, [s. l.], v. 20, n. 8, p. 832–844, 1998.

HORVATH, Dragos. A Virtual Screening Approach Applied to the Search for Trypanothione. [s. l.], v. 2623, n. 96, p. 2412–2423, 1997.

HOU, Lin et al. Antitumor effects of Isatin on human neuroblastoma cell line (SH-SY5Y) and the related mechanism. **European Journal of Pharmacology**, [s. l.], v. 589, n. 1–3, p. 27–31, 2008.

HOUSTON, Douglas R.; WALKINSHAW, Malcolm D. Consensus docking: Improving the reliability of docking in a virtual screening context. **Journal of Chemical Information and Modeling**, [s. l.], v. 53, n. 2, p. 384–390, 2013.

HSU, Kai-Cheng et al. iGEMDOCK: a graphical environment of enhancing GEMDOCK using pharmacological interactions and post-screening analysis. **BMC bioinformatics**, [s. l.], v. 12 Suppl 1, n. Suppl 1, p. S33, 2011.

HUANG, Niu; SHOICHET, Brian K.; IRWIN, John J. Benchmarking sets for molecular docking. **Journal of Medicinal Chemistry**, [s. l.], v. 49, n. 23, p. 6789–6801, 2006.

HUANG, Sheng-You; GRINTER, Sam Z.; ZOU, Xiaoqin. Scoring functions and their evaluation methods for protein-ligand docking: recent advances and future directions. **Physical chemistry chemical physics : PCCP**, [s. l.], v. 12, n. 40, p. 12899–908, 2010. Disponível em: <http://www.ncbi.nlm.nih.gov/pubmed/20730182>

HUI-FANG, Liu et al. Evaluation of various inverse docking schemes in multiple targets

identification. **Journal of Molecular Graphics and Modelling**, [s. l.], v. 29, n. 3, p. 326–330, 2010. Disponível em: <http://dx.doi.org/10.1016/j.jmgm.2010.09.004>

JAIN, Ajay N. Bias, reporting, and sharing: Computational evaluations of docking methods. **Journal of Computer-Aided Molecular Design**, [s. l.], v. 22, n. 3–4, p. 201–212, 2008.

JAIN, SANMATI K.; AGRAWAL, AVANTIKA. De novo Drug Design: An overview. **Indian J. Pharm. Sci.**, [s. l.], v. 66, n. November, p. 721–728, 2004.

JONES, Gareth; WILLETT, Peter; GLEN, Robert C. A genetic algorithm for flexible molecular overlay and pharmacophore elucidation. **Journal of Computer-Aided Molecular Design**, [s. l.], v. 9, n. 6, p. 532–549, 1995.

KAPETANOVIC, I. M. Computer aided darug discovery and development: in silico-chemico-biological approach. **Chemico Biological Interactions**, [s. l.], v. 171, n. 2, p. 165–176, 2008.

KIM, Sunghwan et al. PubChem Substance and Compound databases. **Nucleic acids research**, [s. l.], v. 44, n. D1, p. D1202-13, 2016. Disponível em: <http://www.ncbi.nlm.nih.gov/pubmed/26400175%5Cnhttp://www.pubmedcentral.nih.gov/articlerender.fcgi?artid=PMC4702940>

KONG, De Xin et al. Historical variation of structural novelty in a natural product library. **Chemistry and Biodiversity**, [s. l.], v. 8, p. 1968–1977, 2011.

KORB, Oliver; STUTZLE, Thomas; EXNER, Thomas E. PLANTS: Application of Ant Colony Optimization to Structure-Based Drug Design. **Theoretical Computer Science**, [s. l.], v. 49, n. September, p. 84–96, 2009.

KORKMAZ, Selcuk; ZARARSIZ, Gokmen; GOKSULUK, Dincer. MLViS: A web tool for machine learning-based virtual screening in early-phase of drug discovery and development. **PLoS ONE**, [s. l.], v. 10, n. 4, p. 1–15, 2015.

KRISTENSEN, Thomas G.; NIELSEN, Jesper; PEDERSEN, Christian N. S. METHODS FOR SIMILARITY-BASED VIRTUAL SCREENING. **Computational and Structural Biotechnology Journal**, [s. l.], v. 5, n. 6, p. e201302009, 2013. Disponível em: <http://www.pubmedcentral.nih.gov/articlerender.fcgi?artid=3962175&tool=pmcentrez&rendertype=abstract>

KUKOL, Andreas. Consensus virtual screening approaches to predict protein ligands. **European Journal of Medicinal Chemistry**, [s. l.], v. 46, n. 9, p. 4661–4664, 2011. Disponível em: <http://dx.doi.org/10.1016/j.ejmech.2011.05.026>

KUNTZ, Irwin D. et al. A geometric approach to macromolecule-ligand interactions. **Journal of Molecular Biology**, [s. l.], v. 161, n. 2, p. 269–288, 1982. Disponível em: <http://linkinghub.elsevier.com/retrieve/pii/002228368290153X>

LASKOWSKI, R. A. et al. **PROCHECK: a program to check the stereochemical quality of protein structuresJournal of Applied Crystallography**, 1993.

LÄTTI, Sakari; NIINIVEHMAS, Sanna; PENTIKÄINEN, Olli T. Rocker: Open source, easy-to-use tool for AUC and enrichment calculations and ROC visualization. **Journal of Cheminformatics**, [s. l.], v. 8, n. 1, p. 45, 2016. Disponível em: <http://jcheminf.springeropen.com/articles/10.1186/s13321-016-0158-y>

LAVECCHIA, Antonio. Machine-learning approaches in drug discovery: Methods and applications. **Drug Discovery Today**, [s. l.], v. 20, n. 3, p. 318–331, 2015. Disponível em: <http://dx.doi.org/10.1016/j.drudis.2014.10.012>

LEACH, Andrew R. et al. Three-dimensional pharmacophore methods in drug discovery. **Journal of Medicinal Chemistry**, [s. l.], v. 53, n. 2, p. 539–558, 2010.

LEELANANDA, Sumudu P.; LINDERT, Steffen. Computational methods in drug discovery. **Beilstein Journal of Organic Chemistry**, [s. l.], v. 12, p. 2694–2718, 2016.

LEISINGER, Klaus Michael; GARABEDIAN, Laura Faden; WAGNER, Anita Katharina. Improving access to medicines in low and middle income countries: Corporate responsibilities in context. **Southern Med Review**, [s. l.], v. 5, n. 2, p. 3–8, 2012.

LI, Guo Bo et al. ID-score: A new empirical scoring function based on a comprehensive set of descriptors related to protein-ligand interactions. **Journal of Chemical Information and Modeling**, [s. l.], v. 53, n. 3, p. 592–600, 2013.

LI, Hui; ROBERTSON, Andrew D.; JENSEN, Jan H. Very fast empirical prediction and rationalization of protein pKa values. **Proteins: Structure, Function, and Bioinformatics**, [s. l.], v. 61, n. 4, p. 704–721, 2005. Disponível em: <http://doi.wiley.com/10.1002/prot.20660>

LI, Liwei; WANG, Bo; MEROUEH, Samy O. Support vector regression scoring of receptor-ligand complexes for rank-ordering and virtual screening of chemical libraries. **Journal of Chemical Information and Modeling**, [s. l.], v. 51, n. 9, p. 2132–2138, 2011.

LI, Zhengfu et al. Adaptive molecular docking method based on information entropy genetic algorithm. **Applied Soft Computing**, [s. l.], v. 26, p. 299–302, 2015. Disponível em: <http://linkinghub.elsevier.com/retrieve/pii/S1568494614005080>

LIAN, Zhi Min; SUN, Juan; ZHU, Hai Liang. Design, synthesis and antibacterial activity of isatin derivatives as FtsZ inhibitors. **Journal of Molecular Structure**, [s. l.], v. 1117, p. 8–16, 2016. Disponível em: <http://dx.doi.org/10.1016/j.molstruc.2016.03.036>

LINDORFF-LARSEN, Kresten et al. Improved side-chain torsion potentials for the Amber ff99SB protein force field. **Proteins: Structure, Function and Bioinformatics**, [s. l.], v. 78, n. 8, p. 1950–1958, 2010.

LIONTA, Evanthia et al. Structure-Based Virtual Screening for Drug Discovery: Principles, Applications and Recent Advances. **Current Topics in Medicinal Chemistry**, [s. l.], v. 14, n. 16, p. 1923–1938, 2014. Disponível em: <http://www.ncbi.nlm.nih.gov/pubmed/25262799>

LIU, Jie; WANG, Renxiao. On Classification of Current Scoring Functions. **Journal of Chemical Information and Modeling**, [s. l.], v. 55, n. 3, p. 475–482, 2015. Disponível em: <http://dx.doi.org/10.1021/ci500731a>

LIU, Shengchao et al. Practical Model Selection for Prospective Virtual Screening. **Journal of Chemical Information and Modeling**, [s. l.], p. acs.jcim.8b00363, 2018. Disponível em: <http://www.ncbi.nlm.nih.gov/pubmed/30500183%0Ahttp://pubs.acs.org/doi/10.1021/acs.jcim.8b00363>

LIU, YU; LI, WENTAO; MA, RUIXIN. Particle Swarm Optimization on Flexible Docking. **International Journal of Biomathematics**, [s. l.], v. 05, n. 05, p. 1250044, 2012.

LOURENÇO, Bruno Luis Alves et al. Virtual Screening and Molecular Docking for Arylalkylamine-N-Acetyltransferase (aaNAT) Inhibitors, a Key Enzyme of Aedes (Stegomyia) aegypti (L.) Metabolism. **Computational Molecular Bioscience**, [s. l.], v. 05, n. 03, p. 35–44, 2015. Disponível em: <http://www.scirp.org/journal/doi.aspx?DOI=10.4236/cmb.2015.53005>

MA, Xiao et al. Comparative Analysis of Machine Learning Methods in Ligand-Based Virtual Screening of Large Compound Libraries. **Combinatorial Chemistry & High Throughput Screening**, [s. l.], v. 12, n. 4, p. 344–357, 2009. Disponível em: <http://www.eurekaselect.com/openurl/content.php?genre=article&issn=1386-2073&volume=12&issue=4&spage=344>

MACKERELL, A. D. et al. All-Atom Empirical Potential for Molecular Modeling and Dynamics Studies of Proteins †. **The Journal of Physical Chemistry B**, [s. l.], v. 102, n. 18, p. 3586–3616, 1998. Disponível em: <http://pubs.acs.org/doi/10.1021/jp973084f>

MAIA, Eduardo Habib Bechelane et al. Octopus : a platform for the virtual high-throughput screening of a pool of compounds against a set of molecular targets. **Journal of Molecular**

Modeling, [s. l.], p. 23–26, 2017. Disponível em: <http://dx.doi.org/10.1007/s00894-016-3184-9>

MAITHRI G et al. Computational Drug Design and Molecular Dynamic Studies-A Review. **International Journal of Biomedical Data Mining**, [s. l.], v. 6, n. 1, p. 1–7, 2016.

MALDONADO, Ana G. et al. Molecular similarity and diversity in chemoinformatics: From theory to applications. **Molecular Diversity**, [s. l.], v. 10, n. 1, p. 39–79, 2006.

MARTI-RENOM, M. A. et al. COMPARATIVE PROTEIN STRUCTURE MODELING OF GENES AND GENOMES. **Rev. Annual Review of Biophysics and Biomolecular Structure**, [s. l.], v. 29, p. 291–325, 2000.

MARTIN, Yvonne C. et al. Glossary of terms used in computational drug design, part II (IUPAC Recommendations 2015). **Pure and Applied Chemistry**, [s. l.], v. 88, n. 3, p. 239–264, 2016.

MATHEUS, Maria Eline et al. Isatins inhibit cyclooxygenase-2 and inducible nitric oxide synthase in a mouse macrophage cell line. **European Journal of Pharmacology**, [s. l.], v. 556, n. 1–3, p. 200–206, 2007.

MCGANN, Mark. FRED pose prediction and virtual screening accuracy. **Journal of Chemical Information and Modeling**, [s. l.], v. 51, n. 3, p. 578–596, 2011.

MEIER, René et al. ParaDockS: A framework for molecular docking with population-based metaheuristics. **Journal of Chemical Information and Modeling**, [s. l.], v. 50, n. 5, p. 879–889, 2010.

MELO, F.; FEYTMANS, E. Assessing protein structures with a non-local atomic interaction energy. **Journal of molecular biology**, [s. l.], v. 277, p. 1141–1152, 1998.

MELVILLE, James L.; BURKE, Edmund K.; HIRST, Jonathan D. Machine Learning in Virtual Screening. **Current Topics in Medicinal Chemistry**, [s. l.], v. 12, n. 4, p. 332–343, 2009. Disponível em: <http://openurl.ingenta.com/content/xref?genre=article&issn=1386-2073&volume=12&issue=4&spage=332%5Cnpapers2://publication/doi/10.2174/138620709788167980>

MÉNDEZ, Raúl et al. Assessment of blind predictions of protein-protein interactions: Current status of docking methods. **Proteins: Structure, Function and Genetics**, [s. l.], v. 52, n. 1, p. 51–67, 2003.

MENG, Xuan-Yu et al. Molecular docking: a powerful approach for structure-based drug discovery. **Current computer-aided drug design**, [s. l.], v. 7, n. 2, p. 146–157, 2011.

Disponível em: <http://www.eurekaselect.com/openurl/content.php?genre=article&issn=1573-4099&volume=7&issue=2&spage=146>

MONTES, Matthieu; MITEVA, Maria A.; VILLOUTREIX, Bruno O. Structure-based virtual ligand screening with LigandFit: Pose prediction and enrichment of compound collections. **Proteins: Structure, Function, and Bioinformatics**, [s. l.], v. 68, n. 3, p. 712–725, 2007. Disponível em: <http://doi.wiley.com/10.1002/prot.21405>

MORGAN, Steve et al. The cost of drug development: A systematic review. **Health Policy**, [s. l.], v. 100, n. 1, p. 4–17, 2011. Disponível em: <http://dx.doi.org/10.1016/j.healthpol.2010.12.002>

MORRIS, Garrett M. Gm Garrett M. et al. AutoDock4 and AutoDockTools4: Automated docking with selective receptor flexibility. **Journal of Computational Chemistry**, [s. l.], v. 30, n. 16, p. 2785–2791, 2009. Disponível em: <http://doi.wiley.com/10.1002/jcc.21256>

MUGUMBATE, Grace et al. Target identification of Mycobacterium tuberculosis phenotypic hits using a concerted chemogenomic, biophysical, and structural approach. **Frontiers in Pharmacology**, [s. l.], v. 8, n. SEP, p. 1–13, 2017.

MYSINGER, Michael M. et al. Directory of useful decoys, enhanced (DUD-E): Better ligands and decoys for better benchmarking. **Journal of Medicinal Chemistry**, [s. l.], v. 55, n. 14, p. 6582–6594, 2012.

NG, Marcus C. K.; FONG, Simon; SIU, Shirley W. I. PSOVina: The hybrid particle swarm optimization algorithm for protein-ligand docking. **Journal of bioinformatics and computational biology**, [s. l.], v. 13, n. 3, p. 1541007, 2015. Disponível em: <http://www.ncbi.nlm.nih.gov/pubmed/25800162>

NIKOLAEV, Dmitrii M. et al. A Comparative Study of Modern Homology Modeling Algorithms for Rhodopsin Structure Prediction. **ACS Omega**, [s. l.], v. 3, n. 7, p. 7555–7566, 2018. Disponível em: <http://pubs.acs.org/doi/10.1021/acsomega.8b00721>

NUNES, Renata Rachide et al. Brazilian malaria molecular targets (BraMMT): selected receptors for virtual high-throughput screening experiments. **Memórias do Instituto Oswaldo Cruz**, [s. l.], v. 114, n. 0, p. 1–10, 2019. Disponível em: <http://www.scielo.br/scielo.php?script=sci_arttext&pid=S0074-02762019000100313&lng=en&tlng=en>

O'BOYLE, Noel M. et al. Open Babel: An Open chemical toolbox. **Journal of Cheminformatics**, [s. l.], v. 3, n. 10, 2011.

OGLIC, Dino et al. Active Search for Computer-aided Drug Design. **Molecular Informatics**, [s. l.], v. 37, n. 1, p. 1–15, 2018.

OGRIZEK, Mitja et al. Molecular dynamics to enhance structure-based virtual screening on cathepsin B. **Journal of Computer-Aided Molecular Design**, [s. l.], n. MAY, p. 707–712, 2015.

ONUFRIEV, Alexey V.; ALEXOV, Emil. Protonation and pK changes in protein–ligand binding. **Quarterly Reviews of Biophysics**, [s. l.], v. 46, n. 02, p. 181–209, 2013. Disponível em: <http://www.journals.cambridge.org/abstract_S0033583513000024>

OUYANG, Xuchang; HANDOKO, Stephanus Daniel; KWOH, Chee Keong. Cscore : a Simple Yet Effective Scoring Function for Protein – Ligand Binding Affinity Prediction Using Modified Cmac Learning Architecture. [s. l.], v. 9, p. 1–14, 2011.

PADHY, BM; GUPTA, YK. Drug repositioning: Re-investigating existing drugs for new therapeutic indications. **Journal of Postgraduate Medicine**, [s. l.], v. 57, n. 2, p. 153, 2011. Disponível em: <http://www.jpgmonline.com/text.asp?2011/57/2/153/81870>

PARK, Hwangseo; EOM, Jae Won; KIM, Yang Hee. Consensus scoring approach to identify the inhibitors of AMP-activated protein kinase α2 with virtual screening. **Journal of Chemical Information and Modeling**, [s. l.], v. 54, n. 7, p. 2139–2146, 2014.

PAUL, Steven M. et al. How to improve R&D productivity: the pharmaceutical industry's grand challenge. **Nature reviews. Drug discovery**, [s. l.], v. 9, n. 3, p. 203–14, 2010. Disponível em: <http://www.ncbi.nlm.nih.gov/pubmed/20168317>

PENCE, Harry E.; WILLIAMS, Antony. ChemSpider: An Online Chemical Information Resource. **Journal of Chemical Education**, [s. l.], v. 87, n. 11, p. 1123–1124, 2010. Disponível em: <http://pubs.acs.org/doi/abs/10.1021/ed100697w>

PETERSON, Yuri K. et al. The Discovery of Geranylgeranyltransferase-I Inhibitors with Novel Scaffolds by the Means of Quantitative Structure-Activity Relationship Modeling, Virtual Screening, and Experimental Validation. **J Med Chem.**, [s. l.], v. 52, n. 14, p. 83–88, 2009.

PETTERSEN, Eric F. et al. UCSF Chimera—A visualization system for exploratory research and analysis. **Journal of Computational Chemistry**, [s. l.], v. 25, n. 13, p. 1605–1612, 2004. Disponível em: <http://dx.doi.org/10.1002/jcc.20084>

PETUKH, Marharyta; STEFL, Shannon; ALEXOV, Emil. The role of protonation states in ligand-receptor recognition and binding. **Current pharmaceutical design**, [s. l.], v. 19, n. 23, p. 4182–90, 2013. Disponível em:

<http://www.pubmedcentral.nih.gov/articlerender.fcgi?artid=3625499&tool=pmcentrez&rendertype=abstract>

POLGÁR, Tímea et al. Impact of ligand protonation on virtual screening against β-secretase (BACE1). **Journal of Chemical Information and Modeling**, [s. l.], v. 47, n. 6, p. 2366–2373, 2007.

POLI, Giulio; MARTINELLI, Adriano; TUCCINARDI, Tiziano. Reliability analysis and optimization of the consensus docking approach for the development of virtual screening studies. **Journal of Enzyme Inhibition and Medicinal Chemistry**, [s. l.], v. 31, n. sup2, p. 167–173, 2016. Disponível em: <https://www.tandfonline.com/doi/full/10.1080/14756366.2016.1193736>

RAMACHANDRAN, G. N.; RAMAKRISHNAN, C.; SASISEKHARAN, V. Stereochemistry of polypeptide chain configurations. **Journal of molecular biology**, [s. l.], v. 7, p. 95–99, 1963.

RAMACHANDRAN, Srinivas; DOKHOLYAN, Nikolay V. Homology Modeling: Generating Structural Models to Understand Protein Function and Mechanism. In: [s.l.] : Springer Science+Business Media, 2012. p. 97–116.

RAREY, Matthias et al. A fast flexible docking method using an incremental construction algorithm. **Journal of molecular biology**, [s. l.], v. 261, n. 3, p. 470–89, 1996. Disponível em: <http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.132.7748&rep=rep1&type=pdf%5Cnhttp://www.ncbi.nlm.nih.gov/pubmed/8780787>

REDDY, Rallabandi Harikrishna et al. Structure-Based Virtual Screening of Protein Tyrosine Phosphatase Inhibitors: Significance, Challenges, and Solutions. **J. Microbiol. Biotechnol.**, [s. l.], v. 27, p. 878–895, 2017.

RICHMOND, Nicola J. et al. GALAHAD: 1. Pharmacophore identification by hypermolecular alignment of ligands in 3D. **Journal of Computer-Aided Molecular Design**, [s. l.], v. 20, n. 9, p. 567–587, 2006.

RIPPHAUSEN, Peter; NISIUS, Britta; BAJORATH, Jürgen Jurgen. State-of-the-art in ligand-based virtual screening. **Drug Discovery Today**, [s. l.], v. 16, n. 9–10, p. 372–376, 2011.

RODRIGUES, R. P. et al. Estratégias de Triagem Virtual no Planejamento de Fármacos. **Revista Virtual de Quimica**, [s. l.], v. 4, n. 6, p. 739–776, 2012.

ROGNAN, Didier. Structure-based approaches to target fishing and ligand profiling. **Molecular Informatics**, [s. l.], v. 29, n. 3, p. 176–187, 2010.

ROHILLA, Akshay; KHARE, Garima; TYAGI, Anil K. Virtual Screening , pharmacophore development and structure based similarity search to identify inhibitors against IdeR , a transcription factor of Mycobacterium tuberculosis. [s. l.], n. October 2016, p. 1–14, 2017.

RUIZ-CARMONA, Sergio et al. rDock: A Fast, Versatile and Open Source Program for Docking Ligands to Proteins and Nucleic Acids. **PLoS Computational Biology**, [s. l.], v. 10, n. 4, p. e1003571, 2014. Disponível em: <http://dx.plos.org/10.1371/journal.pcbi.1003571>

ŠALI, A. MODELLER: A Program for Protein Structure Modeling Release 9.12, r9480. **Rockefeller University**, [s. l.], p. 779–815, 2013. Disponível em: <http://scholar.google.com/scholar?hl=en&btnG=Search&q=intitle:MODELLER+A+Program+for+Protein+Structure+Modeling#6>

SALI, Andrej; SALI, Andrej. Statistical potential for assessment and prediction of protein structures. **Protein Science**, [s. l.], p. 2507–2524, 2006.

SAN DIEGO: ACCELRYS SOFTWARE INC. **Discovery Studio Modeling Environment, Release 4.1**. 2015.

SARGSYAN, Karen; GRAUFFEL, Cédric; LIM, Carmay. How Molecular Size Impacts RMSD Applications in Molecular Dynamics Simulations. **Journal of Chemical Theory and Computation**, [s. l.], v. 13, n. 4, p. 1518–1524, 2017.

SCHNECKE, V.; KUHN, L. A. Virtual screening with solvation and ligand-induced complementarity. **Perspectives in Drug Discovery and Design**, [s. l.], v. 20, p. 171–190, 2000.

SCHNEIDER, P.; TANRIKULU, Y.; SCHNEIDER, G. Self-organizing maps in drug discovery: compound library design, scaffold-hopping, repurposing. **Current medicinal chemistry**, [s. l.], v. 16, n. 3, p. 258–66, 2009. Disponível em: <http://www.ncbi.nlm.nih.gov/pubmed/19149576>

SCHREINER, Eduard et al. Stereochemical errors and their implications for molecular dynamics simulations. **BMC bioinformatics**, [s. l.], v. 12, n. 1, p. 190, 2011. Disponível em: <http://www.pubmedcentral.nih.gov/articlerender.fcgi?artid=3124434&tool=pmcentrez&rendertype=abstract>

SCI TEGIC ACCELRYS INC. **The MDL Drug Data Report (MDDR) database**. 2019. Disponível em: <http://accelrys.com/products/collaborative-science/databases/bioactivity-databases/mddr.html>. Acesso em: 22 mar. 2019.

SEELIGER, Daniel; DE GROOT, Bert L. Conformational transitions upon ligand binding: Holo-structure prediction from apo conformations. **PLoS Computational Biology**, [s. l.], v. 6, n. 1,

2010.

SENGUPTA, Soumi; BANDYOPADHYAY, Sanghamitra. Application of support vector machines in virtual screening. **International Journal for Computational Biology**, [s. l.], v. 1, n. 1, p. 56–62, 2014.

SHIN, Woong Hee et al. LigDockCSA: Protein-ligand docking using conformational space annealing. **Journal of Computational Chemistry**, [s. l.], v. 32, n. 15, p. 3226–3232, 2011.

SHIN, Woong Hee et al. GalaxyDock2: Protein-ligand docking using beta-complex and global optimization. **Journal of Computational Chemistry**, [s. l.], v. 34, n. 30, p. 2647–2656, 2013.

SILVA, Cândida et al. Enhancing Scoring Performance of Docking-Based Virtual Screening Through Machine Learning. **Current Bioinformatics**, [s. l.], v. 11, n. 4, p. 408–420, 2016. Disponível em: <http://www.eurekaselect.com/openurl/content.php?genre=article&issn=1574-8936&volume=11&issue=4&spage=408>

SILVERMAN, Richard B. **The Organic Chemistry of Drug Design and Drug Action: Second Edition**. 2nd. ed. Evanston: Elsevier, 2004.

SITKOFF, Doree; SHARP, Kim A.; HONIG, Barry. Accurate calculation of hydration free energies using macroscopic solvent models. **Journal of Physical Chemistry**, [s. l.], v. 98, n. 7, p. 1978–1988, 1994.

SOARES, José António Baptista Machado. **Estudo in silico da relação estrutura-actividade de piperazinas nos receptores 5-HT 1A e 5-HT 2A**. 2011. Universidade do Porto, [s. l.], 2011.

SPITZER, Russell; JAIN, Ajay N. Surflex-Dock: Docking benchmarks and real-world application. **Journal of Computer-Aided Molecular Design**, [s. l.], v. 26, n. 6, p. 687–699, 2012.

SRIDHAR, Devi. Improving Access to Essential Medicines: How Health Concerns can be Prioritised in the Global Governance System. **Public Health Ethics**, [s. l.], v. 1, n. 2, p. 83–88, 2008. Disponível em: <https://academic.oup.com/phe/article-lookup/doi/10.1093/phe/phn012>

SRIVASTAVA, Hemant Kumar et al. Comparison of computational methods to model DNA minor groove binders. **Journal of Chemical Information and Modeling**, [s. l.], v. 51, n. 3, p. 558–571, 2011.

STERLING, Teague; IRWIN, John J. ZINC15–Ligand Discovery for Everyone. **Journal of**

Chemical Information and Modeling, [s. l.], v. 55, n. 11, p. 2324–2337, 2015.

STEVENS, Hilde; HUYS, Isabelle. Innovative Approaches to Increase Access to Medicines in Developing Countries. **Frontiers in Medicine**, [s. l.], v. 4, n. December, p. 1–6, 2017. Disponível em: <http://journal.frontiersin.org/article/10.3389/fmed.2017.00218/full>

STEWART, J. J. P. **MOPAC2016™**. 2016.

STEWART, James J. P. et al. **MOPAC2012**Colorado Springs, CO, USA, CO, USAStewart Computational Chemistry, , 2012. Disponível em: <http://openmopac.net>

STEWART, James J. P. Optimization of parameters for semiempirical methods VI: more modifications to the NDDO approximations and re-optimization of parameters. **J. Mol. Model**, [s. l.], v. 19, n. 1, p. 1–32, 2013. Disponível em: <http://link.springer.com/10.1007/s00894-012-1667-x>

SURABHI, Surabhi; SINGH, BK. COMPUTER AIDED DRUG DESIGN: AN OVERVIEW. **Journal of Drug Delivery and Therapeutics**, [s. l.], v. 8, n. 5, p. 504–509, 2018. Disponível em: <http://jddtonline.info/index.php/jddt/article/view/1894>

SWANSON, Jessica M. J. et al. optimizing the poisson dielectric boundary with explicit solvent forces and energies: Lessons learned with atom-centered dielectric functions. **Journal of Chemical Theory and Computation**, [s. l.], v. 3, n. 1, p. 170–183, 2007.

TAN, Chunhu; YANG, Lijiang; LUO, Ray. How well does Poisson-Boltzmann implicit solvent agree with explicit solvent? A quantitative analysis. **Journal of Physical Chemistry B**, [s. l.], v. 110, n. 37, p. 18680–18687, 2006.

TANCHUK, Vsevolod Yu et al. A New, Improved Hybrid Scoring Function for Molecular Docking and Scoring Based on AutoDock and AutoDock Vina. **Chemical Biology and Drug Design**, [s. l.], v. 87, n. 4, p. 618–625, 2016.

TANG, Christopher L. et al. Calculation of pKas in RNA: On the Structural Origins and Functional Roles of Protonated Nucleotides. **Journal of Molecular Biology**, [s. l.], v. 366, n. 5, p. 1475–1496, 2007.

TARANTO, Alex Gutterres et al. Octopus: A Virtual High Thoughput Screening Plataform for Multi-Compouds and Targets. In: XVIIISIMPÓSIO BRASILEIRO DE QUÍMICA TEÓRICA 2015, Pirenópolis - GO -Brasil. **Anais**... Pirenópolis - GO -Brasil: Editora da UnB, 2015.

TAYARANI, Ali et al. Artificial Neural Networks Analysis Used to Evaluate the Molecular Interactions between Selected Drugs and Human Cyclooxygenase2 Receptor. **Iranian**

Journal of Basic Medical Sciences, [s. l.], v. 16, n. 11, p. 1196–1202, 2016. Disponível em: <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3909632/>

TEN BRINK, Tim; EXNER, Thomas E. Influence of Protonation, Tautomeric, and Stereoisomeric States on Protein−Ligand Docking Results. **Journal of Chemical Information and Modeling**, [s. l.], v. 49, n. 6, p. 1535–1546, 2009. Disponível em: <http://pubs.acs.org/doi/abs/10.1021/ci800420z>

TEN BRINK, Tim; EXNER, Thomas E. pKa based protonation states and microspecies for protein–ligand docking. **Journal of Computer-Aided Molecular Design**, [s. l.], v. 24, n. 11, p. 935–942, 2010. Disponível em: <http://link.springer.com/10.1007/s10822-010-9385-x>

THOMSEN, R.; CHRISTENSEN, M. H. MolDock: a new technique for high accuracy molecular docking. **J. Med. Chem.**, [s. l.], v. 49, p. 3315–3321, 2006.

TIETZE, Simon; APOSTOLAKIS, Joannis. GlamDock: Development and validation of a new docking tool on several thousand protein-ligand complexes. **Journal of Chemical Information and Modeling**, [s. l.], v. 47, n. 4, p. 1657–1672, 2007.

TODA, Kinya; GOTO, Junichi; HIRAYAMA, Noriaki. A Novel de novo Drug Design: Application of Pseudo-Molecular Probe in Construction of Novel Compounds based on Target Structure. **Ryoka Systems**, [s. l.], p. 2–3, 2008.

TRESADERN, Gary; BEMPORAD, Daniele; HOWE, Trevor. A comparison of ligand based virtual screening methods and application to corticotropin releasing factor 1 receptor. **Journal of Molecular Graphics and Modelling**, [s. l.], v. 27, n. 8, p. 860–870, 2009.

TRIBALLEAU, Nicolas et al. Virtual Screening Workflow Development Guided by the “Receiver Operating Characteristic” Curve Approach. Application to High-Throughput Docking on Metabotropic Glutamate Receptor Subtype 4. **Journal of Medicinal Chemistry**, [s. l.], v. 48, n. 7, p. 2534–2547, 2005. Disponível em: <http://pubs.acs.org/doi/abs/10.1021/jm049092j>

TRIPOS, International. **SYBYL-X 1.2**St. LouisTripos International, 1699 South Hanley Rd., St. Louis, Missouri, 63144, USA, , 2011.

TROTT, Oleg; OLSON, Arthur J. AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. **Journal of Computational Chemistry**, [s. l.], v. 31, n. 16, p. NA-NA, 2009. Disponível em: <http://doi.wiley.com/10.1002/jcc.21334>

TUCCINARDI, Tiziano et al. Extensive consensus docking evaluation for ligand pose prediction and virtual screening studies. **Journal of Chemical Information and Modeling**, [s.

l.], v. 54, n. 10, p. 2980–2986, 2014.

UNIVERSIDADE ESTADUAL DE FEIRA DE SANTANA. **NatProDB**. 2006.

VAN DE WATERBEEMD, H. et al. Glossary of terms used in computational drug design (IUPAC Recommendations 1997). **Pure and Applied Chemistry**, [s. l.], v. 69, n. 5, p. 1137–1152, 1997. Disponível em: <https://www.degruyter.com/view/j/pac.2016.88.issue-3/pac-2012-1204/pac-2012-1204.xml>

VENKATESHWARLU, E. et al. STUDY OF ANTI-INFLAMMATORY, ANALGESIC AND ANTIPYRETIC ACTIVITY OF NOVEL ISATIN DERIVATIVES. **Asian Journal of Pharmaceutical and Clinical Research**, [s. l.], v. 5, n. 4, p. 4–7, 2012.

VERDONK, Marcel L. et al. Improved protein-ligand docking using GOLD. **Proteins: Structure, Function and Genetics**, [s. l.], v. 52, p. 609–623, 2003.

VERLI, Hugo. **Bioinformática da Biologia à flexibilidade molecular**. Porto Alegre.

VERSCHUEREN, Erik et al. Protein-peptide complex prediction through fragment interaction patterns. **Structure**, [s. l.], v. 21, n. 5, p. 789–797, 2013. Disponível em: <http://dx.doi.org/10.1016/j.str.2013.02.023>

WANG, Junmei; CIEPLAK, Piotr; KOLLMAN, Peter A. How well does a restrained electrostatic potential (RESP) model perform in calculating conformational energies of organic and biological molecules? **Journal of Computational Chemistry**, [s. l.], v. 21, n. 12, p. 1049–1074, 2000. Disponível em: <http://doi.wiley.com/10.1002/1096-987X%28200009%2921%3A12%3C1049%3A%3AAID-JCC3%3E3.0.CO%3B2-F>

WANG, Tao et al. Fragment-Based Drug Discovery and Molecular Docking in Drug Design. [s. l.], p. 11–25, 2015.

WEBB, Benjamin; SALI, Andrej. Using MODELLER. [s. l.], n. June, p. 1–37, 2016.

WERMUTH, Camille G. Similarity in drugs: reflections on analogue design. **Drug Discovery Today**, [s. l.], v. 11, n. 7–8, p. 348–354, 2006.

WESTERMAIER, Yvonne; BARRIL, Xavier; SCAPOZZA, Leonardo. Virtual screening: An in silico tool for interlacing the chemical universe with the proteome. **Methods**, [s. l.], v. 71, p. 44–57, 2015. Disponível em: <http://linkinghub.elsevier.com/retrieve/pii/S104620231400259X>

WILLETT, Peter. Similarity-based virtual screening using 2D fingerprints. **Drug Discovery Today**, [s. l.], v. 11, n. 23–24, p. 1046–1053, 2006. Disponível em:

<https://linkinghub.elsevier.com/retrieve/pii/S1359644606004193>

WISHART, David S. et al. DrugBank 5 . 0 : a major update to the DrugBank database for 2018. [s. l.], v. 46, n. November 2017, p. 1074–1082, 2018.

WÓJCIKOWSKI, Maciej; BALLESTER, Pedro J.; SIEDLECKI, Pawel. Performance of machine-learning scoring functions in structure-based virtual screening. **Nature Publishing Group**, [s. l.], n. April, p. 1–10, 2017. Disponível em: <http://dx.doi.org/10.1038/srep46710>

XIA, J. et al. Virtual screening against phosphoglycerate kinase 1 in quest of novel apoptosis inhibitors. **Molecules**, [s. l.], v. 22, n. 6, p. 1–13, 2017.

YURIEV, Elizabeth; RAMSLAND, Paul A. Latest developments in molecular docking: 2010-2011 in review. **Journal of Molecular Recognition**, [s. l.], v. 26, n. October 2012, p. 215–239, 2013. Disponível em: <http://doi.wiley.com/10.1002/jmr.2266>

ZHANG, Gang et al. Virtual Screening of Small Molecular Inhibitors against DprE1. **Molecules**, [s. l.], v. 23, n. 3, p. 524, 2018. Disponível em: <http://www.mdpi.com/1420-3049/23/3/524>

ZHAO, Hongyu; GUO, Zongru. **Medicinal chemistry strategies in follow-on drug discoveryDrug Discovery Today**, 2009.

ZHAO, Wei et al. A statistical framework to evaluate virtual screening. **BMC bioinformatics**, [s. l.], v. 10, p. 225, 2009.

ZHAO, Yong; SANNER, Michel F. FLIPDock: Docking flexible ligands into flexible receptors. **Proteins: Structure, Function, and Bioinformatics**, [s. l.], v. 68, n. 3, p. 726–737, 2007. Disponível em: <http://doi.wiley.com/10.1002/prot.21423>

ZILIAN, David; SOTRIFFER, Christoph A. SFCscoreRF: A random forest-based scoring function for improved affinity prediction of protein-ligand complexes. **Journal of Chemical Information and Modeling**, [s. l.], v. 53, n. 8, p. 1923–1933, 2013.

ZSOLDOS, Zsolt et al. eHiTS: A new fast, exhaustive flexible ligand docking system. **Journal of Molecular Graphics and Modelling**, [s. l.], v. 26, n. 1, p. 198–212, 2007.

Appendix

Appendix A – INPI RUNMOPAC registration

RunMopac registration certificate at the National Institute of Industrial Property (INPI, of the Portuguese language “Instituto Nacional da Propriedade Industrial”), published by Industrial Property Magazine Nº 2422 on June 06 of 2017 on page 67.



REPÚBLICA FEDERATIVA DO BRASIL
Ministério Da Indústria, Comércio Exterior e Serviços
Instituto Nacional da Propriedade Industrial
Diretoria de Patentes, Programas de Computador e Topografias de Circuitos Integrados

Certificado de Registro de Programas de Computador

Processo nº: BR 51 2017 000476-5

O Instituto Nacional da Propriedade Industrial expede o presente certificado de Registro de Programas de Computador, válido por 50 anos a partir de 1º de janeiro subsequente à data de Criação: 06 de novembro de 2016 , em conformidade com o parágrafo 2º , artigo 2º da Lei Nº 9.609, de 19 de Fevereiro de 1998.

Título: **RUNMOPAC - A PLATFORM FOR LIGAND REFINEMENT FOR VIRTUAL SCREENING**

Data de Criação: 06 de novembro de 2016

Titular(es): FUNDAÇÃO DE AMPARO À PESQUISA DO ESTADO DE MINAS GERAIS -FAPEMIG (21.949.888/0001-83), Endereço: AVENIDA JOSÉ CÂNDIDO DA SILVEIRA, 1500 HORTO, BELO HORIZONTE, MG, 31035536
CENTRO FEDERAL DE EDUCAÇÃO TECNOLÓGICA DE MINAS GERAIS (17.220.203/0001-96), Endereço: AV. AMAZONAS, CAMPUS I, Nº 5253, BAIRRO NOVA SUIÇA, BELO HORIZONTE, MG, 30480000
UNIVERSIDADE FEDERAL DE SÃO JOÃO DEL REI (21.186.804/0001-05), Endereço: PRAÇA FREI ORLANDO, 170 - CENTRO , SÃO JOÃO DEL REI, MG, 36307352

Autor(es): ALEX GUTTERRES TARANTO (771.081.916-87)
/ ALINE RODRIGUES GUIMARÃES DE OLIVEIRA (136.413.566-36)
/ ALISSON MARQUES DA SILVA (028.636.156-61)
/ EDUARDO HABIB BACHELANE MAIA (047.922.726-88)
/ VINÍCIUS ALVES CAMPOS (136.444.016-40)

Linguagem: JAVA, SHELL SCRIPT, XML

Campo de Aplicação: BL-07, FQ-13, FQ-14, FQ-16, FQ-17

Tipo Programa: GI-06, GI-07, SM-01, TC-01

Expedido em: 21 de junho de 2017

Aprovado por Julio Cesar Castelo Branco Reis Moreira

Julio Cesar Castelo Branco Reis Moreira
Diretor de Patentes, Programas de Computador e Topografias de Circuitos Integrados

Appendix B – INPI MolAr registration

MolAr registration certificate at the National Institute of Industrial Property (INPI, of the Portuguese language “Instituto Nacional da Propriedade Industrial”), published by Industrial Property Magazine Nº 2422 on June 06 of 2017 on page 67.



REPÚBLICA FEDERATIVA DO BRASIL
MINISTÉRIO DA ECONOMIA
INSTITUTO NACIONAL DA PROPRIEDADE INDUSTRIAL
DIRETORIA DE PATENTES, PROGRAMAS DE COMPUTADOR E TOPOGRAFIAS DE CIRCUITOS INTEGRADOS

Certificado de Registro de Programa de Computador

Processo Nº: **BR512018052442-7**

O Instituto Nacional da Propriedade Industrial expede o presente certificado de registro de programa de computador, válido por 50 anos a partir de 1° de janeiro subsequente à data de 01/03/2018, em conformidade com o §2°, art. 2° da Lei 9.609, de 19 de Fevereiro de 1998.

Título: MoLar - Molecular Architect

Data de publicação: 01/03/2018

Data de criação: 01/03/2018

Titular(es): CENTRO FEDERAL DE EDUCAÇÃO TECNOLÓGICA DE MINAS GERAIS

Autor(es): ALISSON MARQUES DA SILVA; ALEX GUTTERRES TARANTO; EDUARDO HABIB BECHELANE MAIA; FERNANDO DE PILLA VAROTTI; RENATA RACHIDE NUNES

Linguagem: JAVA; PYTHON; SHELL SCRIPT

Campo de aplicação: BL-07; FQ-13; FQ-14; FQ-16; FQ-17

Tipo de programa: GI-01; GI-07; SM-01; TC-01

Algoritmo hash: SHA-512

Resumo digital hash:
3fe7651d7476e4919d784d6f09dce4b8f0de97be6f8a88032113c6ecb64187592069cd6109784563e732cf9851243977d71b3ad5d8ecb25e825aa95b784f8cbb

Expedido em: 02/01/2019

Aprovado por:
Liane Elizabeth Caldeira Lage
Diretora de Patentes, Programas de Computador e Topografias de Circuitos Integrados

Appendix C – INPI patent application

INSTITUTO NACIONAL DA PROPRIEDADE INDUSTRIAL

09/04/2019 870190033950 11:12

29409161902884883

Pedido nacional de Invenção, Modelo de Utilidade, Certificado de Adição de Invenção e entrada na fase nacional do PCT

Número do Processo: BR 10 2019 007156 7

Dados do Depositante (71)

Depositante 1 de 3

Nome ou Razão Social: UNIVERSIDADE FEDERAL DE SÃO JOÃO DEL REI
Tipo de Pessoa: Pessoa Jurídica
CPF/CNPJ: 21186804000105
Nacionalidade: Brasileira
Qualificação Jurídica: Instituição de Ensino e Pesquisa
Endereço: Praça Frei Orlando, 170 Centro
Cidade: São João Del Rei
Estado: MG
CEP: 36307-352
País: Brasil
Telefone: (32) 3379 2594
Fax:
Email: seipi@ufsj.edu.br

PETICIONAMENTO ELETRÔNICO

Esta solicitação foi enviada pelo sistema Peticionamento Eletrônico em 09/04/2019 às 11:12, Petição 870190033950

Depositante 2 de 3

Nome ou Razão Social:	FUNDAÇÃO DE AMPARO Á PESQUISA DO ESTADO DE MINAS GERAIS
Tipo de Pessoa:	Pessoa Jurídica
CPF/CNPJ:	21949888000183
Nacionalidade:	Brasileira
Qualificação Jurídica:	Órgão Público
Endereço:	Av. José Cândido da Silveira, N° 1.500, Horto
Cidade:	Belo Horizonte
Estado:	MG
CEP:	31035-536
País:	BRASIL
Telefone:	
Fax:	
Email:	

Depositante 3 de 3

Nome ou Razão Social:	CENTRO FEDERAL DE EDUCAÇÃO TECNOLÓGICA DE MINAS GERAIS
Tipo de Pessoa:	Pessoa Jurídica
CPF/CNPJ:	17220203000196
Nacionalidade:	Brasileira
Qualificação Jurídica:	Órgão Público
Endereço:	Avenida Amazonas, 5253, Bairro Nova Suiça
Cidade:	Belo Horizonte
Estado:	MG
CEP:	30421-169
País:	BRASIL
Telefone:	
Fax:	
Email:	

Dados do Pedido

Natureza Patente: 10 - Patente de Invenção (PI)

Título da Invenção ou Modelo de Utilidade (54): USO DA ATIVIDADE ANTIMALÁRICA DO COMPOSTO CLIOQUINOL (5-CHLORO-7-IODO-8-QUINOLINOL, 5-CHLORO-8-HYDROXY-7-IODOQUINOLINE, CLIOQUINOL, IODOCHLORHYDROXYQUIN)

Resumo: A malária é uma doença parasitária que acomete grande parte da população mundial. Na ausência de uma vacina eficaz, seu tratamento depende do uso de quimioterápicos e a necessidade de novos tratamentos é reforçada pela crescente resistência identificada aos antimaláricos atuais. Assim, o presente estudo teve como objetivo avaliar a atividade antimalárica in vitro dos compostos contra a cepa P. falciparum (W2). Para tanto o Clioquinol foi testado utilizando o teste tradicional. Além disso, foi realizado o ensaio de citotoxicidade utilizando a linhagem celular WI-26VA4 pelo teste de MTT para verificação da viabilidade celular. Experimentalmente verificou-se que não apresenta citotoxicidade, com IC50 > 100. Desta forma, os resultados sugerem que estes compostos podem ser usados como antimaláricos. A presente invenção demonstra Clioquinol (5-Chloro-7-iodo-8-quinolinol, 5-Chloro-8-hydroxy-7-iodoquinoline, Clioquinol, Iodochlorhydroxyquin) como agente antimalárico.

PETICIONAMENTO ELETRÔNICO Esta solicitação foi enviada pelo sistema Peticionamento Eletrônico em 09/04/2019 às 11:12, Petição 870190033950

Appendix D – Published paper

J Mol Model (2017) 23:26
DOI 10.1007/s00894-016-3184-9

ORIGINAL PAPER

Octopus: a platform for the virtual high-throughput screening of a pool of compounds against a set of molecular targets

Eduardo Habib Bechelane Maia[1,2] · Vinícius Alves Campos[1,2] · Bianca dos Reis Santos[1,2] · Marina Santos Costa[1,2] · Iann Gabriel Lima[1,2] · Sandro J. Greco[3] · Rosy I. M. A. Ribeiro[1] · Felipe M. Munayer[3] · Alisson Marques da Silva[3] · Alex Gutterres Taranto[1]

Received: 31 July 2016 / Accepted: 6 December 2016
© Springer-Verlag Berlin Heidelberg 2017

Abstract Octopus is an automated workflow management tool that is scalable for virtual high-throughput screening (vHTS). It integrates MOPAC2016, MGLTools, PyMOL, and AutoDock Vina. In contrast to other platforms, Octopus can perform docking simulations of an unlimited number of compounds into a set of molecular targets. After generating the ligands in a drawing package in the Protein Data Bank (PDB) format, Octopus can carry out geometry refinement using the semi-empirical method PM7 implemented in MOPAC2016. Docking simulations can be performed using AutoDock Vina and can utilize the Our Own Molecular Targets (OOMT) databank. Finally, the proposed software compiles the best binding energies into a standard table. Here, we describe two successful case studies that were verified by biological assay. In the first case study, the vHTS process was carried out for 22 (phenylamino)urea derivatives. The vHTS process identified a metalloprotease with the PDB code 1GKC as a molecular target for derivative LE&007. In a biological assay, compound LE&007 was found to inhibit 80% of the activity of this enzyme. In the second case study, compound Tx001 was submitted to the Octopus routine, and the results suggested that *Plasmodium falciparum* ATP6 (PfATP6) as a molecular target for this compound. Following an antimalarial assay, Tx001 was found to have an inhibitory concentration (IC_{50}) of 8.2 μM against PfATP6. These successful examples illustrate the utility of this software for finding appropriate molecular targets for compounds. Hits can then be identified and optimized as new antineoplastic and antimalarial drugs. Finally, Octopus has a friendly Linux-based user interface, and is available at www.drugdiscovery.com.br.

Keywords Structure-based drug design · Docking · Virtual screening

This paper belongs to Topical Collection Brazilian Symposium of Theoretical Chemistry (SBQT 2015)

✉ Alex Gutterres Taranto
labqf_ufsj@ufsj.edu.br

[1] Universidade Federal de São João del Rei—Campus Centro-Oeste, Divinópolis, MG, Brazil

[2] Centro Federal de Educação Tecnológica de Minas Gerais—Campus Divinópolis, Divinópolis, MG, Brazil

[3] Universidade Federal do Espírito Santo—UFES, Vitória, ES, Brazil

Introduction

The innovation process in the pharmaceutical industry is driven by the release of new drugs onto the market, as this process often involves modifying the structure of a known drug in an attempt to produce a new drug that is as active or more active towards a target receptor. [1]. First, pharmaceutical companies search for "hits;" a hit is a compound that shows activity towards a specific target under study, which is evaluated by performing biological and toxicological assays and structure–activity relationship (SAR) studies. The hit then becomes a lead compound. In this context, even though investment in the innovation process has increased, the availability of new drugs on the market has not followed suit. This is partly because the regulatory requirements associated with the approval of a new drug have also been increased in order to prevent pharmacological accidents (such as the birth defects resulting from thalidomide use by pregnant women). Furthermore, the high cost of the biological assays and the other methodologies used are limiting factors, especially for startups. This makes it critical to develop new approaches to identifying lead compounds [2].

Published online: 07 January 2017

Appendix E – Paper submitted to Frontiers in Chemistry



Methods in Computer Drug Design: from Classical to Artificial Intelligence

Eduardo Habib Bechelane Maia[1,2*], Letícia Cristina Assis[1], Tiago Alves de Oliveira[2], Alisson Marques da Silva[2] and Alex Gutterres Taranto[1]

[1] Laboratory of Pharmaceutical Medicinal chemistry-Federal University of São João Del Rei,Campus Dona Lindu, Divinópolis, MG, Brazil.

[2] Federal Center for Technological Education of Minas Gerais – CEFET-MG, Campus Divinopólis, MG, Brazil.

*** Correspondence Author:**
habib@cefetmg.br

Abstract

The drug development process is a major challenge for the pharmaceutical industry since it takes a substantial amount of time and money to move through all the phases of development of a new drug. One extensively used method to minimize the cost and time of the drug development process is computer-aided drug design (CADD). This method allows better focusing of the experiments, which can reduce the time and costs involved in the research of new drugs. In this context, structure-based virtual screening (SBVS) is robust and useful and one of the most promising *in silico* techniques for drug design. The method attempts to predict the best interaction mode of two molecules to form a stable complex, and it uses scoring functions to estimate, through mathematical methods, the force of noncovalent interactions between a ligand and a molecular target. Thus, the scoring functions are the main reason for the failure of SBVS software. Many software programs are used to perform SBVS, and since they use different algorithms, it is possible to obtain different results for the same input in different software. In the last decade, a new technique of SBVS called consensus virtual screening (CVS) has been used in some studies to increase the accuracy of SBVS and reduce the false positives obtained in SBVS experiments. An indispensable condition to the realization of SBVS is the availability of the 3D structure of the target protein, some virtual databases, such as the Protein Data Bank, have been created to store the 3D structures of molecules. However, sometimes it is not possible to experimentally obtain the 3D structure. In this situation, the homology modeling methodology allows the prediction of the 3D structure of a protein from its amino acid sequence. This review will present an overview of the challenges involved in the use of CADD to perform SBVS, the areas where CADD tools support SBVS, a comparison between the most often used tools, and the techniques currently used in an attempt to accelerate and reduce the cost.

Keywords: SBVS, Homology Modeling, Consensus Virtual Screening, Scoring Functions, computer-aided drug design.

Appendix F – Download history of the first software version

Name	School	Country
Abhijit Debnath	PDMCOP	India
Aijaz Ahamad Malik	Mahidol University	Thailand
amir tabatabaei	university of alberta	Canada
ANA LUCIA SANTOS DE MATOS ARAUJO	UNIVERSIDADE FEDERAL DE JUIZ DE FORA	BRAZIL
Andrei Santos Siqueira	Universidade Federal do Pará	Brazil
Antonio Rescifina	University	Italy
Arijit	University of Hyderabad	India
Ashutosh Trivedi	University of Lucknow	India
Bakku Ranjith Kumar	University of Tsukuba	Japan
cbrlab	Jagannath University	Bangladesh
David Garcia	University of Tennessee Knoxville	United States
Diego Carvalho	IIBCE-MEC	Uruguay
Dimitrios Trafalis	University of Athens	Greece
Divya Gopalakrishnan	Vellore Institute of Technology	India
Eduardo Jose Azevedo Correa	EPAMIG	Brazil
Ethayathulla Abdul samath	All India Institute of Medical	India

Name	School	Country
	Sciences New Delhi	
Fabio Polticelli	University of Roma Tre	Italy
Gaurao Vishwanathji Dhoke	RWTH	Germany
Gilles Berger	UNiversite Libre de Bruxelles	Belgium
HamidZahednasab	University of Tehran	Iran
jainul fathima	kalasalingam university	India
Jonathan Helander	University of Washington	USA
Jose Vicente Roig Genovés	Universidad Católica de Valencia	Valencia
Julijus Bogomolovas	UCSD	USA
JULIO ANDERSON FONSECA ARVELLOS	UNIVERSIDADE FEDERAL DE JUIZ DE FORA	BRASIL
khairul anuar	universivicenteti Malaya	Malaysia
Kotha	sri venkateswara university	india
Lata Ganji	Prin. K. M. Kundnani College of Pharmacy	India

Name	School	Country
Lucas Rolim Medaglia	Universidade Estadual de Londrina	Brasil
Luiz Fernando Belo de Araújo	Universidade Federal do Oeste da Bahia	Brasil
Md Tabish Rehman	King Saud University	Saudi Arabia
Nalini Schaduangrat	Mahidol University	Thailand
Nikolaos Nikoleousakos	NATIONAL KAPODISTRIAN UNIVERSITY OF ATHENS	GREECE
Patrick Chirdon	Chirdon Family	USA
prasad gandham	ICRISAT	India
Rafael Conceição de Souza	Evandro Chagas Institute	Brazil
rakshandha v	sastra university	india
Santanu Sasidharan	NIT-Warangal	India
Sergii Vakal	Wenzhou Medical University	China
Sergii Vakal	Taras Shevchenko National University of Kyiv	Ukraine
Shafi Ullah Khan	Monash University	Malaysia

Name	School	Country
Siming Nie	Northeast Forestry University	China
SRINIVASAN E	VELLORE INSTITUTE OF TECHNOLOGY	INDIA
Stefania Mazzini	University of Milano	Italy
Stephen Molnar	Molecular Modeling	US
subhash chandra	Kumaun University	India
SUDHIR KUMAR	CCS HARYANA AGRICULTURAL UNIVERSITY	INDIA
Suleyman Selim Cinaroglu	Acibadem University	Turkey
Sung-Jin Kim	Kyung Hee University School of Dentistry	KOREA
Tacio Vinicio Amorim Fernandes	UFRJ	Brazil
Taffarel Melo Torres	Universidade Federal Rural do Semi-árido	Brazil
VIBHA MISHRA	BANARAS HINDU UNIVERSITY	INDIA
wang tai jin	sichuan university	china

Name	School	Country
Yanne Szirovicza Chagas Lopes	PUC-Rio	Brasil
YAROSLAV FALETROV	Belarusian State University	Belarus
Yunier Perera Sardiña	Andres Bello University	Chile
zhongfeisheng	SIMM	China

Appendix G – MolAr default parameters

grid.in

```
compute_grids              yes
grid_spacing               0.4
output_molecule            no
contact_score              no
energy_score               yes
energy_cutoff_distance     9999
atom_model                 a
attractive_exponent        6
repulsive_exponent         12
distance_dielectric        yes
dielectric_factor          4
bump_filter                yes
bump_overlap               0.75
receptor_file              RECEPTOR_PDB_CODE.rec.mol2
box_file                   RECEPTOR_PDB_CODE.box.pdb
vdw_definition_file        <FOLDER>/dock6/parameters/vdw_AMBER_parm99.defn
score_grid_prefix          grid
```

showbox.in

```
Y
8.0     #Caixa
selected_spheres.sph
1
RECEPTOR_PDB_CODE.box.pdb
```

flex.in

```
conformer_search_type                   flex
user_specified_anchor                   no
limit_max_anchors                       no
min_anchor_size                         5
pruning_use_clustering                  yes
pruning_max_orients                     1000
```

```
pruning_clustering_cutoff                100
pruning_conformer_score_cutoff           100.0
pruning_conformer_score_scaling_factor   1.0
use_clash_overlap                        no
write_growth_tree                        no
use_internal_energy                      yes
internal_energy_rep_exp                  12
internal_energy_cutoff                   100.0
ligand_atom_file                         RECEPTOR_PDB_CODE.lig.mol2
limit_max_ligands                        no
skip_molecule                            no
read_mol_solvation                       no
calculate_rmsd                           yes
use_rmsd_reference_mol                   yes
rmsd_reference_filename                  RECEPTOR_PDB_CODE.lig.mol2
use_database_filter                      no
orient_ligand                            yes
automated_matching                       yes
receptor_site_file                       selected_spheres.sph
max_orientations                         1000
critical_points                          no
chemical_matching                        no
use_ligand_spheres                       no
bump_filter                              no
score_molecules                          yes
contact_score_primary                    no
contact_score_secondary                  no
grid_score_primary                       yes
grid_score_secondary                     no
grid_score_rep_rad_scale                 1
grid_score_vdw_scale                     1
grid_score_es_scale                      1
grid_score_grid_prefix                   grid
multigrid_score_secondary                no
dock3.5_score_secondary                  no
```

```
continuous_score_secondary                no
footprint_similarity_score_secondary      no
pharmacophore_score_secondary             no
descriptor_score_secondary                no
gbsa_zou_score_secondary                  no
gbsa_hawkins_score_secondary              no
SASA_score_secondary                      no
amber_score_secondary                     no
minimize_ligand                           yes
minimize_anchor                           yes
minimize_flexible_growth                  yes
use_advanced_simplex_parameters           no
simplex_max_cycles                        1
simplex_score_converge                    0.1
simplex_cycle_converge                    1.0
simplex_trans_step                        1.0
simplex_rot_step                          0.1
simplex_tors_step                         10.0
simplex_anchor_max_iterations             500
simplex_grow_max_iterations               500
simplex_grow_tors_premin_iterations       0
simplex_random_seed                       0
simplex_restraint_min                     no
atom_model                                all
vdw_defn_file
<FOLDER>/dock6/parameters/vdw_AMBER_parm99.defn
flex_defn_file                            <FOLDER>/dock6/parameters/flex.defn
flex_drive_file                           <FOLDER>/dock6/parameters/flex_drive.tbl
ligand_outfile_prefix                     RECEPTOR_PDB_CODE.flex
write_orientations                        no
num_scored_conformers                     100
write_conformations                       no
cluster_conformations                     yes
cluster_rmsd_threshold                    2.0
rank_ligands                              yes
```

```
max_ranked_ligands                  500

rigid.in
conformer_search_type                rigid
use_internal_energy                  yes
internal_energy_rep_exp              12
internal_energy_cutoff               100.0
ligand_atom_file                     RECEPTOR_PDB_CODE.lig.mol2
limit_max_ligands                    no
skip_molecule                        no
read_mol_solvation                   no
calculate_rmsd                       yes
use_rmsd_reference_mol               yes
rmsd_reference_filename              RECEPTOR_PDB_CODE.lig.mol2
use_database_filter                  no
orient_ligand                        yes
automated_matching                   yes
receptor_site_file                   selected_spheres.sph
max_orientations                     1000
critical_points                      no
chemical_matching                    no
use_ligand_spheres                   no
bump_filter                          no
score_molecules                      yes
contact_score_primary                no
contact_score_secondary              no
grid_score_primary                   yes
grid_score_secondary                 no
grid_score_rep_rad_scale             1
grid_score_vdw_scale                 1
grid_score_es_scale                  1
grid_score_grid_prefix               grid
multigrid_score_secondary            no
dock3.5_score_secondary              no
continuous_score_secondary           no
```

footprint_similarity_score_secondary no
pharmacophore_score_secondary no
descriptor_score_secondary no
gbsa_zou_score_secondary no
gbsa_hawkins_score_secondary no
SASA_score_secondary no
amber_score_secondary no
minimize_ligand yes
simplex_max_iterations 1000
simplex_tors_premin_iterations 0
simplex_max_cycles 1
simplex_score_converge 0.1
simplex_cycle_converge 1.0
simplex_trans_step 1.0
simplex_rot_step 0.1
simplex_tors_step 10.0
simplex_random_seed 0
simplex_restraint_min yes
simplex_coefficient_restraint 10.0
atom_model all
vdw_defn_file
<FOLDER>/dock6/parameters/vdw_AMBER_parm99.defn
flex_defn_file <FOLDER>/dock6/parameters/flex.defn
flex_drive_file <FOLDER>/dock6/parameters/flex_drive.tbl
ligand_outfile_prefix RECEPTOR_PDB_CODE.rigid
write_orientations no
num_scored_conformers 40
write_conformations no
cluster_conformations yes
cluster_rmsd_threshold 2.0
rank_ligands yes
max_ranked_ligands 500

tv.in
conformer_search_type flex

```
user_specified_anchor                   no
limit_max_anchors                       no
min_anchor_size                         5
pruning_use_clustering                  yes
pruning_max_orients                     1000
pruning_clustering_cutoff               100
pruning_conformer_score_cutoff          100.0
pruning_conformer_score_scaling_factor  1.0
use_clash_overlap                       no
write_growth_tree                       no
use_internal_energy                     yes
internal_energy_rep_exp                 12
internal_energy_cutoff                  100.0
ligand_atom_file                        allLigands.mol2
limit_max_ligands                       no
skip_molecule                           no
read_mol_solvation                      no
calculate_rmsd                          yes
use_rmsd_reference_mol                  yes
rmsd_reference_filename                 RECEPTOR_PDB_CODE.lig.mol2
use_database_filter                     no
orient_ligand                           yes
automated_matching                      yes
receptor_site_file                      selected_spheres.sph
max_orientations                        3000
critical_points                         no
chemical_matching                       no
use_ligand_spheres                      no
bump_filter                             no
score_molecules                         yes
contact_score_primary                   no
contact_score_secondary                 no
grid_score_primary                      yes
grid_score_secondary                    no
grid_score_rep_rad_scale                1
```

```
grid_score_vdw_scale                    1
grid_score_es_scale                     1
grid_score_grid_prefix                  grid
multigrid_score_secondary               no
dock3.5_score_secondary                 no
continuous_score_secondary              no
footprint_similarity_score_secondary    no
pharmacophore_score_secondary           no
descriptor_score_secondary              no
gbsa_zou_score_secondary                no
gbsa_hawkins_score_secondary            no
SASA_score_secondary                    no
amber_score_secondary                   no
minimize_ligand                         yes
minimize_anchor                         yes
minimize_flexible_growth                yes
use_advanced_simplex_parameters         no
simplex_max_cycles                      1
simplex_score_converge                  0.1
simplex_cycle_converge                  1.0
simplex_trans_step                      1.0
simplex_rot_step                        0.1
simplex_tors_step                       10.0
simplex_anchor_max_iterations           500
simplex_grow_max_iterations             500
simplex_grow_tors_premin_iterations     0
simplex_random_seed                     0
simplex_restraint_min                   no
atom_model                              all
vdw_defn_file
<FOLDER>/dock6/parameters/vdw_AMBER_parm99.defn
flex_defn_file                          <FOLDER>/dock6/parameters/flex.defn
flex_drive_file                         <FOLDER>/dock6/parameters/flex_drive.tbl
ligand_outfile_prefix                   RECEPTOR_PDB_CODE.tv
write_orientations                      no
```

num_scored_conformers 100
write_conformations no
cluster_conformations yes
cluster_rmsd_threshold 2.0
rank_ligands yes
max_ranked_ligands 500

amber.in
conformer_search_type flex
user_specified_anchor no
limit_max_anchors no
min_anchor_size 5
pruning_use_clustering yes
pruning_max_orients 1000
pruning_clustering_cutoff 100
pruning_conformer_score_cutoff 100.0
pruning_conformer_score_scaling_factor 1.0
use_clash_overlap no
write_growth_tree no
use_internal_energy yes
internal_energy_rep_exp 12
internal_energy_cutoff 100.0
ligand_atom_file RECEPTOR_PDB_CODE.amber_score.mol2
limit_max_ligands no
skip_molecule no
read_mol_solvation no
calculate_rmsd yes
use_rmsd_reference_mol RECEPTOR_PDB_CODE.ranked.mol2
use_database_filter no
orient_ligand no
bump_filter no
score_molecules yes
contact_score_primary no
contact_score_secondary no
grid_score_primary no

```
grid_score_secondary                    no
multigrid_score_primary                 no
multigrid_score_secondary               no
dock3.5_score_primary                   no
dock3.5_score_secondary                 no
continuous_score_primary                no
continuous_score_secondary              no
footprint_similarity_score_primary      no
footprint_similarity_score_secondary    no
pharmacophore_score_primary             no
pharmacophore_score_secondary           no
descriptor_score_primary                no
descriptor_score_secondary              no
gbsa_zou_score_primary                  no
gbsa_zou_score_secondary                no
gbsa_hawkins_score_primary              no
gbsa_hawkins_score_secondary            yes
gbsa_hawkins_score_rec_filename         RECEPTOR_PDB_CODE.rec.mol2
gbsa_hawkins_score_solvent_dielectric   78.5
gbsa_hawkins_use_salt_screen            no
gbsa_hawkins_score_gb_offset            0.09
gbsa_hawkins_score_cont_vdw_and_es      yes
gbsa_hawkins_score_vdw_att_exp          6
gbsa_hawkins_score_vdw_rep_exp          12
grid_score_rep_rad_scale                1.0
SASA_score_primary                      no
amber_score_primary                     yes
amber_score_receptor_file_prefix        RECEPTOR_PDB_CODE
amber_score_movable_region              ligand
amber_score_minimization_rmsgrad        0.01
amber_score_before_md_minimization_cycles   100
amber_score_md_steps                    3000
amber_score_after_md_minimization_cycles    100
amber_score_gb_model                    5
amber_score_nonbonded_cutoff            18.0
```

amber_score_temperature 310.0
amber_score_abort_on_unprepped_ligand yes
atom_model all
vdw_defn_file
<FOLDER>/dock6/parameters/vdw_AMBER_parm99.defn
flex_defn_file <FOLDER>/dock6/parameters/flex.defn
flex_drive_file <FOLDER>/dock6/parameters/flex_drive.tbl
ligand_outfile_prefix amber
write_orientations no
num_primary_scored_conformers_rescored 1
num_secondary_scored_conformers 1
rank_primary_ligands no
rank_secondary_ligands no

tv_amber.in
conformer_search_type flex
user_specified_anchor no
limit_max_anchors no
min_anchor_size 5
pruning_use_clustering yes
pruning_max_orients 1000
pruning_clustering_cutoff 100
pruning_conformer_score_cutoff 100.0
pruning_conformer_score_scaling_factor 1.0
use_clash_overlap no
write_growth_tree no
use_internal_energy yes
internal_energy_rep_exp 12
internal_energy_cutoff 100.0
ligand_atom_file allLigandsAmber.mol2
limit_max_ligands no
skip_molecule no
read_mol_solvation no
calculate_rmsd yes
use_rmsd_reference_mol yes

```
rmsd_reference_filename                 RECEPTOR_PDB_CODE.amber_score.mol2
use_database_filter                     no
orient_ligand                           no
bump_filter                             no
score_molecules                         yes
contact_score_primary                   no
contact_score_secondary                 no
grid_score_primary                      no
grid_score_secondary                    no
multigrid_score_primary                 no
multigrid_score_secondary               no
dock3.5_score_primary                   no
dock3.5_score_secondary                 no
continuous_score_primary                no
continuous_score_secondary              no
footprint_similarity_score_primary      no
footprint_similarity_score_secondary    no
pharmacophore_score_primary             no
pharmacophore_score_secondary           no
descriptor_score_primary                no
descriptor_score_secondary              no
gbsa_zou_score_primary                  no
gbsa_zou_score_secondary                no
gbsa_hawkins_score_primary              no
gbsa_hawkins_score_secondary            yes
gbsa_hawkins_score_rec_filename         RECEPTOR_PDB_CODE.rec.mol2
gbsa_hawkins_score_solvent_dielectric   78.5
gbsa_hawkins_use_salt_screen            no
gbsa_hawkins_score_gb_offset            0.09
gbsa_hawkins_score_cont_vdw_and_es      yes
gbsa_hawkins_score_vdw_att_exp          6
gbsa_hawkins_score_vdw_rep_exp          12
grid_score_rep_rad_scale                1.0
SASA_score_primary                      no
amber_score_primary                     yes
```

```
amber_score_receptor_file_prefix                RECEPTOR_PDB_CODE
amber_score_movable_region                      ligand
amber_score_minimization_rmsgrad                0.01
amber_score_before_md_minimization_cycles       100
amber_score_md_steps                            3000
amber_score_after_md_minimization_cycles        100
amber_score_gb_model                            5
amber_score_nonbonded_cutoff                    18.0
amber_score_temperature                         310.0
amber_score_abort_on_unprepped_ligand           yes
atom_model                                      all
vdw_defn_file
<FOLDER>/dock6/parameters/vdw_AMBER_parm99.defn
flex_defn_file                                  <FOLDER>/dock6/parameters/flex.defn
flex_drive_file                                 <FOLDER>/dock6/parameters/flex_drive.tbl
ligand_outfile_prefix                           RECEPTOR_PDB_CODE.tv_amber
write_orientations                              no
num_primary_scored_conformers_rescored          1
num_secondary_scored_conformers                 1
rank_primary_ligands                            no
rank_secondary_ligands                          no
```